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Arbitrage-Consistent Neural Smoothing of the Implied Volatility Surface: A State-Space Approach

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Master's Thesis in Econometrics and Operations Research

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Abstract

This paper develops an arbitrage-aware framework for forecasting implied volatility surface dynamics based on European SPX options from January 2010 to December 2023. The model combines tensor-product spline smoothing, mean-reverting Ornstein-Uhlenbeck state-space dynamics, and an arbitrage-aware neural correction targeting systematic contract-level errors in the tails of the surface. Empirically, the neural smoother delivers the lowest forecast errors among benchmark models, reducing full-contract mean absolute error by approximately 66% for aggregate forecasts and 71% for deep out-of-the-money put forecasts relative to the most restrictive benchmark. At the preferred penalty value, no-arbitrage penalization reduces total violations by approximately 89% with only a limited loss in forecast accuracy relative to the unpenalized specification. The gains extend to risk-neutral density forecasts, where the neural smoother performs best under both log-density and continuous ranked probability scores, although Value-at-Risk coverage tests indicate that tail dynamics remain difficult to capture.

Keywords: Implied volatility, No-arbitrage, State-space modeling, Spline smoothing, Neural network, Risk-neutral density.

AI acknowledgment

Due to the rapid development and increasing availability of AI-based tools, acknowledging their use is important to ensure transparency and uphold academic integrity. In this paper, ChatGPT was used to support language refinement, code debugging, verification of code logic and implementation, and LaTeX formatting. QuillBot was used for spelling checks and minor language corrections. All final decisions regarding the content, methodology, analysis, interpretation, and conclusion remain the responsibility of the author.

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1. Introduction

Option prices contain key information about market sentiment in financial markets, with values depending on various contract characteristics, including time to maturity and moneyness. Since direct comparisons across contracts are not straightforward, prices are often expressed in terms of implied volatility by inverting the Black and Scholes (1973) option pricing formula. Across maturities and levels of moneyness, this produces an implied volatility surface. Under the assumptions of the Black-Scholes model, this surface should be flat, as options written on the same underlying asset should yield a single, constant implied volatility. However, empirical evidence consistently contradicts this implication: implied volatilities vary systematically across strike prices and times to expiration, and these cross-sectional differences are not random but follow common patterns (e.g., Bollen & Whaley, 2004; Rubinstein, 1985). More specifically, for a given maturity, deep in- or out-of-the-money options tend to have higher implied volatilities than at-the-money options, producing the volatility smile (Dumas et al., 1998). Beyond moneyness, implied volatility also varies across maturities, giving rise to the volatility term structure (e.g., Y. Chen et al., 2018; Mixon, 2007).

Beyond these stylized facts, Carr and Wu (2016) show that volatility risk and risk premia can be extracted from volatility surfaces. Since these surfaces are constructed from option prices, they embed market expectations and risk compensation across different states of the world, while also making deviations from the log-normal Black-Scholes benchmark apparent. Unlike historical volatility measures, implied volatility provides a forward-looking measure of the market’s assessment of future risk (Castrén, 2005). As a result, the shape of the surface can reveal how investors price downside tail risk and how uncertainty is distributed across horizons. In particular, elevated levels and steep downside skew are consistent with greater fear of rare disaster events and higher associated risk premia (Bollerslev & Todorov, 2011). This also explains why out-of-the-money put options are priced at a premium, as they provide insurance against disaster states, implying higher volatility relative to at-the-money options (Seo & Wachter, 2019).

Moreover, the implied volatility surface is not static. Early two-stage methods exploit cross-sectional and time-series information to show that its dynamics can be modeled and predicted statistically (Gonçalves & Guidolin, 2006). Furthermore, since option contracts written on the same underlying asset are expected to exhibit strong co-movement across strikes and maturities, this dependence naturally carries over to the implied volatility surface (van der Wel et al., 2016). This motivates representing the shared variation by a small number of common factors, as in Cont and da Fonseca (2002), who model implied volatility surface dynamics using principal component analysis, and Fengler et al. (2007), who introduce a dynamic semiparametric factor model. Given these time-varying dynamics of the surface and the common factors that govern its evolution, a state-space framework provides a natural modeling approach by separating the observed implied volatility surface from the latent factors that drive its movements over time, as frequently explored in the literature (e.g., S. Chen et al., 2016; Wang et al., 2017).

However, a few limitations commonly arise in the existing literature. First, the literature often reports region-specific difficulties in modeling implied volatility surfaces. Dynamic factor models fit the center of the surface relatively well but struggle in the tails and corners, where residuals can deviate from white-noise behavior and display autocorrelation and heteroskedasticity (e.g., van der Wel et al., 2016; van Vliet, 2017). Similarly, Ulrich et al. (2023) document large prediction errors for short-maturity out-of-the-money put options, suggesting that the downside-tail region is particularly difficult to model accurately. This is especially important in light of the earlier discussion on tail-risk premia: deep out-of-the-money puts are central to downside-risk pricing, where rare-disaster concerns and insurance demand are most strongly reflected.

Second, enforcing no-arbitrage restrictions is nontrivial when surfaces are modeled or forecasted directly. Existing literature often ignores arbitrage consistency during the cleaning of data, and standard smoothing and forecasting methods do not generally yield an arbitrage-free surface, since no-arbitrage constraints become high-dimensional and nonlinear in option-price and implied volatility space, especially in dynamic settings (e.g., Cont & Vuletić, 2023; Dellaportas & Mijatović, 2014; Fengler, 2009).

Recent advances in machine learning address these limitations by exploring how implied volatility surface models can remain flexible while also allowing economic restrictions to be incorporated. For example, Zeng and Klabjan (2019) develop an adaptive support vector regression approach for surface modeling, while Akerer et al. (2020) use neural networks to smooth implied volatility surfaces. Similarly, Zheng et al. (2021) show that financial domain knowledge can be embedded in neural network specifications, thereby incorporating no-arbitrage conditions directly into the training objective. At the same time, most of these approaches are primarily concerned with either static surface fitting or direct surface prediction, thereby not explicitly combining learning-based flexibility with latent factor dynamics in a state-space framework. This leaves room for a hybrid approach that retains the interpretability and temporal structure of dynamic factor models while using machine learning as a flexible correction layer.

Motivated by these limitations and advances, this paper examines whether a three-part modeling framework can reduce localized forecasting errors, particularly in the corners of the surface, while improving economic consistency with no-arbitrage restrictions. First, the cross-sectional shape of the surface is represented through a spline-based specification following Fengler and Hin (2015), allowing the surface to be summarized by a small number of time-varying factors in line with factor-based implied volatility literature (e.g., Fengler et al., 2007; van der Wel et al., 2016). Second, in line with Wang et al. (2017), the time-series evolution of these latent factors is modeled in state-space form using mean-reverting Ornstein-Uhlenbeck dynamics, preserving the interpretability of dynamic factor models while allowing the surface to evolve over time. Third, the paper incorporates a neural smoothing correction for the residual component of the surface by adjusting the approach of Akerer et al. (2020). This learning-based step is designed to capture local nonlinear structure that remains unexplained by the prior model specification, with particular emphasis on the corners and deep out-of-the-money put region, where the literature documents systematic fitting and forecasting difficulties.

Furthermore, this paper addresses the arbitrage limitations from two complementary perspectives. First, an iterative no-arbitrage filtering procedure is introduced during the data-cleaning stage. This follows Moussa (2026), who emphasizes no-arbitrage preprocessing for European option quotes, and Ait-Sahalia and Duarte (2003), who show that shape restrictions are central to economically consistent option prices. By applying these restrictions before estimation, this paper extends the arbitrage-free implied volatility surface literature with a data construction step designed to prevent the model from learning arbitrage-contaminated surface dynamics (e.g., Cont & Vuletić, 2023; Dellaportas & Mijatović, 2014; Fengler, 2009). Second, inspired by the recent advances in machine learning (e.g., Akerer et al., 2020; Zheng et al., 2021), the third step of the modeling approach leverages soft no-arbitrage penalties, thereby promoting predicted surfaces that are both locally flexible and more consistent with arbitrage-free option pricing restrictions.

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature and positions this study relative to existing research. Section 3 describes the data, including the construction and cleaning of the option chain, and introduces the iterative no-arbitrage algorithm. Section 4 presents the methodology, including the setup and evaluation metrics. The empirical results are reported in Section 5, after which Section 6 provides a discussion and directions for further research. Finally, Section 7 concludes the study.

2. Literature review

A first strand of literature models the implied volatility surface as a high-dimensional object driven by a small number of common sources of variation. Skiadopoulos et al. (2000) show that principal components explain a large share of the variation in the S&P 500 implied volatility surface, while Fengler et al. (2003) obtain similar evidence using common principal components across maturity slices. Bedendo and Hodges (2009) further show that state-space methods can capture volatility skew dynamics, and van der Wel et al. (2016) estimate dynamic factor models for the full volatility surface. These findings motivate a low-dimensional representation of surface dynamics, in which latent factors summarize the dominant movements across strikes and maturities. However, because these specifications primarily capture global variation, they can leave systematic local errors in short maturities, tail contracts, and surface corners (e.g., van der Wel et al., 2016; van Vliet, 2017). This limitation is central to the present paper, which retains the interpretability of dynamic factor models while targeting localized contract-level forecast errors. Relative to this literature, this paper combines tensor-product splines with state-space latent dynamics, providing a flexible but economically interpretable foundation for implied volatility forecasting.

A second strand of literature highlights why local regions of the implied volatility surface are economically important rather than merely statistically difficult to fit. Option prices contain information about stochastic volatility, jumps, jump-risk premia, and volatility-risk premia, as shown by Bates (1996), Pan (2002), and Eraker (2004). This information is particularly relevant in index options, where downside-risk compensation and volatility-risk premia are strongly reflected in option prices (e.g., Bakshi & Kapadia, 2003; Bollerslev et al., 2011), especially out-of-the-money puts (Bondarenko, 2014). Consequently, because tail-region option prices are closely linked to rare-disaster fears and downside-risk compensation, forecast errors in this region may affect the measurement of economically meaningful downside-risk information (e.g., Bollerslev & Todorov, 2011; Seo & Wachter, 2019). This motivates an evaluation approach that looks beyond aggregate surface-level forecast errors, since average performance can hide economically important misspecification in precisely the contracts most closely linked to downside-risk pricing. Accordingly, the present paper contributes a forecasting framework aimed at reducing systematic region-specific residuals and contract-level forecast errors. Particular attention is given to short-maturity out-of-the-money puts, where errors have been shown to be especially pronounced (Ulrich et al., 2023).

A third strand of literature examines the use of machine learning for option-based applications. Early contributions show that neural networks can learn option-pricing relations and improve derivative pricing performance (e.g., Hutchinson et al., 1994; Lajbcygier & Connor, 1997). Garcia and Gençay (2000) further show that incorporating financial restrictions improves out-of-sample pricing performance. More recent studies use deep learning to approximate option prices and implied volatilities efficiently (e.g., Culkun & Das, 2017; Liu et al., 2019). Within this literature, machine learning methods have increasingly been used to combine flexibility with economic restrictions. Akerer et al. (2020) propose a neural smoothing method with soft arbitrage penalties, Zheng et al. (2021) embed financial domain knowledge into neural networks for surface prediction, and Ning et al. (2023) combine variational autoencoders with arbitrage-free SDE-based models to generate implied volatility surfaces. However, these approaches typically use machine learning either as the primary pricing model, a direct surface predictor, or a static smoothing device. As a result, learning-based flexibility is often separated from the interpretable dynamic structure of the surface. Building on this literature, the present paper uses machine learning as a residual correction layer on top of a dynamic state-space prior, combining nonlinear flexibility with economically interpretable surface dynamics.

A fourth strand of literature concerns arbitrage-free smoothing and economically admissible option surfaces. One part of this literature characterizes the no-arbitrage restrictions governing option-price and implied volatility surfaces across strikes and maturities, thereby defining the set of economically admissible surface shapes (Davis & Hobson, 2007; Roper, 2010). These restrictions impose economically meaningful shape conditions on option prices rather than serving merely as technical regularization devices. In particular, Breeden and Litzenberger (1978) link the second derivative of call prices with respect to strike to state prices, providing an economic rationale for convexity restrictions. Another part of the literature enforces the restrictions through arbitrage-free interpolation and SVI-based parameterizations (e.g., Gatheral & Jacquier, 2014; Kahalé, 2004). However, these approaches typically treat arbitrage consistency as a static surface-construction or parameterization problem. The present paper instead considers arbitrage consistency in a dynamic forecasting framework. Economic admissibility is incorporated at two stages: an iterative no-arbitrage filter is applied before estimation, while the soft arbitrage penalty approach of Ackerer et al. (2020) is extended to the dynamic neural correction framework.

In this context, this paper documents the trade-off between statistical accuracy and economic admissibility. Flexible smoothing methods may reduce forecast errors by adapting to local nonlinearities in the observed option surface, but without explicit no-arbitrage constraints, they may generate implied volatility surfaces that violate no-arbitrage restrictions (Ackerer et al., 2020; Zheng et al., 2021). Conversely, imposing an arbitrage-free structure improves admissibility by ruling out static arbitrage (Gatheral & Jacquier, 2014; Roper, 2010), but also restricts the set of feasible surfaces the model can produce. The relevant question is therefore whether improved forecasting accuracy can be achieved without sacrificing economically admissible predictions.

Finally, this paper contributes to the literature on risk-neutral density forecasting and risk-management evaluation. Within this literature, risk-neutral or state-price densities are recovered from option prices to study market expectations and the pricing of future states of the underlying asset (e.g., Aït-Sahalia & Lo, 1998; Bliss & Panigirtzoglou, 2002; Jackwerth & Rubinstein, 1996). Accordingly, the reliability of the resulting densities depends on the accuracy and arbitrage consistency of the underlying implied volatility surface. This motivates evaluating density forecasts using probability integral transforms, likelihood-based tests, and proper scoring rules (Berkowitz, 2001; Diebold et al., 1998; Gneiting & Raftery, 2007). Risk-management evaluation provides a complementary perspective, since Value-at-Risk backtesting assesses whether predicted tail behavior translates into accurate downside-risk forecasts through coverage tests and dynamic quantile-based methods (P. F. Christoffersen, 1998; Engle & Manganelli, 2004; Kupiec, 1995). Relative to this literature, the present paper evaluates whether improvements in implied volatility surface forecasts also carry over to risk-neutral density forecasts and Value-at-Risk performance, given that volatility and tail-quantile forecasting are distinct tasks (Bams et al., 2017).

Taken together, existing literature provides dynamic factor models for surface dynamics, machine learning methods for flexible option pricing and smoothing, and no-arbitrage restrictions for economic consistency. However, these components are usually studied separately. Dynamic models preserve temporal structure but may leave systematic local residuals. Machine learning methods provide flexibility but are often used as standalone black-box models. Arbitrage-free smoothing methods impose economic discipline but are frequently static rather than dynamic. This paper addresses this gap by combining a spline-based state-space prior, mean-reverting factor dynamics, and an arbitrage-aware neural residual correction. The central contribution is therefore to improve localized contract-level implied volatility forecasts while preserving both interpretable surface dynamics and economic admissibility.

3. Data

Reliable volatility surfaces are constructed using historical options market data. First, Subsection 3.1 describes the data sources and index selection, after which Subsection 3.2 discusses the cleaning procedures and introduces an iterative no-arbitrage filtering algorithm. Furthermore, Subsection 3.3 presents the implied volatility surface construction, after which Subsection 3.4 provides a preliminary analysis to describe the structural characteristics and motivate the methodologies throughout this paper. Particular attention is drawn to the consistency and economic plausibility of the option data.

3.1. Data sources and index selection

Grover and Thomas (2012) emphasize that illiquidity can distort observed implied volatilities, consistent with illiquidity discounts in currency options documented by Brenner et al. (2001). Against this background, S&P 500 index options provide a suitable empirical setting for analysis, with the S&P 500 options market being highly liquid, relatively less prone to market frictions, and widely used in empirical options literature. Its broad strike coverage and wide range of maturities make it appropriate for the construction of volatility surfaces.

End-of-day data are retrieved from optionsDX¹, which provides detailed data on European-style SPX index options traded on the Chicago Board Options Exchange (CBOE). This choice of data is consistent with recent work by Zou et al. (2025), who rely on the same data source in an analysis of implied volatility surface dynamics. The dataset contains 15,556,477 pre-filtered observations over the period January 4, 2010, to December 29, 2023. This covers over 13 years for a total of 3499 distinct days, excluding weekends and holidays. After converting the dataframe into long format, that is, stacking puts and calls in separate rows, the dataset contains 31,112,954 observations. Variables include the quote date, underlying price, expiry date, days-to-expiry, bid and ask quotes, and relevant option sensitivities. Bid-ask midpoints, defined as the average of both corresponding quotes, are used as proxies for the true option price instead of transaction prices to reduce issues stemming from bid-ask bounces.

A key input in the analysis is the choice of a risk-free rate. While measures such as the Secured Overnight Funding Rate (SOFR) could be used as a proxy for the risk-free rate, Brace et al. (2024) note that SOFR is an overnight rate, so maturity-specific term rates must be constructed separately, making it less suitable in this context. Historical data on the risk-free rate are thus retrieved from the U.S. Department of Treasury.² Specifically, the daily Treasury Par Yield Curve Rates are used, providing constant-maturity yields across a range of maturities. This approach is consistent with a range of empirical option studies, which rely on Treasury-based risk-free rates (e.g., Sridi & Bilokon, 2023; Yamazaki, 2022). For each observation, the yield corresponding to the maturity closest to the option's time to maturity is selected, after which all rates are converted from percentage terms to decimals before being used as the risk-free rate input.

Finally, data on the monthly annualized S&P 500 dividend yields are obtained from the multpl online database³, consistent with recent literature (e.g., Beare et al., 2025; Pazarbasi et al., 2019). This source provides monthly historical series based on the Standard & Poor's reported current S&P 500 dividend yield,

¹The S&P 500 option data are available via <https://www.optionsdx.com/>.

²The US Treasury database is available via <https://home.treasury.gov/resource-center/data-chart-center/interest-rates/daily-treasury-rate-archives>.

³The S&P 500 monthly dividend yield data are available via <https://www.multpl.com/s-p-500-dividend-yield/table/by-month>.

supplemented with historical estimates from Shiller (2015). As the data are available only at a monthly frequency, the dividend yield is treated as constant within each month. In other words, each monthly observation is assumed to apply uniformly to all trading days in that month, implying no intra-month variation.

3.2. Data cleaning

A sequence of cleaning procedures is applied to obtain a dataset that is reliable and suitable for constructing implied volatility surfaces. This subsection first presents a set of well-established quote-filtering rules from the existing literature, aimed at reducing the influence of illiquid or noisy observations. These filters are then followed by several no-arbitrage constraints, which are implemented through an iterative procedure to ensure that the data are economically consistent.

3.2.1. Initial quote filtering

As a first step, option quotes are filtered following Jiang and Tian (2005), who provide a key filtering procedure in which (i) option quotes below \$3/8 are excluded, as their quotes are strongly affected by tick-size discreteness. For such options, the minimum price increment represents a large relative change, preventing prices from adjusting smoothly to small changes in underlying value, time-to-maturity, or volatility. Next, (ii) options with less than one week to maturity are excluded due to liquidity and market microstructure concerns, while, following Barone-Adesi et al. (2008), options with time-to-maturity above 360 days are discarded to ensure a more homogeneous cross-section of options. Additionally, (iii) in-the-money⁴ options, which are typically more expensive, less liquid, and trade at lower volumes than at-the-money or out-of-the-money options, are removed. In line with van der Wel et al. (2016), in-the-money options are defined as call options with delta greater than 0.5 and put options with delta less than -0.5. The choice to use delta as a measure of moneyness is substantiated in Subsection 3.3, and does not result in a loss of information, since in-the-money options can be matched to out-of-the-money options with the same strike and maturity through put-call parity.⁵ Finally, (iv) implied volatility quotes above 1 are discarded to mitigate potential data-related errors.

3.2.2. No-arbitrage algorithm

Carr and Madan (2005) emphasize that the absence of arbitrage is a fundamental principle in asset pricing and imposes strict cross-sectional restrictions on prices. Since full verification of arbitrage-freeness is difficult in practice, it is natural to impose simple no-arbitrage restrictions to ensure that the observed option prices are economically consistent and do not imply riskless profit opportunities. Imposing these restrictions is essential for preventing analysis from being driven by pricing errors or market microstructure noise and avoiding estimation and calibration on mispriced observations. Let C_t denote the time t call price, P_t the put price, T the maturity in years, $\mathbb{Q}$ the risk-neutral probability measure, r the annualized risk-free rate, and q the annualized dividend yield. Then, using the risk-neutral call option pricing formula, $C_t =$

⁴Note that in-the-money options may be abbreviated by ITM, at-the-money by ATM, out-of-the-money by OTM, and deep out-of-the-money by DOTM.

⁵Let C_t and P_t denote the time t call and put option prices, respectively. Then the put-call parity is given by $C_t - P_t = S_t e^{-q(T-t)} - K e^{-r(T-t)}$. This no-arbitrage condition implies that an ITM call can be replicated by the corresponding OTM put, and vice versa, with the corresponding deltas satisfying $\Delta_t^P = \Delta_t^C - e^{-q(T-t)}$. Information discarded in ITM options is thus already contained in the retained OTM option.

$e^{-r(T-t)}\mathbb{E}_t^{\mathbb{Q}}[\max(S_T - K, 0)]$, the most immediate bounds are the lower bounds for call and put options on an index with a continuous dividend yield:

$$C_t \geq \max(S_t e^{-q(T-t)} - K e^{-r(T-t)}, 0), \quad P_t \geq \max(K e^{-r(T-t)} - S_t e^{-q(T-t)}, 0). \quad (1)$$

The proof proceeds by exploiting the structure of the risk-neutral expectation formula. Let $(\cdot)^+ = \max(\cdot, 0)$, then by Jensen's inequality for convex functions, it follows that $\mathbb{E}_t^{\mathbb{Q}}[(S_T - K)^+] \geq (\mathbb{E}_t^{\mathbb{Q}}[S_T - K])^+$. Since $e^{-r(T-t)} > 0$, multiplying both sides by the discount factor preserves the direction of the inequality, and the factor can equivalently be taken inside the maximum operator. Additionally, under the risk-neutral probability measure $\mathbb{Q}$, the expected stock price grows by the risk-free rate adjusted for the dividend yield, $r - q$, yielding⁶

$$\begin{aligned} C_t &\geq e^{-r(T-t)}(\mathbb{E}_t^{\mathbb{Q}}[S_T - K])^+ \\ &= e^{-r(T-t)}(S_t e^{(r-q)(T-t)} - K)^+ \\ &= (S_t e^{-q(T-t)} - K e^{-r(T-t)})^+. \end{aligned} \quad (2)$$

Similar reasoning applies to the put options. The upper bounds follow analogously from payoff dominance, since $(S_T - K)^+ \leq S_T$. Applying risk-neutral pricing then yields the following result:

$$C_t = e^{-r(T-t)}\mathbb{E}_t^{\mathbb{Q}}[(S_T - K)^+] \leq e^{-r(T-t)}\mathbb{E}_t^{\mathbb{Q}}[S_T] = S_t e^{-q(T-t)}. \quad (3)$$

Taken together, $C_t \leq S_t e^{-q(T-t)}$, implying that the call price cannot exceed the dividend-adjusted present value of the underlying. Similarly, $P_t \leq K e^{-r(T-t)}$, implying that the put price cannot exceed the discounted strike price. Moussa (2026) outlines several other key no-arbitrage bounds for preprocessing European option quotes. First, (i) monotonicity in strike implies that the first derivative of the call price with respect to the strike must satisfy $-e^{-r\tau} \leq \partial C_t / \partial K \leq 0$ for identical time to maturity $\tau = T - t$ (Aït-Sahalia & Duarte, 2003). Operationally, for a fixed quote date t and maturity T , the discrete slope between each pair of adjacent ordered strikes $K_i < K_{i+1}$ must satisfy

$$-e^{-r\tau} \leq \frac{C_t(K_{i+1}, T) - C_t(K_i, T)}{K_{i+1} - K_i} \leq 0. \quad (4)$$

The opposite condition applies to puts, as their prices are nondecreasing in strike. Second, (ii) option prices must be convex in strike K , i.e., $\partial^2 C_t(K, T) / \partial K^2 \geq 0$, which is enforced by sorting the options by strike and ensuring that

$$\frac{C_t(K_i, T) - C_t(K_{i-1}, T)}{K_i - K_{i-1}} \leq \frac{C_t(K_{i+1}, T) - C_t(K_i, T)}{K_{i+1} - K_i}. \quad (5)$$

Analogously, since put options are also convex in strike, the same condition applies. Building on the result of Breeden and Litzenberger (1978), Aït-Sahalia and Duarte (2003) note that local convexity failures imply negative state prices and hence violate no-arbitrage. Moreover, because removing one quote can alter whether nearby quotes violate no-arbitrage conditions, the filtering procedure is applied iteratively to the dataset until no further removals occur, yielding a stable subset that satisfies the specified no-arbitrage restrictions. The resulting filtered dataset forms the basis for the analysis of upcoming sections.⁷

⁶Under the risk-neutral measure, $\mathbb{E}_t^{\mathbb{Q}}[S_T] = S_t e^{(r-q)(T-t)}$, which reflects that the expected return on the underlying equals the risk-free rate adjusted for dividends.

⁷Pseudo-code of the no-arbitrage algorithm and a data reduction table is found in Appendix A.

3.3. Construction of the IV surface

Computation of implied volatility (IV) surfaces relies on the option pricing framework of Black and Scholes (1973), with dividend-related adjustments motivated by the payout analysis of Merton (1973). Although its underlying assumptions are often empirically violated, it serves as a standardized mapping from prices to implied volatilities. This allows for consistent comparison of option prices across strikes and maturities. Accordingly, the European call and put option valuation formulae are given by

$$\begin{aligned} C_t &= S_t e^{-q(T-t)} \Phi(d_1) - K e^{-r(T-t)} \Phi(d_2), & d_1 &= \frac{\log(S_t/K) + (r - q + 0.5\sigma^2)(T-t)}{\sigma\sqrt{T-t}}, \\ P_t &= K e^{-r(T-t)} \Phi(-d_2) - S_t e^{-q(T-t)} \Phi(-d_1), & d_2 &= d_1 - \sigma\sqrt{T-t}. \end{aligned} \quad (6)$$

The notation in Eq. (6) is as follows: S_t is the time t underlying index level, K is the option's strike price, r is the risk-free rate, q is the dividend yield, $T-t$ is the remaining time to maturity, σ is the volatility of the underlying asset's returns, and $\Phi(\cdot)$ is the cumulative distribution function of the standard normal. The implied volatility, studied as the 'implied standard deviation' by Latané and Rendleman (1976), is defined as the value of σ , holding all other inputs fixed, that equates the Black-Scholes option price to the observed market price. Let $\tau = T-t$ denote the remaining time-to-maturity, then $C_t^{BS}(S_t, K, r, q, \tau, \sigma_{IV}) = C_t^{mkt}(K, \tau)$, where σ_{IV} is defined as the volatility value that equates the Black-Scholes price to the observed market price. No closed-form solution exists, as the Black-Scholes pricing formula is not analytically invertible with respect to σ . Consequently, implied volatility is obtained by numerical root finding:

$$f(\sigma) = C_t^{BS}(S_t, K, r, q, \tau, \sigma) - C_t^{mkt}(K, \tau). \quad (7)$$

The implied volatility is then the value σ_{IV} that solves $f(\sigma_{IV}) = 0$. This procedure applies analogously to put options. Once obtained, the implied volatility surface spans the moneyness and maturity dimensions of option contracts for a fixed t , $\sigma(m, \tau)$ for $m \in \mathbb{R}$ and $\tau > 0$. Bollen and Whaley (2004) note that strike-to-spot moneyness, typically defined as the ratio of strike to underlying price, K/S_t , ignores that the probability of finishing in the money also depends on volatility and time to maturity. As a result, this measure does not fully capture how far an option is from ending ITM, making classification comparisons across assets and maturities problematic. To address this, the authors propose to use delta, defined as the sensitivity of the option with respect to changes in the underlying,

$$\Delta_t^c = \frac{\partial C_t}{\partial S_t} = e^{-q(T-t)} \Phi(d_1), \quad \Delta_t^p = \frac{\partial P_t}{\partial S_t} = -e^{-q(T-t)} \Phi(-d_1) = \Delta_t^c - e^{-q(T-t)}. \quad (8)$$

From Eq. (6) and Eq. (8), delta depends both on volatility and time to maturity, reflecting its sensitivity to key option characteristics. Its computation therefore requires a volatility input σ , which, under the sticky strike assumption of Daglish et al. (2007), is taken to be the option's implied volatility and treated as fixed with respect to changes in the underlying price. Furthermore, for options on a non-dividend-paying underlying, the call delta can be interpreted as the risk-neutral probability of finishing in-the-money when using the stock as numéraire, which provides a probabilistic justification for its use as a moneyness measure. However, because dividend yields are accounted for, this interpretation should be viewed as an approximation. As a result, for classification purposes, the data are divided into 24 Δ -moneyness-maturity⁸ buckets following van der Wel et al. (2016). Four maturity groups are defined using cutoffs at 45, 90, and 180 days. In addition, six moneyness categories are distinguished, three for call options and three for put options. Namely, call

⁸From this point onward, moneyness measured using delta will be referred to as Δ -moneyness.

options with $0.375 < \Delta_t^c < 0.5$ are labeled as at-the-money, $0.125 < \Delta_t^c < 0.375$ as out-of-the-money, and $0 < \Delta_t^c < 0.125$ as deep out-of-the-money. Similarly, put options with $-0.125 < \Delta_t^p < 0$ are labeled deep out-of-the-money, $-0.375 < \Delta_t^p < -0.125$ as out-of-the-money, and $-0.5 < \Delta_t^p < -0.375$ as at-the-money. Accordingly, dividing the options in this way helps control for contract characteristics, reduces noise from changing contract compositions, and is in line with literature (e.g., Barone-Adesi et al., 2008; Bollen & Whaley, 2004). The choice of 24 groups balances obtaining a balanced panel of similar contracts over the entire sample period and representing movements in the cross-section of options (van der Wel et al., 2016).

3.4. Summary statistics and preliminary analysis

The data are explored through constructed volatility surfaces and descriptive statistics to motivate the methodology. First, two non-bucketed implied volatility surfaces and summary statistics across buckets illustrate the stylized facts discussed in Section 1. Second, principal component analysis, together with term-structure and moneyness-slope measures, is used to assess whether a small number of common factors captures the main features of the implied volatility surface. Finally, residuals of a preliminary model are analyzed to uncover region-specific systematic structure.

3.4.1. Implied volatility surfaces

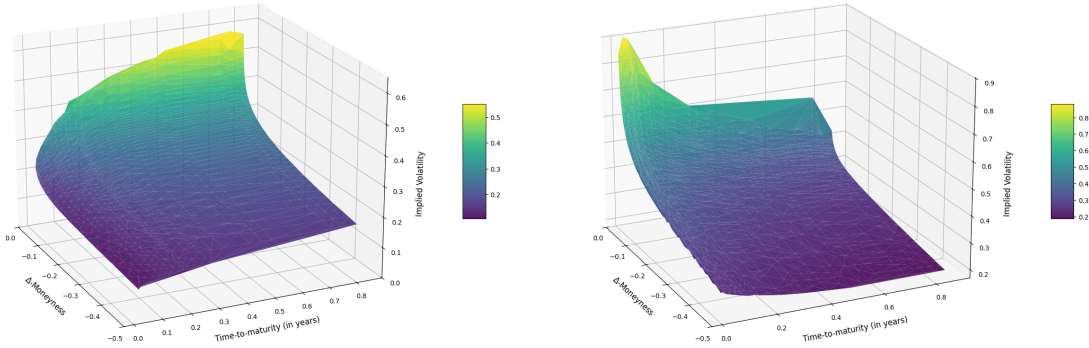


Figure 1: **3D surface plots of SPX European put option implied volatility surfaces (IVS) on March 21, 2016 and March 4, 2020.** The figure displays implied volatility (annualized) across Δ^P -moneyness and time-to-maturity (in years). The left panel corresponds to March 21, 2016, and the right panel to March 4, 2020. Lighter colors indicate higher levels of implied volatility, while darker colors indicate lower levels.

Figure 1 presents the implied volatility surfaces of European SPX put options on March 21, 2016, and March 4, 2020. The figure shows substantial variation across both moneyness and maturity. In both panels, implied volatility rises as Δ_t^p approaches zero, consistent with the volatility skew discussed in Section 1: deep out-of-the-money put options command a premium relative to constant-volatility Black and Scholes (1973) prices. The skew is clearly nonlinear, with implied volatility increasing more sharply near $\Delta_t^p = 0$ than it declines for more negative values, indicating particularly high demand for deep out-of-the-money protection.

Moreover, the term structure differs across the two dates. The left panel exhibits an upward-sloping term structure, with implied volatility increasing with maturity, indicating greater uncertainty at longer horizons. In contrast, the right panel displays a downward-sloping term structure, with implied volatility decreasing with maturity, consistent with heightened short-term uncertainty during the COVID-19 period. This suggests

that, unlike the volatility skew, the slope of the term structure varies between calm and stressful periods. Importantly, the 2020 surface is not merely an upward shift of the 2016 surface. It is more sharply curved, particularly at short maturities, implying that stressed market conditions affect not only the overall level of implied volatility but also its distribution across moneyness and horizon. In this sense, the right panel reflects a stronger concentration of risk in short-dated options, as markets assign substantial weight to immediate uncertainty and jump risk while expecting stress to decay over longer maturities.

3.4.2. Summary statistics

		7–45 days		45–90 days		90–180 days		180–360 days	
		Mean	Std.	Mean	Std.	Mean	Std.	Mean	Std.
DOTM Put	Mid-Quote	3.754	3.977	6.530	6.883	11.330	11.872	17.579	17.887
	IV	0.305	0.136	0.341	0.138	0.371	0.141	0.370	0.125
	Δ^p	-0.042	0.034	-0.042	0.035	-0.041	0.036	-0.042	0.036
	K/S	0.864	0.086	0.765	0.122	0.671	0.151	0.586	0.161
	Maturity	24.004	10.614	65.209	12.639	131.090	26.433	267.924	52.242
OTM Put	Mid-Quote	22.377	15.531	35.953	23.510	65.332	38.253	97.425	55.828
	IV	0.191	0.080	0.203	0.066	0.226	0.062	0.236	0.053
	Δ^p	-0.233	0.072	-0.233	0.072	-0.233	0.072	-0.232	0.072
	K/S	0.966	0.021	0.940	0.030	0.912	0.043	0.877	0.057
	Maturity	24.283	10.471	65.344	12.658	131.900	26.773	272.455	52.459
ATM Put	Mid-Quote	46.935	27.229	71.031	40.145	128.492	62.584	188.456	89.682
	IV	0.168	0.075	0.169	0.057	0.187	0.052	0.195	0.043
	Δ^p	-0.437	0.036	-0.437	0.036	-0.434	0.035	-0.431	0.034
	K/S	0.994	0.005	0.991	0.008	0.989	0.012	0.986	0.019
	Maturity	24.322	10.403	65.302	12.665	131.454	26.915	271.568	52.636
ATM Call	Mid-Quote	42.072	23.553	60.928	33.295	105.837	48.631	146.374	64.944
	IV	0.158	0.070	0.156	0.052	0.173	0.048	0.179	0.040
	Δ^c	0.439	0.036	0.438	0.036	0.438	0.036	0.435	0.034
	K/S	1.007	0.006	1.011	0.009	1.022	0.014	1.036	0.023
	Maturity	24.212	10.454	65.352	12.614	131.538	26.848	272.832	52.605
OTM Call	Mid-Quote	16.698	12.242	24.406	17.394	43.189	27.542	61.432	38.310
	IV	0.141	0.062	0.136	0.046	0.150	0.042	0.156	0.035
	Δ^c	0.240	0.073	0.240	0.073	0.245	0.073	0.246	0.073
	K/S	1.026	0.015	1.043	0.022	1.069	0.031	1.107	0.045
	Maturity	24.018	10.609	65.428	12.614	129.778	26.531	271.654	52.408
DOTM Call	Mid-Quote	2.338	2.312	3.477	3.546	5.251	5.506	7.007	7.646
	IV	0.135	0.062	0.129	0.045	0.141	0.041	0.145	0.032
	Δ^c	0.052	0.033	0.051	0.034	0.047	0.036	0.044	0.036
	K/S	1.061	0.039	1.104	0.056	1.174	0.091	1.279	0.128
	Maturity	24.027	10.717	65.170	12.524	129.088	26.461	266.505	52.472

Table 1: **Summary statistics for S&P 500 index options by moneyness and maturity.** The table reports the mean and standard deviation of the option’s mid-quote in US dollars, implied volatility, Black-Scholes delta, strike-to-spot ratio, and time to expiration in days. Options are grouped by delta-based moneyness categories and maturity buckets: 7-45 days, 45-90 days, 90-180 days, 180-360 days. DOTM, OTM, and ATM denote deep out-of-the-money, out-of-the-money, and at-the-money options, respectively. Mid-Quote is the average of bid and ask quotes, maturity is measured in calendar days, and K/S denotes the strike-to-spot ratio. Statistics are computed across all option observations within each group over the full sample period.

Table 1 reports summary statistics for observations in each moneyness-maturity group over the full sample period. It presents the mean and standard deviation of five variables across the 24 buckets: mid-quote, implied volatility, Δ , moneyness K/S , and time-to-maturity. Several patterns emerge. First, mid-quotes rise as options move toward the at-the-money region and, within each moneyness category, increase with times to maturity. This is consistent with options closer to the strike being more likely to finish with a non-zero payoff, while longer maturities contain more time value by allowing a wider range of future prices.

Moreover, several stylized facts of the implied volatility surface stand out. For put options, implied volatility decreases monotonically with regular moneyness, measured by K/S , within maturity buckets: DOTM puts have the highest implied volatilities, followed by OTM and ATM puts, reflecting the volatility skew and the premium investors pay for downside protection. On the call side, implied volatilities are lower and vary much less across moneyness categories, indicating that surface asymmetry is driven primarily by the put area rather than a symmetric smile. The term structure is also upward sloping on average, with implied volatility increasing with maturity for most moneyness groups, indicating greater long-horizon uncertainty. This underscores that the overall shape of the surface is largely shaped by downside risk pricing.

Finally, the strike-to-spot ratios vary systematically across maturities within each delta bucket. For put options, K/S declines with maturity, whereas for call options it increases with maturity. This reflects an important feature of delta-based classification: maintaining a given option in the same delta region at longer maturities requires more extreme strike-to-spot ratios, since time value affects the mapping between delta and moneyness. Thus, the table not only summarizes pricing patterns across contracts but also shows how the geometry of the implied volatility surface depends jointly on metrics such as the strike, delta, and maturity.

3.4.3. Factor decomposition

Table 13 in Appendix C1 shows strong cross-correlations between short- and long-term implied volatility buckets, with the strength of co-movement gradually weakening for more distant moneyness groups. Since all correlations exceed 0.85, the results indicate substantial co-movement across the surface, suggesting that the implied volatility surface may be represented by a small number of common factors. Principal component analysis on a balanced panel of 24 daily implied volatility series across maturity-moneyness buckets reinforces this conclusion: as shown in Table 15 in Appendix C3, the first principal component explains roughly 95% of total variation, while the first three components jointly account for nearly 99%. This implies that the implied volatility surface is highly compressible and can be represented parsimoniously by three common factors, whose economic interpretation may provide insight into the key drivers of surface dynamics.

Along these lines, P. Christoffersen et al. (2018) identify three economically relevant dimensions of the implied volatility surface: the overall level, the moneyness slope or skew, and the maturity slope or term structure. Similarly, Cont and da Fonseca (2002) interpret the main principal components as level, skew, and curvature factors, reflecting common shifts, opposite movements across moneyness, and changes in surface convexity. It is therefore natural to relate the estimated principal components to simple measures of the surface's level, skew, and term structure. Let $\mathcal{S}$ and $\mathcal{L}$ denote the short- and long-maturity buckets, respectively, and let $\mathcal{C}$ and $\mathcal{P}$ denote the deep out-of-the-money call and put buckets. With $\log \sigma_{i,t}$ denoting the observed log implied volatility in bucket i at time t , the corresponding maturity and moneyness spreads are defined as:

$$\text{Spread}_t^{\text{mat}} = \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} \log \sigma_{i,t} - \frac{1}{|\mathcal{L}|} \sum_{i \in \mathcal{L}} \log \sigma_{i,t}, \quad \text{Spread}_t^{\text{mon}} = \frac{1}{|\mathcal{C}|} \sum_{i \in \mathcal{C}} \log \sigma_{i,t} - \frac{1}{|\mathcal{P}|} \sum_{i \in \mathcal{P}} \log \sigma_{i,t}. \quad (9)$$

Here, $\text{Spread}_t^{\text{mat}}$ and $\text{Spread}_t^{\text{mon}}$ denote the spreads along maturity and moneyness, respectively, and $|\cdot|$

denotes the cardinality of the corresponding set.⁹ Since principal components are purely statistical constructs and not uniquely tied to specific economic interpretations, the associated economic variables may differ slightly across specifications.

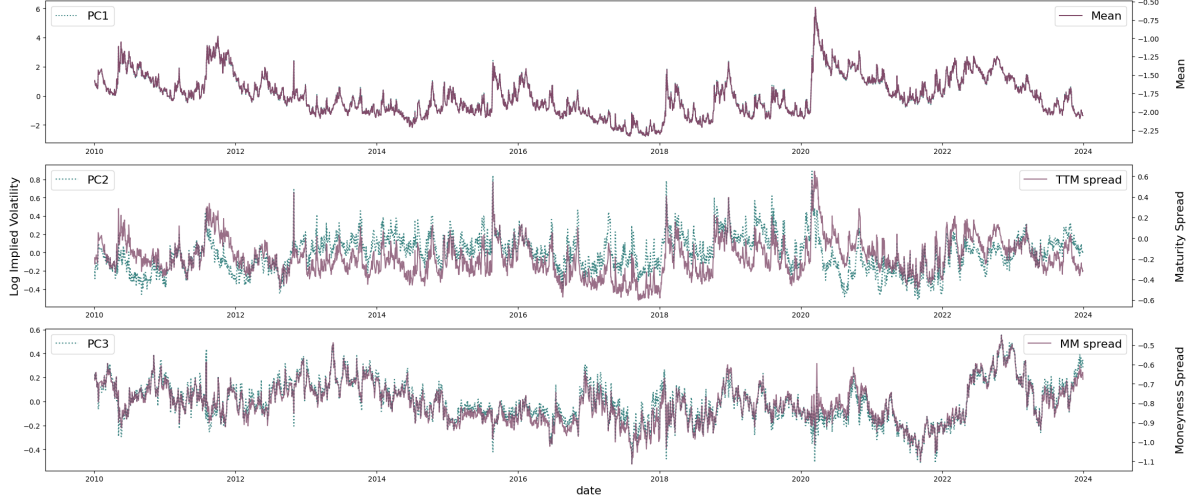


Figure 2: **Time series comparison of the first three principal components of the log implied volatility series of all 24 moneyness-maturity buckets and summary measures.** The top panel shows PC1 together with the cross-sectional mean log implied volatility. The middle panel shows PC2 alongside the maturity spread, defined as the difference between average log implied volatility in the short- and long-term maturity buckets. The bottom panel shows PC3 alongside the moneyness spread, defined as the difference between average log implied volatility in the DOTM call and DOTM put buckets.

Figure 2 compares the first three principal components with simple summary measures of the implied volatility surface. PC1 tracks the cross-sectional mean almost perfectly, indicating that the dominant source of variation is a common level shift across all moneyness-maturity buckets. PC2 co-moves strongly with the maturity spread, suggesting that it captures changes in the term structure, while PC3 aligns with the moneyness spread, reflecting variation in skew. Overall, the results indicate that the surface is effectively low-dimensional and can be represented by a small number of economically interpretable factors related to moneyness and maturity, motivating a parsimonious factor structure.

Gonçalves and Guidolin (2006) exploit this factor structure using a cross-sectional five-factor specification consisting of a level component, linear and quadratic moneyness effects, a time-to-maturity effect, and a moneyness-maturity interaction term. Following this approach, daily cross-sectional regressions are estimated across moneyness-maturity buckets, indexed by i , to characterize variation across the surface. For each trading day t , the coefficient vector $\beta_t = (\beta_{0,t}, \beta_{1,t}, \beta_{2,t}, \beta_{3,t}, \beta_{4,t})^\top$ is obtained by ordinary least squares, producing a time series of estimated factors $\{\hat{\beta}_t\}_{t=1}^n$. The dynamic behavior of these factors is then modeled using a first-order vector autoregression, yielding the following model set-up:

$$\begin{aligned} \log \sigma_{i,t} &= \beta_{0,t} + \beta_{1,t} m_{i,t} + \beta_{2,t} m_{i,t}^2 + \beta_{3,t} \tau_{i,t} + \beta_{4,t} m_{i,t} \tau_{i,t} + \varepsilon_{i,t}, \quad \varepsilon_{i,t} \sim N(0, s_i^2), \\ \hat{\beta}_t &= \mu + \Phi \hat{\beta}_{t-1} + u_t, \quad u_t \sim N(0, \Omega). \end{aligned} \quad (10)$$

Here, $\mu \in \mathbb{R}^p$ is a vector of intercepts, $\Phi \in \mathbb{R}^{p \times p}$ governs dependence on past coefficient values, and $u_t \sim N(0, \Omega)$ denotes the innovation vector. Table 19 in Appendix C8 relates these estimated factors to the principal components. PC1 is strongly correlated with the baseline level, indicating that it captures the

⁹Alongside the spreads, the cross-sectional mean is computed to resemble the overall level, defined as $\overline{IV}_t = \frac{1}{N_t} \sum_{i=1}^{N_t} \log \sigma_{i,t}$.

overall level of the surface, while the remaining components are more diffusely associated with slope and curvature effects.

Furthermore, the coefficient paths reported in Figure 8 in Appendix C5 evolve smoothly around their long-run mean, $(I_p - \Phi)^{-1}\mu$, and display substantial persistence. This is supported by Table 17, where the autocorrelations remain high across several lags, while the PACFs are dominated by the first lag and decline sharply thereafter. The ADF test rejects a unit root for all series, providing evidence for stationarity, and the VAR estimates in Table 18 show significant persistence and interdependence across coefficients. Finally, the estimated VAR(1) process is stable, meaning that all eigenvalues of Φ lie strictly inside the unit circle, that is, $|\lambda_k(\Phi)| < 1, \forall k = 1, \dots, p$. It follows that $\Phi^h \rightarrow 0$ as $h \rightarrow \infty$, which implies that shocks decay over time and the process admits a stationary representation. Motivated by these findings, the system in Eq. (10) is re-estimated in a one-step state-space framework, which serves as a benchmark in the empirical analysis.

3.4.4. Residual decomposition

To assess the limitations of global polynomial specifications often encountered in the literature, the residuals of the preliminary VAR state-space model are examined across economically relevant regions of the moneyness-maturity space. This diagnostic step is motivated by the fact that lower-order parametric specifications can provide a reasonable global fit while still producing systematic local errors, especially in the tails of the moneyness distribution. Let $\mathcal{R} \subseteq \mathcal{M} \times \mathcal{T}$ denote a region of the moneyness-maturity domain, and let $\mathcal{T}_{test}$ denote the set of test-period dates. For each testing sample date $t \in \mathcal{T}_{test}$, define $\mathcal{I}_t(\mathcal{R}) = \{i : (m_{i,t}, \tau_{i,t}) \in \mathcal{R}\}$. For each region, the average absolute residual is computed as:

$$\bar{\varepsilon}_{\mathcal{R}} = \left(\sum_{t \in \mathcal{T}_{test}} |\mathcal{I}_t(\mathcal{R})| \right)^{-1} \sum_{t \in \mathcal{T}_{test}} \sum_{i \in \mathcal{I}_t(\mathcal{R})} |\hat{\varepsilon}_{i,t}|. \quad (11)$$

This statistic measures the average magnitude of the model pricing error within a given region. The heatmaps in Appendix D2 show that the largest residuals occur in the deep out-of-the-money put region, especially at longer maturities. While the model captures the central surface relatively well, it leaves systematic errors in the far-left tail, where risk premia are most pronounced. To understand why this occurs, Figure 9 in Appendix C9 compares observed, predicted, and residual put implied volatility surfaces for two representative dates. The baseline VAR five-factor state-space model fails mainly in the low-moneyness region, where observed implied volatility rises sharply as delta moves toward zero, while the predicted surface can slope in the wrong direction, producing systematic residuals rather than small random errors. This pattern can be understood from the local slope of the fitted surface implied by Eq. (10) when maturity is held fixed:

$$\frac{\partial \widehat{\log \sigma_{i,t}}}{\partial m_{i,t}} = \hat{\beta}_{1,t} + 2\hat{\beta}_{2,t}m_{i,t} + \hat{\beta}_{4,t}\tau_{i,t}. \quad (12)$$

Hence, the steepness of the fitted put wing is governed by the time-varying coefficient combination of $\hat{\beta}_{1,t}$, $\hat{\beta}_{2,t}$, and $\hat{\beta}_{4,t}$. If this combination is positive in the left-tail region, the fitted surface decreases precisely where the observed put surface increases. The residual surfaces in Figure 9 show that this limitation is not isolated to a few contracts but occurs systematically across the relevant region of the surface. Importantly, this is not only a limitation of the specific VAR state-space implementation but reflects a broader restriction of state-space models that rely on low-dimensional polynomial cross-sectional representations. This motivates the use of an additional residual correction that can preserve the structure of the baseline model while allowing greater local flexibility in the regions where systematic errors remain.

4. Methodology

The modeling framework addresses the high-dimensional problem of forecasting the implied volatility surface through a low-dimensional dynamic factor structure, augmented with learning-based corrections to capture residual nonlinearities and enforce arbitrage consistency. Let $t \in \mathbb{T}$ index time, and let the discrete moneyness-maturity grid be denoted by $\mathcal{M} \times \mathcal{T} \subseteq \mathbb{R} \times \mathbb{R}_+$. Let $\sigma_t(m, \tau)$, $\sigma_t : \mathcal{M} \times \mathcal{T} \rightarrow \mathbb{R}_+$ denote the implied volatility at grid point (m, τ) , and define $\mathcal{S}_t = \{\sigma_t(m, \tau)\}_{(m, \tau) \in \mathcal{M} \times \mathcal{T}}$ as the implied volatility surface. Furthermore, $IV_t = \text{vec}(\mathcal{S}_t)$, and $\mathcal{F}_t = \sigma(\mathcal{S}_s : s \leq t)$ denotes the natural filtration generated by its history. The objective is to model the conditional evolution of the surface, that is, the conditional distribution of $IV_{t+1}|\mathcal{F}_t$.

The preliminary evidence motivates a staged modeling framework that separates the estimation task into three components, allowing each part of the model to address a distinct feature of the data:

1. **Cross-sectional representation (surface compression):** In Subsection 4.1, the raw implied volatility quotes are mapped onto a moneyness-maturity grid $\mathcal{M} \times \mathcal{T}$, yielding a discretized representation of the implied volatility surface. The vectorized surface $IV_t = \text{vec}(\mathcal{S}_t) \in \mathbb{R}^{|\mathcal{M}| \cdot |\mathcal{T}|}$ is then projected onto a lower-dimensional shape basis $M_t \in \mathbb{R}^{|\mathcal{M}| \cdot |\mathcal{T}| \times p}$, so that the dominant cross-sectional variation is summarized by the time-varying coefficient vector $\beta_t \in \mathbb{R}^p$.
2. **Latent dynamics (dynamic factor evolution):** In Subsection 4.2, the dynamics of the implied volatility surface over time are modeled in state-space form. Specifically, the latent coefficients follow a mean-reverting Ornstein-Uhlenbeck process $d\beta_t = K(\Theta - \beta_t)dt + \Gamma dW_t^{\mathbb{P}}$, with correlation structure $dW_t^{\mathbb{P}}(dW_t^{\mathbb{P}})^{\top} = Rdt$.
3. **Residual structure (nonlinear smoothing):** In Subsection 4.3, the remaining systematic variation not captured by the linear state-space specification is isolated through the residual process. A learning-based map $\omega^{nn}(\cdot; \theta_1)$ is then estimated on this residual component in total-variance space in accordance with Akerer et al. (2020), yielding a corrected representation of the prior implied volatilities $\sigma_t^{\theta}(m, \tau)$, which accounts for several no-arbitrage restrictions.

Furthermore, Subsection 4.4 describes the two-step estimation procedure, Subsection 4.5 defines the benchmark specifications, and Subsection 4.6 presents the evaluation methods.

4.1. Cross-sectional representation (surface compression)

Since the analysis of Subsection 3.4 indicates that the volatility surface is characterized by a set of low-dimensional common factors, this section improves upon the aforementioned five-factor specification within a state-space formulation. First, the filtered option data are mapped onto a discrete moneyness-maturity grid to obtain a set of representative contracts. Next, a parametric spline basis structure is imposed to summarize the cross-sectional variation in implied volatility, which serves as the measurement equation linking observed data to the underlying latent coefficients. This procedure compresses the high-dimensional implied volatility surface into a lower-dimensional set of factors that capture its dominant cross-sectional features.

4.1.1. Cross-sectional compression

To obtain a time-consistent lower-dimensional cross-sectional representation of the implied volatility surface, the filtered option data are mapped onto a discrete Δ -moneyness-maturity grid. This procedure compresses the high-dimensional cross-section of option contracts into a small set of representative observations.

Inspired by Gonçalves and Guidolin (2006), the option contract closest to the midpoint of each bucket is selected for every day t . Let $\mathcal{Q}_t = \{q_{i,t}\}_{i=1}^{N_t}$ denote the contracts available on day t , where each contract $q_{i,t}$ is characterized by delta-moneyness $\Delta_{i,t}$, time-to-maturity $\tau_{i,t}$, and implied volatility $\sigma_{i,t}$. The moneyness-maturity space is divided into $G = 24$ buckets, $\{B_g\}_{g=1}^G$, formed by six delta-moneyness and four maturity categories. For each bucket g , the set of available contracts is denoted by $C_{g,t} = \{i \in \{1, \dots, N_t\} : (\Delta_{i,t}, \tau_{i,t}) \in B_g\}$, where N_t denotes the number of raw option contracts available on date t . Furthermore, let $\bar{\Delta}_g$ and $\bar{\tau}_g$ denote the midpoints of the delta and maturity intervals for bucket g , respectively, and let σ_{Δ}^2 and σ_{τ}^2 denote the variances of moneyness and maturity. Then, the selected contract in bucket g at time t is defined as

$$i_{g,t}^* = \arg \min_{i \in C_{g,t}} \left[\frac{(\Delta_{i,t} - \bar{\Delta}_g)^2}{\sigma_{\Delta}^2} + \frac{(\tau_{i,t} - \bar{\tau}_g)^2}{\sigma_{\tau}^2} \right]. \quad (13)$$

The selected contract thus minimizes the distance to the bucket center in terms of both Δ -moneyness and maturity. Since the values of delta inherently vary on a different scale than the time-to-maturity values, both metrics are scaled by their variances to ensure comparability within the distance metric. This ensures that the selected contract is the most central observation within the bucket and therefore most representative of the intended maturity-moneyness category, reducing compositional noise from changing contract availability. The resulting set of representative contracts serves as an input to the subsequent measurement equation.

4.1.2. Tensor spline-based cross-sectional representation

Delta depends on both the option pricing model and the implied volatility value itself. It is therefore not appropriate as a direct explanatory variable when implied volatility is the dependent variable, as this may introduce endogeneity. In this framework, delta is used only to allocate contracts to moneyness-maturity buckets. Inspired by Gonçalves and Guidolin (2006), the following time-adjusted measure of moneyness is utilized:

$$m_{i,t} = \frac{\ln\left(\frac{K_i}{\exp((r_t - q_t)\tau_{i,t})S_t}\right)}{\sqrt{\tau_{i,t}}}. \quad (14)$$

Here, K_i denotes the strike price, S_t the underlying price, r_t the risk-free rate, q_t the dividend yield, and $\tau_{i,t}$ the time to maturity in years. This measure expresses moneyness relative to the forward price, while its sign identifies calls and puts as out-of-the-money or in-the-money.

In line with Zou et al. (2025), implied volatilities are observed at each time $t = 1, \dots, n$ on time-varying grids of moneyness values $\mathcal{M}_t \in \mathbb{R}^{\kappa_t}$ and times to maturity $\mathcal{T}_t \in \mathbb{R}_+^{\nu_t}$. In particular, $\sigma_t(m_{j,t}, \tau_{h,t})$ denotes the implied volatility of an option with moneyness $m_{j,t}$, $j = 1, \dots, \kappa_t$ and time to maturity $\tau_{h,t}$, $h = 1, \dots, \nu_t$. Since not every moneyness-maturity combination is necessarily observed, the number of available implied volatility quotes at date t , denoted by N_t , satisfies $N_t \leq \kappa_t \cdot \nu_t$. For notational convenience, let $(m_{i,t}, \tau_{i,t})$, $i = 1, \dots, N_t$, denote the observed moneyness and time-to-maturity pairs at date t . Stacking the observed implied volatilities into the vector $IV_t \in \mathbb{R}^{N_t}$ then yields

$$IV_t = (\sigma_t(m_{1,t}, \tau_{1,t}), \sigma_t(m_{2,t}, \tau_{2,t}), \dots, \sigma_t(m_{N_t,t}, \tau_{N_t,t}))^\top. \quad (15)$$

This setup accommodates a time-varying number of option contracts and changes in the composition of contracts over time, both of which are typical features of option data. Because an option's expiration date is fixed, its time-to-maturity declines naturally over time, causing its location on the maturity grid to change.

To obtain a stable representation for estimation purposes, the aforementioned compressed setting replaces the original cross-section with a set of representative contracts at each date. The compression retains six moneyness categories and four distinct maturity categories at each date. Consequently, although the dimension of the raw option universe may change over time, the compressed representation fixes its cross-sectional dimension at $N = 24$ for every t . The subsequent model descriptions, however, will retain the general formulation with time-varying cross-sectional dimensions.

Since the implied volatility surface is highly nonlinear and varies locally across moneyness and maturity, a tensor-product spline basis following Fengler and Hin (2015) is used to capture this variation. This allows the volatility skew to differ across short- and long-dated options, which cannot be adequately represented by fixed linear, quadratic, or interaction terms. Let $S(p_1, p_2; \xi, \nu) = S(p_1; \xi) \otimes S(p_2; \nu)$ denote the bivariate spline space formed as the tensor product of two univariate spline spaces. The first space is defined over the moneyness knot sequence ξ with degree p_1 , and the second over the maturity knot sequence ν with degree p_2 . More precisely, $S(p_1; \xi) = \{s \in \mathbb{C}^{p_1-1} : s \text{ is a polynomial of degree } p_1 \text{ on each subinterval in } \xi\}$, where $\mathbb{C}^p$ is the set of p -times continuously differentiable functions. The cross-sectional component is then modeled as a time-varying function $f_t(m, \tau; \theta) \in S(p_1, p_2; \xi, \nu)$, yielding:

$$\log \sigma_t(m_{i,t}, \tau_{i,t}) = \beta_{0,t} + \underbrace{\sum_{j=1}^{K_m} \beta_{j,t}^{(m)} \Phi_j(m_{i,t})}_{\text{moneyness effect}} + \underbrace{\sum_{h=1}^{K_\tau} \beta_{h,t}^{(\tau)} \Psi_h(\tau_{i,t})}_{\text{maturity effect}} + \underbrace{\sum_{j=1}^{K_m} \sum_{h=1}^{K_\tau} \beta_{jh,t}^{(m,\tau)} \Phi_j(m_{i,t}) \Psi_h(\tau_{i,t})}_{\text{interaction effect}} + \varepsilon_{i,t}. \quad (16)$$

The resulting specification decomposes log implied volatility into several components. The term $\beta_{0,t}$ is a time-varying intercept capturing the overall level of the implied volatility surface at time t . The functions $\{\Phi_j(\cdot)\}_{j=1}^{K_m}$ and $\{\Psi_h(\cdot)\}_{h=1}^{K_\tau}$ capture variation along the moneyness and maturity dimensions as predetermined spline basis functions, with K_m and K_τ denoting the number of basis functions in each dimension. The final component allows the two dimensions to interact through products of the corresponding moneyness and maturity basis functions. The remaining term, $\varepsilon_{i,t}$, is an idiosyncratic error term, typically assumed to satisfy $\varepsilon_{i,t} \sim N(0, s_i^2)$.

For each day t , the N_t observed log implied volatilities across the available moneyness-maturity combinations are stacked into a single cross-sectional vector, yielding the following measurement equation:

$$\log IV_t = M_t \beta_t + \varepsilon_t, \quad \varepsilon_t \sim N(0, H_t). \quad (17)$$

Here, $M_t \in \mathbb{R}^{N_t \times p}$, whose i -th row is given by $M_{i,t} = (1, \phi_{i,t}^\top, \psi_{i,t}^\top, \phi_{i,t}^\top \otimes \psi_{i,t}^\top) \in \mathbb{R}^{1 \times p}$, with $\phi_{i,t} = \phi(m_{i,t}) = (\Phi_1(m_{i,t}), \dots, \Phi_{K_m}(m_{i,t}))^\top$, and $\psi_{i,t} = \psi(\tau_{i,t}) = (\Psi_1(\tau_{i,t}), \dots, \Psi_{K_\tau}(\tau_{i,t}))^\top$. Furthermore, $\beta_t = (\beta_{0,t}, \{\beta_{j,t}^{(m)}\}_{j=1}^{K_m}, \{\beta_{h,t}^{(\tau)}\}_{h=1}^{K_\tau}, \{\beta_{jh,t}^{(m,\tau)}\}_{j=1, h=1}^{K_m, K_\tau})^\top$ denotes a vector of time-varying coefficients, with $p = 1 + K_m + K_\tau + K_m K_\tau$, and the innovation term $\varepsilon_t \in \mathbb{R}^{N_t}$ is assumed to be normally distributed with diagonal covariance matrix $H_t = \text{diag}(s_1^2, \dots, s_{N_t}^2) \in \mathbb{R}^{N_t \times N_t}$. In line with Zou et al. (2025), M_t is treated as an exogenous input rather than a stochastic component. That is, the shape terms are taken as given at each point in time and are not modeled as part of the state dynamics.

4.2. Latent dynamics (dynamic factor evolution)

Building on the cross-sectional representation, the time variation in the underlying parameters is modeled using a continuous-time Ornstein-Uhlenbeck process, capturing mean reversion and persistence. The

continuous-time dynamics are subsequently discretized to obtain a tractable state transition equation suitable for estimation within the state-space framework.

4.2.1. State update equation

The dynamic behavior of the time-varying surface coefficients is modeled with an Ornstein-Uhlenbeck process for three reasons. First, this choice is consistent with Wang et al. (2017) and the empirical evidence in Subsection 3.4: the estimated coefficients appear stationary and mean-reverting, with gradually decaying ACFs, sharply cutting-off PACFs, and a stable VAR indicating that shocks fade over time. Second, the OU process imposes structured mean-reverting dynamics, so that the discrete-time transition and innovation covariance are generated by a set of economically interpretable parameters, rather than being left unrestricted as in a VAR. Finally, because the OU process admits a closed-form solution over a single time step, it can be written conveniently in state-space form.

Accordingly, the latent coefficients are assumed to follow the following continuous-time mean-reverting dynamics under the physical probability measure $\mathbb{P}$:

$$d\beta_t = K(\Theta - \beta_t)dt + \Gamma dW_t^{\mathbb{P}}, \quad dW_t^{\mathbb{P}}(dW_t^{\mathbb{P}})^{\top} = Rdt. \quad (18)$$

This state equation describes the instantaneous evolution of the coefficient vector β_t . The diagonal matrix $K = \text{diag}(k_1, \dots, k_p) \in \mathbb{R}^{p \times p}$ contains the mean reversion rates. Each parameter $k_i > 0$ describes how quickly coefficient $i = 1, \dots, p$ is pulled back toward its long-run mean θ_i , collected in the vector $\Theta = (\theta_1, \dots, \theta_p)^{\top} \in \mathbb{R}^p$. The diagonal matrix $\Gamma = \text{diag}(\gamma_1, \dots, \gamma_p) \in \mathbb{R}^{p \times p}$ contains the volatility parameters, where γ_i governs the shock intensity of coefficient i . Finally, $W_t^{\mathbb{P}} = (W_{1,t}^{\mathbb{P}}, \dots, W_{p,t}^{\mathbb{P}})^{\top} \in \mathbb{R}^p$ denotes the vector of Brownian motions driving the latent coefficients, with quadratic covariance $dW_t^{\mathbb{P}}(dW_t^{\mathbb{P}})^{\top} = Rdt$, where $R = (\rho_{ij})_{i,j=1}^p$ is the Brownian correlation matrix.

4.2.2. Discretization of the Ornstein-Uhlenbeck process

To operationalize the state update dynamics in Eq. (18) and to derive the discrete-time representation of the stochastic differential equation, consider the instantaneous evolution of coefficient i :

$$d\beta_{i,t} = k_i(\theta_i - \beta_{i,t})dt + \gamma_i dW_{i,t}^{\mathbb{P}}. \quad (19)$$

Here, $dW_{i,t}^{\mathbb{P}}dW_{j,t}^{\mathbb{P}} = p_{ij}dt$, $i, j = 1, \dots, p$, implying that shocks to different factors move together contemporaneously, correlated with correlation coefficient p_{ij} . To eliminate $\beta_{i,t}$ from the drift term, Eq. (19) is first rearranged and then multiplied by the integrating factor $e^{k_i t}$. Applying Itô's product rule¹⁰ allows the left-hand side to be written as the differential of a transformed process. This yields an exact differential equation, which can be integrated directly and leads to a recursive representation of the latent coefficients. Consequently, Eq. (19) can be expressed more compactly as:

$$\begin{aligned} e^{k_i t} d\beta_{i,t} + k_i e^{k_i t} \beta_{i,t} dt &= k_i \theta_i e^{k_i t} dt + \gamma_i e^{k_i t} dW_{i,t}^{\mathbb{P}} \\ \implies d(e^{k_i t} \beta_{i,t}) &= k_i \theta_i e^{k_i t} dt + \gamma_i e^{k_i t} dW_{i,t}^{\mathbb{P}}. \end{aligned} \quad (20)$$

Consequently, the first term on the right-hand side is deterministic and therefore can be evaluated analytically using standard integration techniques, unlike in Eq. (19) where the drift term still depended on the stochastic state variable $\beta_{i,t}$. Next, both sides are integrated over $[t-1, t]$ to obtain an expression isolated for $\beta_{i,t}$:

¹⁰Itô's product rule: $d(e^{k_i t} \beta_{i,t}) = e^{k_i t} d\beta_{i,t} + k_i e^{k_i t} \beta_{i,t} dt$.

$$\begin{aligned}
\int_{t-1}^t d(e^{k_i s} \beta_{i,s}) &= \int_{t-1}^t k_i \theta_i e^{k_i s} ds + \int_{t-1}^t \gamma_i e^{k_i s} dW_{i,s}^{\mathbb{P}} \\
&= \theta_i (e^{k_i t} - e^{k_i(t-1)}) + \int_{t-1}^t \gamma_i e^{k_i s} dW_{i,s}^{\mathbb{P}}.
\end{aligned} \tag{21}$$

Applying the fundamental theorem of calculus yields an expression for the integrated left-hand side, which, after rearranging and dividing by $e^{k_i t}$, allows the current state $\beta_{i,t}$ to be expressed recursively in terms of $\beta_{i,t-1}$. For the stochastic component, the constant γ_i can be taken outside the integral, while the exponential terms combine to produce a decaying kernel. Combining these steps leads to the following expression:

$$\begin{aligned}
e^{k_i t} \beta_{i,t} - e^{k_i(t-1)} \beta_{i,t-1} &= \theta_i (e^{k_i t} - e^{k_i(t-1)}) + \int_{t-1}^t \gamma_i e^{k_i s} dW_{i,s}^{\mathbb{P}} \\
\implies \beta_{i,t} &= \theta_i (1 - e^{-k_i}) + e^{-k_i} \beta_{i,t-1} + \gamma_i \int_{t-1}^t e^{-k_i(t-s)} dW_{i,s}^{\mathbb{P}}.
\end{aligned} \tag{22}$$

The next step is to characterize the distribution of the stochastic integral. By properties of Itô integration, an integral of the form $\int_a^b H_s dW_s$ has mean zero if H_s is adapted and square-integrable. Moreover, when the integrand is deterministic, the integral is normally distributed with variance equal to the integral of the squared integrand. In this case, the integrand $H_s = e^{-k_i(t-s)}$ is deterministic and therefore adapted and is square integrable over the interval $[t-1, t]$. Hence, the stochastic integral has mean zero and, by Itô isometry¹¹, its variance is given by:

$$\text{Var}\left(\int_{t-1}^t e^{-k_i(t-s)} dW_{i,s}^{\mathbb{P}}\right) = \int_{t-1}^t e^{-2k_i(t-s)} ds = \frac{1 - e^{-2k_i}}{2k_i}. \tag{23}$$

Since the stochastic integral is normally distributed with mean zero and variance given by Eq. (23), it can be expressed as a scaled standard normal random variable. Specifically, any normally distributed random variable with mean zero and variance σ_i^2 can be written as $\sigma_i z_{i,t}$, where $z_{i,t} \sim N(0, 1)$. Defining $\xi_{i,t} = z_{i,t}$ and $\sigma_i = \sqrt{\frac{1 - e^{-2k_i}}{2k_i}}$, Eq. (22) becomes:

$$\beta_{i,t} = \theta_i (1 - e^{-k_i}) + e^{-k_i} \beta_{i,t-1} + \gamma_i \sqrt{\frac{1 - e^{-2k_i}}{2k_i}} \xi_{i,t}, \quad \xi_{i,t} \sim N(0, 1). \tag{24}$$

The discrete-time transition equation for $\beta_{i,t}$ in Eq. (24) makes clear that $\beta_{i,t}$ consists of three components: a mean-reversion term $\theta_i(1 - e^{-k_i})$, a persistence term $e^{-k_i} \beta_{i,t-1}$, and a Gaussian innovation term. In this form, the dynamics are directly comparable to an AR(1) process, where e^{-k_i} governs persistence and $\gamma_i \sqrt{\frac{1 - e^{-2k_i}}{2k_i}}$ determines the conditional standard deviation of the shock. Rewriting (24) back into multivariate form yields:

$$\beta_t = (I_p - \Phi)\Theta + \Phi\beta_{t-1} + \eta_t, \quad \eta_t \sim N(0, Q). \tag{25}$$

Here, $\eta_t \in \mathbb{R}^p$ with distribution $N(0, Q)$ denotes the innovation vector with covariance matrix $Q = \Omega\Omega^\top \in \mathbb{R}^{p \times p}$, obtained via the representation $\eta_t = \Omega\xi_t$, with $\xi_t \sim N(0, I_p)$. Furthermore, $\Phi = \text{diag}(e^{-k_1}, \dots, e^{-k_p}) \in \mathbb{R}^{p \times p}$ collects the persistence parameters, while the entries of Q are given by $Q_{ij} = \rho_{ij} \gamma_i \gamma_j \frac{1 - e^{-(k_i + k_j)}}{k_i + k_j}$, for $i, j = 1, \dots, p$. The term $(I_p - \Phi)\Theta$ is the discrete-time intercept implied by continuous-time mean reversion. It ensures that the latent factor vector β_t reverts to the long-run mean Θ , with speed of adjustment governed by $I_p - \Phi$.

¹¹Itô isometry: $\mathbb{E}\left[\left(\int_a^b H_s dW_s\right)^2\right] = \mathbb{E}\left[\int_a^b H_s^2 ds\right]$. When H_s is deterministic, the expression simplifies to $\int_a^b H_s^2 ds$.

4.3. Residual structure (nonlinear smoothing)

In light of concerns raised in the literature about poor fit in the corners of the surface and the importance of maintaining an arbitrage-free specification, this subsection outlines a residual-learning correction. While the state-space specification provides an interpretable representation of the main surface dynamics, its linear basis structure may leave local nonlinearities across moneyness and maturity in the residuals. The residual component is therefore used as a flexible correction layer rather than a replacement for the state-space model.

4.3.1. Neural network architecture

Following Akerer et al. (2020), the correction is formulated in total-variance space, allowing smoothness and no-arbitrage penalties to be incorporated directly into the learning objective. This yields a corrected surface that preserves the dynamic factor structure while improving local fit and economic consistency. Let $\omega_t(m, \tau) = \sigma_t^2(m, \tau)\tau$ denote the total variance, where $\sigma_t(m, \tau)$ is the implied volatility at moneyness m , maturity τ , and date t . The prior total-variance surface implied by the state-space model is denoted by $\omega_t^{\text{prior}}(m, \tau; \theta_2)$. Working in total variance is convenient, as static no-arbitrage conditions admit a simpler and more tractable formulation in this space. The corrected surface is then specified as:

$$\omega_t^\theta(m, \tau) = \omega^{nn}(m, \tau, \tilde{\sigma}_t^{\text{prior}}; \theta_1) \times \omega_t^{\text{prior}}(m, \tau; \theta_2), \quad \text{with} \quad \sigma_t^\theta(m, \tau) = \sqrt{\omega_t^\theta(m, \tau)/\tau}. \quad (26)$$

Here, the parameters are contained in $\theta = (\theta_1, \theta_2)$. The mapping $\omega_t^{\text{prior}} : \mathbb{R} \times \mathbb{R}_+ \rightarrow \mathbb{R}_+$ denotes the forecasted prior total-variance surface implied by the fitted state space model with parameters θ_2 . In the implementation, the prior implied volatility surface is floored before constructing the prior total variance using $\tilde{\sigma}_t^{\text{prior}}(m, \tau) = \max\{\sigma_t^{\text{prior}}(m, \tau), \epsilon\}$, with $\epsilon = 0.02$. This stabilizes the correction in total-variance space and prevents near-zero implied volatilities from producing disproportionally large neural-network multipliers.

Furthermore, $\omega_t^{nn} : \mathbb{R} \times \mathbb{R}_+ \times \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is a standard feedforward multilayer neural network that acts as the flexible correction term. At each date t , the network takes moneyness, maturity, and the forecasted prior implied volatility as inputs and outputs a positive scaling factor that adjusts the prior total-variance surface multiplicatively. This allows the correction to depend on both option characteristics and the shape of the model-implied prior surface.

$$\omega^{nn}(m, \tau, \tilde{\sigma}_t^{\text{prior}}; \theta_1) = \bigcirc_{i=1}^{n+1} f_i^{W_i, b_i}(m, \tau, \tilde{\sigma}_t^{\text{prior}}), \quad \text{with} \quad f_i(x) = \begin{cases} g_i(W_i x + b_i), & i < n+1, \\ \exp(W_{n+1} x + b_{n+1}), & i = n+1. \end{cases} \quad (27)$$

The hidden-layer activation functions g_i are chosen as hyperbolic tangent functions, $g_i(x) = \tanh(x)$. The exponential final layer ensures that the neural network correction is strictly positive so that the corrected total variance remains positive. Unlike the bounded specification of Akerer et al. (2020), the exponential formulation does not impose an upper bound on the scaling factor, allowing substantial corrections for prior predictions near zero. The neural network parameter vector $\theta_1 = \{W_1, b_1, W_2, b_2, \dots\}$ contains the set of weight matrices and bias vectors of the neural network and is estimated by minimizing the loss function:

$$\mathcal{L}(\theta) = \mathcal{L}_0(\theta) + \sum_{k=1}^K \lambda_k \mathcal{L}_k(\theta) + \frac{1}{2} \lambda_{reg} \|\theta_1\|_2^2. \quad (28)$$

Here, $\mathcal{L}_0(\theta)$ denotes the prediction error component of the loss function, while the terms $\mathcal{L}_k(\theta)$, with corresponding penalty weights λ_k , impose soft constraints promoting a total variance surface $\{\omega_t^\theta(m, \tau) : (m, \tau) \in$

$\mathbb{R} \times \mathbb{R}_+$ that is free of static arbitrage. The prediction error is defined as the sum of root mean squared error (RMSE) and the mean absolute percentage error (MAPE):

$$\mathcal{L}_0(\theta) = \sqrt{\frac{1}{N_0} \sum_{t \in \mathcal{T}_{train}} \sum_{i=1}^{N_t} (\sigma_t(m_{i,t}, \tau_{i,t}) - \sigma_t^\theta(m_{i,t}, \tau_{i,t}))^2} + \frac{1}{N_0} \sum_{t \in \mathcal{T}_{train}} \sum_{i=1}^{N_t} \frac{|\sigma_t(m_{i,t}, \tau_{i,t}) - \sigma_t^\theta(m_{i,t}, \tau_{i,t})|}{\sigma_t(m_{i,t}, \tau_{i,t})}. \quad (29)$$

Here, $\sigma_t(m_{i,t}, \tau_{i,t})$ denotes the observed implied volatility quote at date t for moneyness $m_{i,t}$ and maturity $\tau_{i,t}$, while $\sigma_t^\theta(m_{i,t}, \tau_{i,t})$ denotes the neural network-produced value. The number of observed moneyness-maturity pairs on training date t is given by N_t , and $N_0 = \sum_{t \in \mathcal{T}_{train}} N_t$ represents the total number of observed quotes in the training sample. This formulation penalizes both absolute and relative deviations, which is useful because implied volatility levels vary across the surface and are typically higher in the wings than near the at-the-money region.

4.3.2. Penalization and regularization

Roper (2010) establishes sufficient conditions for an implied volatility surface to be free from static arbitrage. In the architecture of Subsubsection 4.3.1, not all conditions must be enforced explicitly. Ackerer et al. (2020) note that smoothness and basic structural properties are ensured by the network design itself. In particular, smoothness is satisfied if the network mapping is twice differentiable, positivity is satisfied by restricting the output to be positive, and the value at maturity condition is imposed by working with total variance. This paper focuses on two of the remaining conditions: calendar monotonicity and butterfly convexity, which extend the arbitrage restrictions implemented in the iterative filtering algorithm.

The penalty terms contain sign restrictions that determine whether a violation has occurred at a given moneyness-maturity point. In particular, the calendar condition is defined as $\ell_{cal,t}(k, \tau) = \partial \omega_t^\theta(k, \tau) / \partial \tau \geq 0$, while the butterfly condition is defined through¹²

$$\ell_{but,t}(k, \tau) = \left(1 - \frac{k \cdot \partial \omega_t^\theta(k, \tau) / \partial k}{2\omega_t^\theta(k, \tau)}\right)^2 - \frac{(\partial \omega_t^\theta(k, \tau) / \partial k)^2}{4} \left(\frac{1}{\omega_t^\theta(k, \tau)} + \frac{1}{4}\right) + \frac{\partial^2 \omega_t^\theta(k, \tau) / \partial k^2}{2} \geq 0. \quad (30)$$

No-arbitrage restrictions apply to the implied volatility surface as a whole, rather than only to the observed option contracts. Hence, even when the fitted surface behaves well at the observed quotes, calendar or butterfly arbitrage may still arise between observations or in sparsely populated regions of the surface. To account for this, the arbitrage penalties are evaluated as soft constraints on a synthetic grid covering the relevant maturity and moneyness domain. The resulting loss components are then defined as

- Calendar arbitrage: $\mathcal{L}_1(\theta) = \frac{1}{|\mathcal{T}_{train}| \cdot N_{12}} \sum_{t \in \mathcal{T}_{train}} \sum_{(k_i, \tau_i) \in \mathcal{I}_{12}} \max\{0, -\ell_{cal,t}(k_i, \tau_i)\}$.
- Butterfly arbitrage: $\mathcal{L}_2(\theta) = \frac{1}{|\mathcal{T}_{train}| \cdot N_{12}} \sum_{t \in \mathcal{T}_{train}} \sum_{(k_i, \tau_i) \in \mathcal{I}_{12}} \max\{0, -\ell_{but,t}(k_i, \tau_i)\}$.

The common evaluation grid is given by $\mathcal{I}_{12} = \{(k, \tau) : k \in \mathcal{M}^*, \tau \in \mathcal{T}^*\}$, where $\mathcal{M}^* = \{x^3 : x \in [-(2k_{min})^{1/3}, (2k_{max})^{1/3}]_{100}\}$, and $\mathcal{T}^* = \{\exp(x) : x \in [\log(1/365), \log(\tau_{max} + 1)]_{100}\}$. Here, $[a, b]_x$ denotes an equally spaced grid of x points on the interval $[a, b]$. Moreover, $k_{min} = \min(\mathcal{I}_0^k)$, $k_{max} = \max(\mathcal{I}_0^k)$, and $\tau_{max} = \max(\mathcal{I}_0^\tau)$. The sets $\mathcal{I}_0^k$ and $\mathcal{I}_0^\tau$ denote the unique moneyness values and maturities in the observed sample, and $N_{12} = |\mathcal{I}_{12}|$. At these artificial grid points, the spline basis is recomputed within the

¹²Ackerer et al. (2020) stress that the no-arbitrage conditions are formulated with respect to log-forward moneyness, $k = \log(K / (Se^{(r-q)\tau}))$. For this reason, the derivatives entering the calendar and butterfly constraints are evaluated with respect to $k = m\sqrt{\tau}$, rather than directly with respect to the standardized moneyness variable used as an input to the neural network.

differentiable model pipeline, allowing the spline prior and the resulting predicted total variance surface to remain compatible with automatic differentiation in PyTorch. $\ell_{cal,t}$ and $\ell_{but,t}$ are then computed using the `autograd` functionality.

Together, these transformations make the synthetic grid denser around the most actively traded regions. The cubic transformation used to construct $\mathcal{M}^*$ concentrates moneyness points around the at-the-money region, while the exponential transformation used to construct $\mathcal{T}^*$ places more maturity points near short maturities, providing finer local coverage across both dimensions than is available from observed contracts alone. Accordingly, the loss function becomes

$$\mathcal{L}(\theta) = \mathcal{L}_0(\theta) + \lambda_1 \mathcal{L}_1^{cal}(\theta) + \lambda_2 \mathcal{L}_2^{but}(\theta) + \frac{1}{2} \lambda_{reg} \|\theta_1\|_2^2. \quad (31)$$

The term $\frac{1}{2} \lambda_{reg} \|\theta_1\|_2^2$ is a standard L_2 -regularization penalty on the parameter vector, included to discourage large weights and improve out-of-sample fit (Lewkowycz & Gur-Ari, 2020). Here, $\|\theta_1\|_2^2 = \sum_{\ell} (\|W_{\ell}\|_F^2 + \|b_{\ell}\|_2^2)$ denotes the squared Euclidean norm of the parameter vector, obtained by summing the squared Frobenius norms of the layer-specific weight matrices W_{ℓ} and the squared Euclidean norms of the bias vectors b_{ℓ} , where ℓ indexes the neural network layers.

4.4. Two-step estimation procedure

To effectively capture the dynamics of the implied volatility surface, a suitable filtering and estimation framework is required, implemented through a two-step procedure. In the first step, the general linear Gaussian state-space estimation approach of Durbin and Koopman (2012) is employed within a maximum likelihood framework. To address the computational challenges associated with the high dimensionality of the observation vector, a collapsing technique in the spirit of Bräuning and Koopman (2014) is applied. The second step then focuses on neural network parameter learning, incorporating several no-arbitrage constraints and correcting for potential misspecifications in the prior model.

4.4.1. Step 1: Prior model estimation

To apply the Kalman filter and related recursions for maximum likelihood estimation, the compressed systems in Eq. (17) and Eq. (25) are recast into a state-space representation. To ensure consistency with the notation and framework of Durbin and Koopman (2012), the model is specified as a (partially) time-invariant linear Gaussian state-space system:

$$\begin{aligned} y_t &= Z_t \alpha_t + \varepsilon_t, & \varepsilon_t &\sim N(0, H), \\ \alpha_{t+1} &= c + T \alpha_t + \eta_t, & \eta_t &\sim N(0, Q). \end{aligned} \quad (32)$$

Aligning notation, $y_t = \log IV_t$ denotes the vector of log implied volatilities, which is linked to the latent state vector $\alpha_t = \beta_t$ containing the factors. The variable $Z_t = M_t$ then represents the design matrix of observable basis functions, while imposing a time-invariant diagonal covariance matrix H of fixed dimension for estimation purposes. The discretized Ornstein-Uhlenbeck coefficient dynamics can be written in state-space form by defining the constant term $c = (I_p - \Phi)\Theta$, the transition matrix $T = \Phi$, and the innovation vector $\eta_t \sim N(0, Q)$. Moreover, the standard linear Gaussian state-space assumptions are imposed. In particular, the innovation terms ε_t and η_t are assumed to be serially independent, mutually independent, and independent of the initial state α_1 .

Let $Y_s = \{y_1, \dots, y_s\}$ denote the information set available up to time s . The Kalman filter provides recursive

estimates of the latent state vector conditional on the available information. In particular, the filtered state estimate $a_{t|t} = \mathbb{E}(\alpha_t|Y_t)$ and its associated covariance matrix $P_{t|t} = \text{Var}(\alpha_t|Y_t)$, as well as the one-step-ahead predictions $a_{t+1|t} = \mathbb{E}(\alpha_{t+1}|Y_t)$ and $P_{t+1|t} = \text{Var}(\alpha_{t+1}|Y_t)$, are obtained through the following recursions:

$$\begin{aligned} v_t &= y_t - Z_t a_{t|t-1}, & F_t &= Z_t P_{t|t-1} Z_t^\top + H, \\ a_{t|t} &= a_{t|t-1} + P_{t|t-1} Z_t^\top F_t^{-1} v_t, & P_{t|t} &= P_{t|t-1} - P_{t|t-1} Z_t^\top F_t^{-1} Z_t P_{t|t-1}, \\ a_{t+1|t} &= c + T a_{t|t-1} + K_t v_t, & P_{t+1|t} &= T P_{t|t-1} T^\top + Q - K_t F_t K_t^\top. \end{aligned} \quad (33)$$

Note that $K_t = T P_{t|t-1} Z_t^\top F_t^{-1}$ defines the Kalman gain. Since the model is linear and Gaussian, the conditional distribution of y_t given Y_{t-1} is multivariate normal. Under conditional independence, the log-likelihood can be expressed as a sum of one-step-ahead log densities, $L(Y_n; \theta_2) = \sum_{t=1}^n \ell(y_t; \theta_2)$, where $\ell(y_t; \theta_2) = \log f(y_t|Y_{t-1}; \theta_2)$. Using the prediction error decomposition implied by the Kalman filter, the log-likelihood can be written as

$$L(Y_n; \theta_2) = -\frac{Nn}{2} \log(2\pi) - \frac{1}{2} \sum_{t=1}^n \log |F_t| - \frac{1}{2} \sum_{t=1}^n v_t^\top F_t^{-1} v_t. \quad (34)$$

The log-likelihood is then evaluated using the prediction error decomposition. For a given parameter vector θ_2 , the Kalman filter generates the prediction errors $v_t = y_t - Z_t a_{t|t-1}$ and their covariance matrices $F_t = Z_t P_{t|t-1} Z_t^\top + H$. Together, v_t and F_t characterize the conditional distribution of the observations and are used to construct the log-likelihood, which is evaluated at each iteration of the optimization routine. The maximum likelihood estimates of θ_2 are then obtained by numerically maximizing this function.

However, as noted by Bräuning and Koopman (2014), direct maximum likelihood estimation becomes computationally costly when the panel dimension increases. The main difficulty is the Kalman gain $K_t = T P_{t|t-1} Z_t^\top F_t^{-1}$, which requires inversion of the $N \times N$ covariance matrix F_t at each likelihood evaluation. This inversion can be particularly demanding in high-dimensional settings. To address this issue, a collapsed filtering approach is employed. Specifically, using standard GLS identities, the projection matrix $A^* = (Z_t^\top H^{-1} Z_t)^{-1} Z_t^\top H^{-1}$ is introduced together with the complementary transformation $A^+ = B(I_N - Z_t A^*)$, where B is a suitably chosen $(N - p) \times N$ matrix. The matrix A^* projects the observation vector into a lower-dimensional state-relevant component, while the complementary transformation A^+ captures the part orthogonal to the information needed for inference on the latent state:

$$\begin{aligned} y_t^* &= (Z_t^\top H^{-1} Z_t)^{-1} Z_t^\top H^{-1} y_t, \\ \implies y_t^* &= \alpha_t + \varepsilon_t^*, \quad \varepsilon_t^* \sim N(0, H_t^*), \quad H_t^* = (Z_t^\top H^{-1} Z_t)^{-1}. \end{aligned} \quad (35)$$

The remaining component of y_t can be discarded, as it is independent of both the initial state α_1 and the transformed observations y_t^* . In particular, it can be shown that $\mathbb{E}(\alpha_t|y_1, \dots, y_s) = \mathbb{E}(\alpha_t|y_1^*, \dots, y_s^*)$ and $\text{Var}(\alpha_t|y_1, \dots, y_s) = \text{Var}(\alpha_t|y_1^*, \dots, y_s^*)$, implying that the transformed observations y_t^* are sufficient for inference on α_t . The original high-dimensional observation equation can therefore be replaced by an equivalent lower-dimensional representation without losing information. This reduces the computational cost associated with inverting the $N \times N$ covariance matrix F_t , while preserving the full probabilistic structure of the model. The log-likelihood can then be expressed in terms of the transformed system:

$$L(Y_n; \theta_2) = C + L(Y_n^*; \theta_2) - \frac{1}{2} \sum_{t=1}^n \log \left(\frac{|H|}{|H_t^*|} \right) - \frac{1}{2} \sum_{t=1}^n e_t^\top H^{-1} e_t. \quad (36)$$

Here, C is a parameter-independent constant. The log-likelihood $L(Y_n^*; \theta_2)$ is obtained by applying the

Kalman filter to the reduced model. The remainder component is captured by $e_t = y_t - Z_t y_t^*$, which accounts for the part of the observation orthogonal to the column space of Z_t in the GLS sense. The maximum likelihood estimator of the parameter vector is then defined as

$$\hat{\theta}_2 = \arg \max_{\theta_2 \in \Psi} L(Y_n; \theta_2), \quad \text{s.t.} \quad \begin{cases} H \succ 0, \\ Q \succeq 0, \\ |\lambda_j(T)| < 1 \quad \forall j. \end{cases} \quad (37)$$

The parameter space is defined as $\Psi = \{\theta_2 : H \succ 0, Q \succeq 0, \rho(T) < 1\}$, where H is required to be positive definite, Q positive semi-definite, and the spectral radius of the transition matrix T is strictly less than one, ensuring stationarity and stability of the state process. Since the state update dynamics in Eq. (32) are stationary under the constraints, initialization is based on the unconditional distribution of the state vector. Specifically, the initial state is assumed to follow $\alpha_1 \sim N(a_1, P_1)$, where the mean and covariance correspond to the steady-state moments of the process. In particular, the unconditional mean is given by $a_1 = (I_p - T)^{-1}c$, while the unconditional covariance matrix satisfies $\text{vec}(P_1) = (I_{p^2} - (T \otimes T))^{-1} \text{vec}(Q)$.

4.4.2. Step 2: Neural network learning

Conditional on the previously estimated structural parameters $\hat{\theta}_2$, the neural network is trained to recover the remaining flexibility needed to match the observed implied volatility surface while adhering to economically meaningful shape restrictions. The objective function therefore combines a data-fitting term, penalty terms promoting the desired shape properties, and a standard L_2 regularization to stabilize estimation. Let ϑ denote the admissible parameter space, then the objective function is specified as

$$\hat{\theta}_1 = \arg \min_{\theta_1 \in \vartheta} \left[\mathcal{L}_0(\theta_1; \hat{\theta}_2) + \sum_{k=1}^K \lambda_k \mathcal{L}_k(\theta_1; \hat{\theta}_2) + \frac{1}{2} \lambda_{reg} \|\theta_1\|_2^2 \right]. \quad (38)$$

The objective is minimized using the Adam optimizer of Kingma and Ba (2015), which is well-suited to high-dimensional and non-convex optimization. The neural network consists of three hidden layers with 80 neurons in each layer, remaining flexible enough to reproduce a broad range of implied volatility surfaces. The network parameters are initialized following Glorot and Bengio (2010) so that signals propagated through the network neither explode nor vanish across layers. In particular, the weights and biases, W_i and b_i , are initialized as Gaussian random variables with mean zero and standard deviation $(\eta_{i-1} + \eta_i)^{-1/2}$, where η_i denotes the output dimension of layer i . The parameters of the final output layer are then set to zero so that the network starts from a neutral multiplicative correction before learning deviations from the prior surface. After initialization, training starts with a learning rate of 0.001. Weight decay is set to 10^{-6} , corresponding to the implemented L_2 -regularization term. The network is trained for a maximum of 400 epochs, with the learning rate adjusted adaptively based on the evolution of the average loss. Every 100 epochs, the current loss is compared with its value 100 epochs earlier. If the relative improvement is below 0.3%, the learning rate is reduced by a factor of 2, allowing the optimization procedure to switch gradually from larger initial updates to finer parameter adjustments. In addition, an early-stopping criterion is evaluated every 200 epochs. If the loss has not improved by at least 0.3% over this window, training is terminated. This procedure stabilizes estimation while avoiding unnecessary iterations once the objective has flattened.

The neural network is estimated under several penalty specifications, with $\lambda_k = \lambda \in \{0, 10, 20, 50, 100\}$. The case $\lambda = 0$ represents the purely statistical fit, while larger values place increasing weight on the no-arbitrage penalties. This allows the trade-off between forecast accuracy and economic admissibility to be evaluated.

4.5. Benchmark specifications

Several benchmark specifications are constructed and reported in Table 2 to assess the contribution of each modeling layer in the proposed framework. Rather than evaluating the full model against only a single baseline, the benchmark sequence introduces additional structure step-by-step. This layered design makes it possible to determine whether improvements in forecasting performance arise from the estimation procedure, the specification of the latent time-series dynamics, the flexibility of the cross-sectional representation, or the addition of the neural residual smoother.

4.5.1. Model specifications

Model	Measurement equation	State update / correction equation
Basic two-step VAR model	$\log \sigma_{i,t} =$ $\beta_{0,t} + \beta_{1,t}m_{i,t} + \beta_{2,t}m_{i,t}^2 + \beta_{3,t}\tau_{i,t}$ $+ \beta_{4,t}m_{i,t}\tau_{i,t} + \varepsilon_{i,t}, \quad \varepsilon_{i,t} \sim N(0, s_i^2)$	$\hat{\beta}_t = \mu + \Phi\hat{\beta}_{t-1} + u_t$ $u_t \sim N(0, \Omega)$
VAR five-factor state-space model	$\log IV_t = M_t\beta_t + \varepsilon_t$ $\varepsilon_t \sim N(0, H_t)$	$\beta_t = \mu + \Phi\beta_{t-1} + u_t$ $u_t \sim N(0, \Omega)$
Ornstein–Uhlenbeck five-factor model	$\log IV_t = M_t\beta_t + \varepsilon_t$ $\varepsilon_t \sim N(0, H_t)$	$d\beta_t = K(\Theta - \beta_t)dt + \Gamma dW_t^{\mathbb{P}}$ $dW_t^P (dW_t^P)^\top = R dt$ $\beta_t = (I_p - \Phi)\Theta + \Phi\beta_{t-1} + \eta_t$ $\eta_t \sim N(0, Q)$
Ornstein–Uhlenbeck tensor spline model	$\log \sigma_t(m_{i,t}, \tau_{i,t}) =$ $\beta_{0,t} + \sum_{j=1}^{K_m} \beta_{j,t}^{(m)} \Phi_j(m_{i,t}) + \sum_{h=1}^{K_\tau} \beta_{h,t}^{(\tau)} \Psi_h(\tau_{i,t})$ $+ \sum_{j=1}^{K_m} \sum_{h=1}^{K_\tau} \beta_{j,h,t}^{(m,\tau)} \Phi_j(m_{i,t}) \Psi_h(\tau_{i,t}) + \varepsilon_{i,t}$ $\varepsilon_{i,t} \sim N(0, s_i^2)$	$d\beta_t = K(\Theta - \beta_t)dt + \Gamma dW_t^{\mathbb{P}}$ $dW_t^P (dW_t^P)^\top = R dt$ $\beta_t = (I_p - \Phi)\Theta + \Phi\beta_{t-1} + \eta_t$ $\eta_t \sim N(0, Q)$
Neural smoother model	$\omega_t(m, \tau) = \sigma_t^2(m, \tau)\tau$ $\omega_t^\theta(m, \tau) =$ $\omega^{nn}(m, \tau, \tilde{\sigma}_t^{prior}; \theta_1) \times \omega_t^{prior}(m, \tau; \theta_2)$ $\sigma_t^\theta(m, \tau) = \sqrt{\frac{\omega_t^\theta(m, \tau)}{\tau}}$	$\omega_t^{prior}(m, \tau; \theta_2)$ from spline state-space model $\omega^{nn}(m, \tau, \tilde{\sigma}_t^{prior}; \theta_1)$ neural multiplicative correction

Table 2: **Measurement and state update equations for the competing implied volatility surface models.** The table summarizes the observation equation and the dynamic state update equation for each model specification used in the forecasting exercise. The measurement equation defines how observed log implied volatilities or total variance are linked to the latent factors, spline coefficients, or smoothing components. The state update equation describes how the corresponding factor or coefficient process evolves over time. For the neural smoother, the right-hand column reports the prior spline state-space input and neural correction rather than a separate state-update equation.

4.5.2. Sequential contributions

The basic two-step VAR model (M0) serves as the basic benchmark and least flexible cross-sectional specification. It follows the five-factor representation of Gonçalves and Guidolin (2006) and is estimated in two steps: the cross-sectional coefficients are first obtained by ordinary least squares, after which their dynamics are modeled using a VAR(1) specification. Building on this benchmark, the VAR state-space model (M1) retains the same cross-sectional representation and VAR(1) dynamics but estimates the system jointly by state-space maximum likelihood. The Ornstein-Uhlenbeck model (M2) then replaces the discrete-time VAR dynamics with continuous-time structured mean-reversion, while preserving the same measurement equation. The tensor spline model (M3) keeps this state-space and mean-reverting structure but replaces the global polynomial basis with a flexible tensor-product spline representation. Finally, the neural smoother model (M4) builds on M3 by learning a second-stage multiplicative correction in total variance space, reducing the remaining discrepancy between the spline prior and the observed surface.

4.6. Evaluation metrics

The performance of competing models is evaluated along two dimensions. First, statistical criteria are used to assess both the in-sample fit and the out-of-sample forecasting accuracy of the implied volatility surface and related densities. Second, economic criteria are used to examine whether differences in statistical performance translate into economically meaningful improvements in risk measurement and tail-risk evaluation. This distinction is important because a model that fits well statistically does not necessarily provide superior performance from a risk-management or investment perspective.

4.6.1. Statistical criteria: general evaluation

The general statistical evaluation compares the competing models from two complementary perspectives. The first is in-sample fit, which assesses how well each specification explains the data used for estimation. The second is the out-of-sample forecasting performance, which evaluates how accurately the estimated dynamics predict the future implied volatility surface. The in-sample fit of the likelihood-based models is evaluated using the maximized log-likelihood. For model m , the in-sample log-likelihood is defined as:

$$\mathcal{L}_{in}^{(m)} = \sum_{t \in \mathcal{T}_{train}} \log f^{(m)}(y_t | \mathcal{F}_{t-1}; \hat{\theta}^{(m)}). \quad (39)$$

In this expression, $f^{(m)}(y_t | \mathcal{F}_{t-1}; \hat{\theta}^{(m)})$ denotes the conditional density implied by model m for the observed training data. A higher in-sample log-likelihood indicates that the model assigns a greater probability to the realized estimation sample and therefore provides a better statistical fit. However, likelihood values alone are not sufficient for model comparison, since more flexible specifications can mechanically improve the fit by introducing additional parameters. To address this trade-off between goodness of fit and model complexity, the Akaike (1974) information criterion (AIC) and the Schwarz (1978) information criterion (BIC) are also reported:

$$AIC^{(m)} = -2\mathcal{L}_{in}^{(m)} + 2k_m, \quad BIC^{(m)} = -2\mathcal{L}_{in}^{(m)} + k_m \log(n_{train}). \quad (40)$$

Additionally, the Hannan and Quinn (1979) information criterion (HQIC) is considered, defined as $HQIC^{(m)} = -2\mathcal{L}_{in}^{(m)} + 2k_m \log \log(n_{train})$. Here, k_m denotes the number of estimated parameters of model m and n_{train} the number of in-sample dates. All reported information criteria penalize over-parameterization, so lower values indicate a more favorable balance between fit and parsimony.

The second part of the evaluation focuses on out-of-sample forecasting performance. This is assessed by comparing the predicted implied volatility surface with the realized surface observed on the following day. Let $\hat{\sigma}_{i,t|t-1}^{(m)}(m_{i,t}, \tau_{i,t})$ denote the time t forecasted implied volatility at moneyness $m_{i,t}$ and maturity $\tau_{i,t}$, based on information available at time $t - 1$, and let $\sigma_{i,t}(m_{i,t}, \tau_{i,t})$ denote the corresponding realized value. Forecast accuracy is then summarized by the mean absolute error (MAE) and the mean squared error (MSE), computed across all out-of-sample days, moneyness values, and maturities:

$$\begin{aligned} MAE^{(m)} &= \frac{1}{|\mathcal{T}_{test}|} \sum_{t \in \mathcal{T}_{test}} \frac{1}{N_t} \sum_{i=1}^{N_t} \left| \sigma_{i,t}(m_{i,t}, \tau_{i,t}) - \hat{\sigma}_{i,t|t-1}^{(m)}(m_{i,t}, \tau_{i,t}) \right|, \\ MSE^{(m)} &= \frac{1}{|\mathcal{T}_{test}|} \sum_{t \in \mathcal{T}_{test}} \frac{1}{N_t} \sum_{i=1}^{N_t} \left(\sigma_{i,t}(m_{i,t}, \tau_{i,t}) - \hat{\sigma}_{i,t|t-1}^{(m)}(m_{i,t}, \tau_{i,t}) \right)^2. \end{aligned} \quad (41)$$

Forecast errors are averaged within each out-of-sample date, ensuring that each trading day contributes equally to the overall loss measure irrespective of the number of observations. The MAE provides a direct measure of the average error magnitude and is less sensitive to extreme deviations. The MSE, by contrast, places greater weight on larger errors and is therefore informative when a model occasionally produces substantial mispricing of the surface. Lower values of both MAE and MSE indicate more accurate out-of-sample forecasts. Statistically significant differences in forecast performance are assessed using the Diebold-Mariano test, which evaluates whether the two competing models have equal predictive accuracy. Additionally, the Model Confidence Set is used to identify the set of superior models by sequentially eliminating specifications with significantly inferior predictive performance. Both test specifications can be found in Appendix B.

4.6.2. Statistical criteria: risk-neutral density forecasts

The risk-neutral density summarizes how option markets price future index states and is useful for evaluating downside risk, crash insurance, and tail probabilities. Since the models forecast the implied volatility surface rather than the density directly, the forecasted surface is first translated into a cross-section of predicted option prices. For each strike-maturity pair, the forecasted implied volatility is inserted into the Black-Scholes formula, together with the underlying level, risk-free rate, dividend yield, and remaining time to maturity. This yields predicted option prices across strikes for each fixed maturity. The curvature of this call-price function then identifies the risk-neutral density through the Breeden and Litzenberger (1978) relation, which links the second derivative of the call price¹³ with respect to strike to the risk-neutral density:

$$\hat{f}_{t|t-1,T}(K_g, \cdot) = e^{r\tau} \frac{\partial^2 \hat{C}_{t|t-1,T}(K_g, \cdot)}{\partial K^2}. \quad (42)$$

Here, K_g denotes the g -th point on the observed strike grid and T denotes the corresponding option maturity date. This approach does not require imposing a separate parametric distribution of the underlying index. Instead, the density is implied by the shape of the forecasted call price curve itself. Because option prices are observed only on a discrete strike grid, the second derivative is approximated numerically by the unequal-grid central finite-difference expression:

$$\frac{\partial^2 \hat{C}_{t|t-1,T}(K_g, \cdot)}{\partial K^2} \approx \frac{2}{K_{g+1} - K_{g-1}} \left[\frac{\hat{C}_{t|t-1,T}(K_{g+1}, \cdot) - \hat{C}_{t|t-1,T}(K_g, \cdot)}{K_{g+1} - K_g} - \frac{\hat{C}_{t|t-1,T}(K_g, \cdot) - \hat{C}_{t|t-1,T}(K_{g-1}, \cdot)}{K_g - K_{g-1}} \right]. \quad (43)$$

¹³To mitigate the sensitivity of numerical second derivatives to local noise, the predicted call price curve is first smoothed using a cubic smoothing spline before differentiation.

The accuracy of the resulting density forecast is evaluated using two complementary scoring rules. The first is the log-density score, which evaluates the density assigned to the realized future spot price, $LDS_{t|t-1,T} = -\log \hat{f}_{t|t-1,T}(S_T)$. A lower value indicates that the realized outcome lies in a region to which the model assigns relatively high density. This score therefore rewards local sharpness around the realized outcome. The second metric is the continuous ranked probability score (CRPS) (Matheson & Winkler, 1976). This metric evaluates the entire forecasted distribution rather than only the density at the realized point:

$$CRPS_{t|t-1,T} = \int_{-\infty}^{\infty} \left(\hat{G}_{t|t-1,T}(x) - \mathbb{I}\{S_T \leq x\} \right)^2 dx. \quad (44)$$

Here, $\hat{G}_{t|t-1,T}(x)$ denotes the cumulative distribution function associated with the forecasted risk-neutral density. The CRPS measures the distance between the forecasted CDF and the empirical CDF implied by the realized spot price. Lower CRPS values indicate better overall probabilistic accuracy. These two complementary measures balance two aspects of density forecasts: the log-density score rewards assigning high density to the realized outcome, while the CRPS rewards accurate calibration of the full forecasted distribution. Lower values of each respective loss indicate more accurate risk-neutral density forecasts. Statistical superiority in density forecasting is again assessed using pairwise Diebold-Mariano tests.

4.6.3. Economic evaluation: applications to Value-at-Risk backtesting

Following Zou et al. (2025), the economic evaluation is conducted through a Value-at-Risk backtest. Value-at-Risk is used as a lower-tail threshold for the next-period average log implied volatility. Let the forecasted average log implied volatility at date t , based on information available at date $t-1$, be defined as $\tilde{P}_{t|t-1} = \frac{1}{N_t} \sum_{i=1}^{N_t} \log \hat{\sigma}_{t|t-1}(m_{i,t}, \tau_{i,t})$, where N_t is the number of contracts observed on date t . The Value-at-Risk is then given by

$$\widehat{VaR}_{\alpha,t|t-1} = \tilde{P}_{t|t-1} + \frac{z_{\alpha}}{N_t} \sqrt{\iota_{N_t}^{\top} F_t \iota_{N_t}}. \quad (45)$$

Here, z_{α} denotes the α -quantile of the standard normal distribution, and ι_{N_t} is an $N_t \times 1$ vector of ones. The matrix F_t denotes the forecast-error covariance matrix of the vector of contract-level log implied volatilities. It aggregates both latent-factor uncertainty and residual uncertainty and is computed as

$$F_t = M_t P_{t|t-1}^{\beta} M_t^{\top} + H_t. \quad (46)$$

Here, M_t is the design matrix, $P_{t|t-1}^{\beta}$ is the Kalman predicted covariance matrix of the latent coefficient vector as in Eq. (33), and H_t is the diagonal matrix of residual variances. H_t is constructed at the contract level by assigning each contract the estimated variance corresponding to its moneyness-maturity bucket. A Value-at-Risk violation then occurs when the observed average log implied volatility, $\tilde{P}_t = \frac{1}{N_t} \sum_{i=1}^{N_t} \log \sigma_t(m_{i,t}, \tau_{i,t})$, falls below the predicted lower-tail threshold, that is, when $\tilde{P}_t < \widehat{VaR}_{\alpha,t|t-1}$. The backtest evaluates unconditional coverage using the test of Kupiec (1995), and evaluates the conditional coverage of violations using the test of P. F. Christoffersen (1998). Both test specifications are found in Appendix B. In addition, the tick loss is reported to evaluate the accuracy of the predicted VaR quantile:

$$TL_{\alpha,t} = (\alpha - \mathbb{I}\{\tilde{P}_t < \widehat{VaR}_{\alpha,t|t-1}\})(\tilde{P}_t - \widehat{VaR}_{\alpha,t|t-1}). \quad (47)$$

The reported value is the average tick loss. While the violation rate only indicates how often the VaR threshold is breached, the tick loss also accounts for the size of the forecasting error. A lower average tick loss therefore indicates more accurate forecasts of the lower-tail quantile.

5. Empirical results

This section presents the empirical results of the competing implied volatility surface models and situates them with the existing literature. The analysis evaluates whether extensions introduced in the methodology improve model performance, starting from the basic two-step VAR benchmark and moving toward the more flexible spline-based and learning-corrected specifications. The models are first evaluated in- and out-of-sample in Subsection 5.1 using various metrics, followed by an assessment of no-arbitrage penalization in Subsection 5.2. Subsection 5.3 then evaluates risk-neutral density forecasts, while Subsection 5.4 presents a Value-at-Risk backtesting exercise alongside tick losses.

5.1. Statistical performance: general evaluation

The statistical evaluation compares the competing model specifications using both in-sample and out-of-sample criteria. In-sample performance is assessed using likelihood-based measures and information criteria, while out-of-sample performance is evaluated using forecast errors based on the mean squared error and mean absolute error. Particular attention is given to region-specific errors, as these reveal whether model improvements are concentrated in economically important areas of the surface.

5.1.1. In-sample fit

Model	Log-likelihood	AIC	HQIC	BIC	Parameters
VAR	137,498.30	-274,918.60	-274,836.97	-274,694.70	39
OU	144,283.15	-288,468.30	-288,365.74	-288,186.98	49
Spline	146,852.45	-293,320.90	-292,919.02	-292,218.61	192

Table 3: **In-sample model comparison based on likelihood and information criteria using bucketed data.** The table reports the in-sample log-likelihood, Akaike information criterion (AIC), Hannan-Quinn information criterion (HQIC), Bayesian information criterion (BIC), and number of estimated parameters for each model specification. Lower AIC, HQIC, and BIC values indicate better relative model performance after accounting for model complexity and in-sample fit.

Table 3 reports the in-sample fit of the five-factor state-space VAR, five-factor Ornstein-Uhlenbeck, and tensor spline specifications on the bucketed dataset using likelihood-based criteria. The basic two-step five-factor VAR and neural smoother are excluded because they are not estimated within the same likelihood-based framework. Contrary to van der Wel et al. (2016), where spline-based models are not preferred over the general dynamic factor model, the spline model attains the highest log-likelihood and lowest information criterion values, indicating the strongest in-sample performance even after accounting for the substantially larger number of estimated parameters in the spline model. This suggests that the spline basis captures relevant variation in the implied volatility surface rather than only adding flexibility.

The OU specification ranks second across the likelihood-based criteria. Its improvement relative to the VAR benchmark indicates that imposing mean-reverting dynamics on the latent coefficients provides a better description of the in-sample evolution of the surface, consistent with the persistent but stationary factor behavior documented in Subsection 3.4. This is in line with Wang et al. (2017), who find that an OU-based state-space model outperforms two-step VAR and random-walk models, supporting structured mean-reverting dynamics. However, these results should be interpreted as preliminary evidence of relative fit rather than as a direct measure of forecasting performance. Since the criteria are evaluated on the same observations used for estimation, they may favor specifications that capture sample-specific variation.

5.1.2. Out-of-sample performance on bucketed data

Model ID	All regions				Region-specific (DOTM put)			
	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
Full test sample (2019-2023)								
M0 (Basic)	0.00021 ***	0.01009 ***	0.83	0.93	0.00023 ***	0.01109 ***	0.76	0.89
M1 (VAR)	0.00025	0.01083	1.00	1.00	0.00030	0.01247	1.00	1.00
M2 (OU)	0.00024 **	0.01065 **	0.96	0.98	0.00030	0.01146 ***	0.99	0.92
M3 (Spline)	0.00017 ***	0.00819 ***	0.71	0.76	0.00034	0.01077 ***	1.13	0.86
Non-COVID period								
M0 (Basic)	0.00009 ***	0.00730 ***	0.85	0.94	0.00011 ***	0.00866 ***	0.65	0.84
M1 (VAR)	0.00011	0.00779	1.00	1.00	0.00018	0.01027	1.00	1.00
M2 (OU)	0.00011 **	0.00769	0.96	0.99	0.00013 ***	0.00817 ***	0.72	0.80
M3 (Spline)	0.00007 ***	0.00584 ***	0.65	0.75	0.00011 ***	0.00682 ***	0.60	0.66
COVID period (2020-2022)								
M0 (Basic)	0.00027 ***	0.01168 ***	0.83	0.93	0.00030 ***	0.01248 ***	0.79	0.91
M1 (VAR)	0.00032	0.01257	1.00	1.00	0.00037	0.01372	1.00	1.00
M2 (OU)	0.00031 **	0.01235 *	0.96	0.98	0.00040 **	0.01334 **	1.06	0.97
M3 (Spline)	0.00023 ***	0.00954 ***	0.72	0.76	0.00048 *	0.01303	1.28	0.95

Table 4: **Model comparison using bucketed implied volatility surface data.** The table reports aggregate forecast errors and DOTM put region-specific forecast errors for each model specification over the full test sample, the COVID period, and excluding COVID. Forecast accuracy is measured using the mean squared error (MSE) and mean absolute error (MAE). For each sample period, relative MSE and MAE are computed against the VAR benchmark, which is normalized to one. Stars indicate significance from Diebold-Mariano tests comparing each model against the VAR benchmark. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. When a relative error metric is statistically significant, values below one indicate statistically superior forecast performance relative to the VAR benchmark, while values above one indicate significantly inferior performance.

Table 4 reports the out-of-sample forecasting performance of the competing model specifications using bucketed data. Across all reported aggregate metrics, the tensor spline Ornstein-Uhlenbeck model is identified as the dominant specification. It delivers the lowest aggregate forecast errors in all samples, with the largest relative gains excluding COVID, where the aggregate relative MSE and MAE fall to 0.65 and 0.75, respectively. The full-sample pairwise DM tests in Appendix E show that these improvements are statistically significant relative to the competing models under both metrics, while the MCS results reinforce the same conclusion by retaining the tensor spline OU model as the final model across all sample periods and loss functions considered.

The region-specific deep out-of-the-money put results are more mixed than the aggregate evidence. For MAE, the spline model and basic two-step VAR model emerge as the dominant specifications, supported by the pairwise Diebold-Mariano tests and Model Confidence Sets. Under MSE, the spline model does not retain its MAE advantage: it does not consistently deliver statistically significant improvements over competing specifications and is itself outperformed by alternatives. The basic two-step VAR model instead emerges as the strongest-performing model under this metric, as it is retained in every MCS. Moreover, among the remaining models, the OU specification is preferred to the VAR model in most samples, although the absolute performance differences are small. These results differ somewhat from Wang et al. (2017), as the bucketed results show that the restricted OU specification does not uniformly outperform the basic two-step

VAR benchmark, suggesting that any forecasting gains in this application cannot be attributed to structured mean-reverting dynamics alone. Similarly, van Vliet (2017) finds that adding additional dynamic structure improves in-sample fit but does not automatically translate into better out-of-sample performance.

A caveat is that these results are based on bucketed data rather than original contract-level observations. This distinction matters because Ackerer et al. (2020) emphasize that IVS models should generalize to unobserved regions, since market data may be noisy and sparse away from the money. Bucketing may therefore overstate performance by smoothing contract-level noise and local variation. This may explain the relatively strong performance of the basic two-step VAR model: daily re-estimated coefficients can summarize broad movements in the implied volatility surface more effectively than the restrictive mean-reverting OU structure. Consequently, strong bucketed performance indicates broad surface-level accuracy but not necessarily reliable handling of localized contract-level errors. For this reason, less emphasis is placed on the bucketed results, and the full-contract-level analysis is treated as the more informative evaluation.

5.1.3. Out-of-sample performance on full-contract data

Model ID	All regions				Region-specific (DOTM put)			
	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
Full test sample (2019-2023)								
M1 (VAR)	0.21378	0.04168	1.00	1.00	0.49230	0.08339	1.00	1.00
M2 (OU)	0.00999	0.03848	0.05	0.92	0.02686	0.07702	0.05	0.92
M3 (Spline)	0.00752	0.03643 *	0.04	0.87	0.02187	0.08763	0.04	1.05
M4 (Smoother)	0.00058	0.01402 ***	0.00	0.34	0.00142 *	0.02453 ***	0.00	0.29
COVID period (2020-2022)								
M1 (VAR)	0.01189	0.04117	1.00	1.00	0.03191	0.08011	1.00	1.00
M2 (OU)	0.01317	0.04587 ***	1.11	1.11	0.03576	0.09349 ***	1.12	1.17
M3 (Spline)	0.00701 **	0.03536 **	0.59	0.86	0.02133 **	0.08558	0.67	1.07
M4 (Smoother)	0.00079 ***	0.01691 ***	0.07	0.41	0.00201 ***	0.03110 ***	0.06	0.39
Excluding 2023								
M1 (VAR)	0.00997	0.03736	1.00	1.00	0.02664	0.07152	1.00	1.00
M2 (OU)	0.01116	0.04166 ***	1.12	1.12	0.03014	0.08354 ***	1.13	1.17
M3 (Spline)	0.00770	0.03770	0.77	1.01	0.02245	0.09000 ***	0.84	1.26
M4 (Smoother)	0.00067 ***	0.01526 ***	0.07	0.41	0.00167 ***	0.02709 ***	0.06	0.38

Table 5: **Model comparison using full-contract implied volatility surface data.** The table reports aggregate forecast errors and DOTM put region-specific forecast errors for each model specification over the full test sample, the COVID period, and excluding 2023. Forecast accuracy is measured using the mean squared error (MSE) and mean absolute error (MAE). For each sample period, relative MSE and MAE are computed against the VAR benchmark, which is normalized to one. Stars indicate significance from Diebold-Mariano tests comparing each model against the VAR benchmark. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. When a relative error metric is statistically significant, values below one indicate statistically superior forecast performance relative to the VAR benchmark, while values above one indicate significantly inferior performance.

Table 5 reports the out-of-sample forecasting performance of the competing model specifications using the full contract data. The basic two-step VAR model is excluded because its forecast design is not directly comparable to the dynamic specifications evaluated here. Based on the reported metrics, the neural smoother extension of the tensor spline Ornstein-Uhlenbeck model emerges as the strongest specification. It delivers the

lowest aggregate and region-specific MSE and MAE in all reported samples and significantly outperforms the VAR, OU, and spline models in the majority of aggregate and region-specific pairwise Diebold-Mariano tests reported in Appendix E3 and E4. This suggests that the prior models leave systematic contract-level residual variation that is exploited by the neural smoother to improve forecast accuracy beyond the structured surface forecasts. This finding is closely related to Ackerer et al. (2020), who show in S&P 500 option data that a neural network correction to a standard IVS prior can improve interpolation and extrapolation accuracy relative to uncorrected baseline surfaces.

Furthermore, Table 6 reinforces the statistical superiority of the smoother model over the competing specifications. The smoother is retained by the Model Confidence Set as the final model across all sample periods and loss functions in both the aggregate and region-specific tests reported in Appendix E7 and E8. This shows that the smoother model improves forecast accuracy not only on average, but also consistently across different market conditions. This is coherent with S. Chen et al. (2016), who find that nonlinear measurement equations improve both IVS fitting and forecasting accuracy relative to linear dynamic specifications. This therefore suggests that the neural correction adds statistically meaningful predictive content beyond the tensor spline OU surface, especially for the full contract data, where residual errors are concentrated in specific regions.

Sample	MSE			MAE		
	First removed	Last removed	p -value	First removed	Last removed	p -value
Full sample	VAR	Spline	0.000	VAR	Spline	0.000
COVID period	OU	Spline	0.000	OU	Spline	0.000
Excluding 2023	OU	Spline	0.000	OU	VAR	0.000

Table 6: Model confidence set results summary for full-contract out-of-sample forecasts across all reported aggregate samples. For each evaluation sample and loss measure, the table reports the first and last model removed from the model confidence set, together with the corresponding p -values. Results are reported separately for mean squared error (MSE) and mean absolute error (MAE). Across all sample periods and evaluation metrics, the smoother is the final model retained in the MCS, indicating that it remains in the superior set under every reported specification.

The spline prior model provides the clearest aggregate secondary ranking among the remaining specifications, reinforcing the value of a flexible cross-sectional representation. The aggregate error metrics generally favor the spline model, while the Model Confidence Sets generally retain it longer than the competing prior specifications. This advantage is less pronounced in the region-specific results. Under MSE, the spline model remains competitive and is retained longer than competing benchmarks. Under MAE, however, this relative advantage weakens, as the spline model does not consistently outperform the state-space VAR benchmark. This suggests that the relative performance of the OU and state-space VAR models may be influenced by a small number of comparatively large forecast errors. Overall, the results suggest that the loss of spline dominance in this specific region is not simply a ranking fluctuation but reflects the difficulty of representing localized tail dynamics even with a flexible cross-sectional basis.

Overall forecast errors remain larger than in the bucketed setting, confirming that full contract-level forecasting is considerably more difficult because the models must predict noisy individual observations rather than smoothed averages. This illustrates a key limitation in existing literature: conclusions are drawn from bucketed contract surfaces rather than from the full unbalanced contract-level cross-section (e.g. van der Wel et al., 2016). Furthermore, recent implied volatility surface studies emphasize surface-generating neural models, with Zheng et al. (2021) showing that neural networks with embedded financial constraints outper-

form benchmark models for S&P 500 implied volatility surface predictions, and Zhang et al. (2023) obtaining strong forecasts from learned surface features combined with DNN reconstruction. The present results point to a different role for machine learning: the neural component is most useful as a correction to a structured prior forecast, rather than a standalone replacement for the surface model. This is emphasized in Figure 3, which illustrates how the set of prior models already captures a large amount of the variation in average implied volatility, thereby leaving the neural component to correct only remaining systematic deviations.

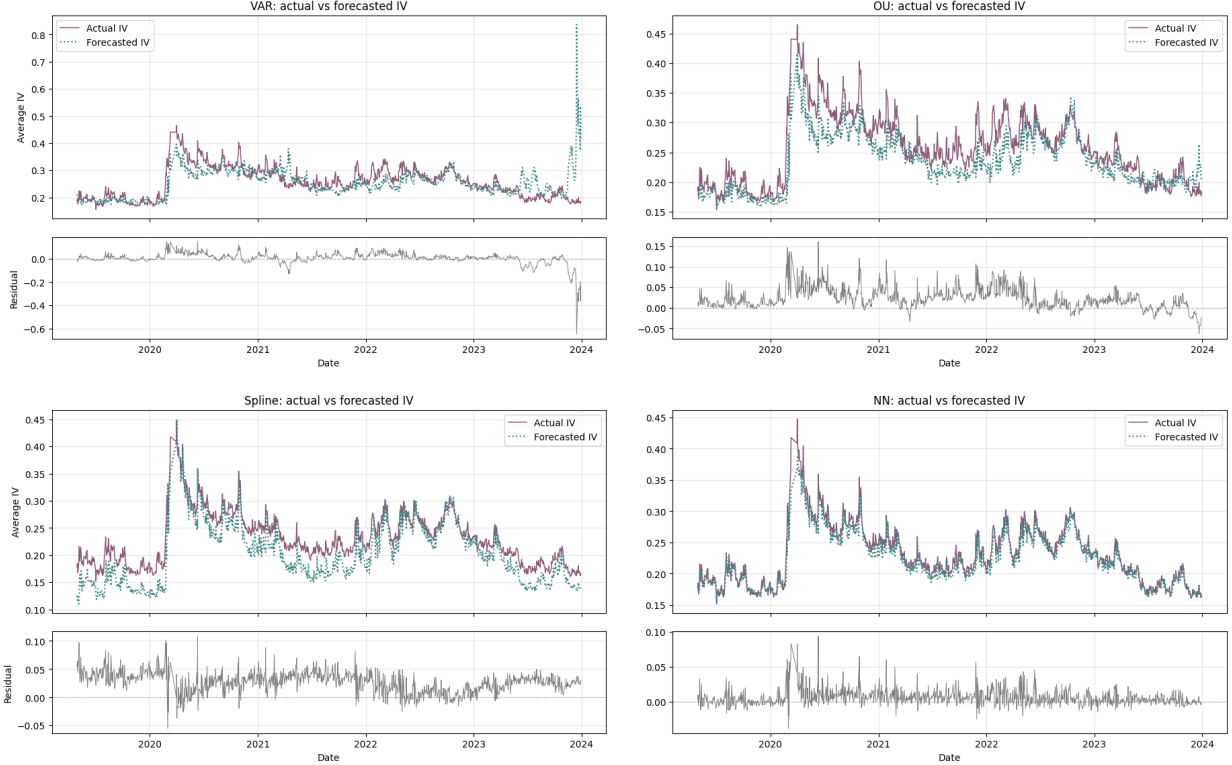


Figure 3: **Actual, forecasted, and residual average implied volatility for the full-contract test sample.** Each panel reports one forecasting model: the VAR benchmark, the Ornstein-Uhlenbeck state-space model, the tensor spline OU model, and the neural-network smoother. For each model, the upper subplot shows the daily cross-sectional average of the observed implied volatility together with the forecasted value, while the lower subplot shows the daily average residual, defined as observed implied volatility minus forecasted implied volatility. The figure is based on the common evaluation dates available for all models so that the displayed paths are directly comparable across specifications.

Figure 3 presents the specification-level average daily implied volatility predictions against the actual observed values. The plot reinforces the conclusions above: the spline model improves visibly over the VAR and OU specifications, tracking the main movements in implied volatility more closely. However, the smoother model clearly emerges as the best fit, as its forecasted series follows the actual implied volatility very closely throughout the sample and leaves visibly smaller forecast errors. The figure also reveals the main drawback of the VAR specification. In the 2023 period, the forecasted series displays mildly explosive behavior, resulting in a small number of very large forecast errors, consistent with the elevated aggregate VAR mean squared error of Table 5. This points to a weakness of the less restricted VAR specification that is not visible in Zou et al. (2025), whose sample ends before 2023. Meanwhile, this reveals an advantage of the OU specification: although it performs worse in aggregate, its structured mean-reverting dynamics help avoid the unstable dynamics observed for the state-space VAR model.

5.1.4. Region-specific residual reductions

The region-specific results in Table 5 reveal a much sharper pattern than in the bucketed evaluation, with forecast errors clearly concentrated in the deep out-of-the-money put region. This is consistent with evidence that surface construction and forecasting errors are particularly severe in tail regions of the implied volatility surface (e.g., Ulrich et al., 2023; van der Wel et al., 2016; van Vliet, 2017). The large differences between MSE and MAE indicate that this region generates disproportionately large outliers rather than small uniform errors across the surface. This issue is not resolved by solely adding flexibility: during the COVID period, the spline model’s MSE in the deep out-of-the-money put region is about 3.04 times larger than its aggregate MSE. This concentration of residuals in the full-contract evaluation explains why the neural smoother is trained on contract-level observations: they reveal tail-region errors that are partly masked by bucketing, allowing the smoother to correct systematic forecast errors left by the parametric tensor spline OU model.

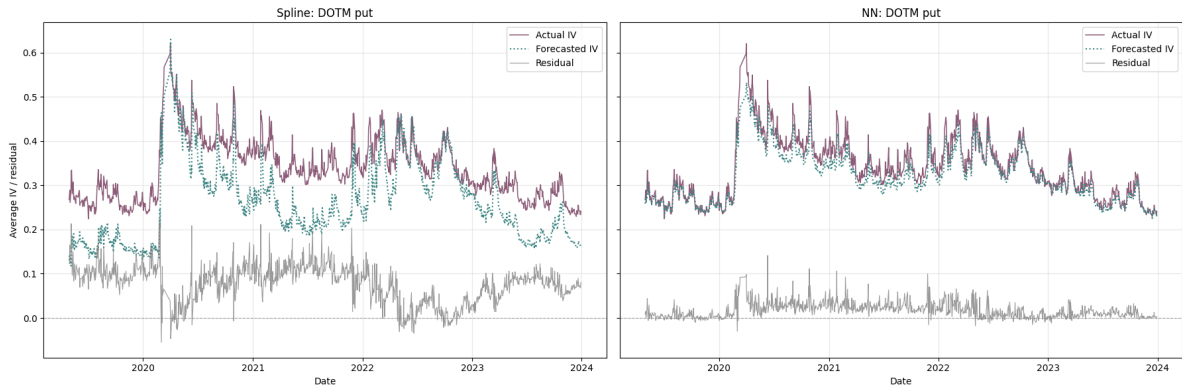


Figure 4: **Actual, forecasted, and residual average implied volatility for the full-contract DOTM put region.** The figure reports daily time series for the deep out-of-the-money put region in the full-contract test sample. The left panel corresponds to the spline-based model, while the right panel corresponds to the neural network smoother. In each panel, the upper series show the cross-sectional average of the observed implied volatility and the corresponding forecasted implied volatility for contracts classified as deep out-of-the-money puts. The lower series shows the associated average residual, computed as the observed implied volatility minus forecasted implied volatility over the same contracts and dates.

Figure 4 shows the forecasted and residual series for the deep out-of-the-money put region. The plot reinforces the evidence observed in Table 5: the spline model has substantially larger forecast errors than its aggregate results, as its forecasted series does not track the observed deep out-of-the-money put implied volatility closely and leaves large residuals around high-volatility periods. This localized weakness is particularly relevant in light of Ulrich and Walther (2020), who show that spline-based volatility surfaces display weaknesses in the short-maturity left tail. In contrast, Figure 4 shows that the neural smoother substantially mitigates this region-specific issue. It follows the observed series more closely and produces residuals that remain visibly tighter around zero. In fact, as shown in Table 5, the neural smoother reduces the average absolute residual in the deep out-of-the-money put region to a level below the aggregate residuals of the benchmark models.

The residual heatmaps in Appendix D5 provide further evidence for this interpretation. In contrast to the bucketed setting, the benchmark residuals are no longer spread relatively evenly across the moneyness-maturity domain but are concentrated primarily in the deep out-of-the-money put region across maturity groups. After applying the neural correction, this concentration is substantially reduced. The deep out-of-the-money put region residuals show a pronounced drop, while the deep out-of-the-money call region and the central part of the surface also improve. This indicates that the smoother is not merely shifting errors away from the left tail but reducing systematic nonlinear structure across the surface.

5.1.5. Neural smoothing: a deeper look

To illustrate the added value of the neural smoother, consider the observed implied volatility surface plot on June 12, 2019, shown in Figure 5. The surface displays an upward slope as it moves toward the deep out-of-the-money put region, consistent with Bollen and Whaley (2004), who find that movements in S&P 500 implied volatilities are most strongly affected by buying pressure for index puts. This shape provides a benchmark for assessing whether the spline prior and neural correction can reproduce the observed surface. Since the derivative in the moneyness direction can become positive as the model coefficients vary over time, the prior may generate a downward slope precisely where the observed surface slopes upward.

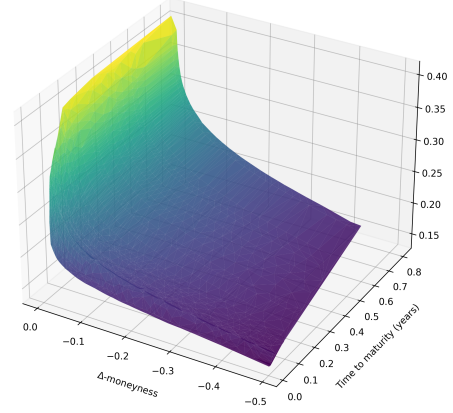


Figure 5: **Observed put option implied volatility surface on June 12, 2019.** The figure displays put option implied volatility across Δ -moneyness and time to maturity.

Figure 6 compares the spline-prior surface with its neural network correction on June 12, 2019. The spline prior misses the deep out-of-the-money region, where it slopes in the opposite direction from the observed surface. The corrected surface then closely resembles the observed surface, indicating that the neural smoother captures local variation in this region while preserving the overall shape of the surface. Importantly, Figure 6 is not a fitted reconstruction: it compares genuine one-step-ahead forecasts, so the improvement reflects predictive rather than purely in-sample smoothing ability.

5.1.6. Robustness checks

The pointwise forecasting analysis already includes several robustness dimensions. The models are evaluated across alternative sample periods, different regions of the implied volatility surface, and multiple loss functions. These comparisons verify that the ranking is not driven by a single market regime, concentrated in one moneyness-maturity region, or dependent on a particular statistical criterion. As an additional robustness exercise, the rolling-window design of Zou et al. (2025) is implemented. Specifically, the models are estimated using a rolling window of 500 observations, corresponding to approximately two years of daily data, after which the next 250 observations, approximately one year, are forecast out-of-sample.

In particular, let y_t denote the daily observation vector and let $R = 500$ and $H = 250$ denote the training and testing window lengths. For rolling block j , the model is estimated on $\mathcal{T}_j^{\text{train}}$ and evaluated on $\mathcal{T}_j^{\text{test}}$, where

$$\mathcal{T}_j^{\text{train}} = \{y_{(j-1)H+1}, \dots, y_{(j-1)H+R}\}, \quad \mathcal{T}_j^{\text{test}} = \{y_{(j-1)H+R+1}, \dots, y_{jH+R}\}. \quad (48)$$

The model parameters are therefore re-estimated as $\hat{\theta}_j = \arg \max_{\theta} \sum_{t \in \mathcal{T}_j^{\text{train}}} \log f(y_t | \mathcal{F}_{t-1}; \theta)$, after which the out-of-sample forecasts are then produced for $t \in \mathcal{T}_j^{\text{test}}$ using the parameters $\hat{\theta}_j$. Ultimately, the out-of-sample performance is computed by pooling forecast errors across all test blocks. The resulting rankings remain largely unchanged under this rolling-window design. In particular, the smoother continues to perform best across the main pointwise forecasting metrics, while the ordering of the benchmark models is broadly preserved. This indicates that the main conclusions are not an artifact of a single fixed training-test split but continue to hold when the estimation sample is updated through time. Appendix F provides a more detailed analysis of the robustness exercise, including heatmaps and pairwise Diebold-Mariano tests.

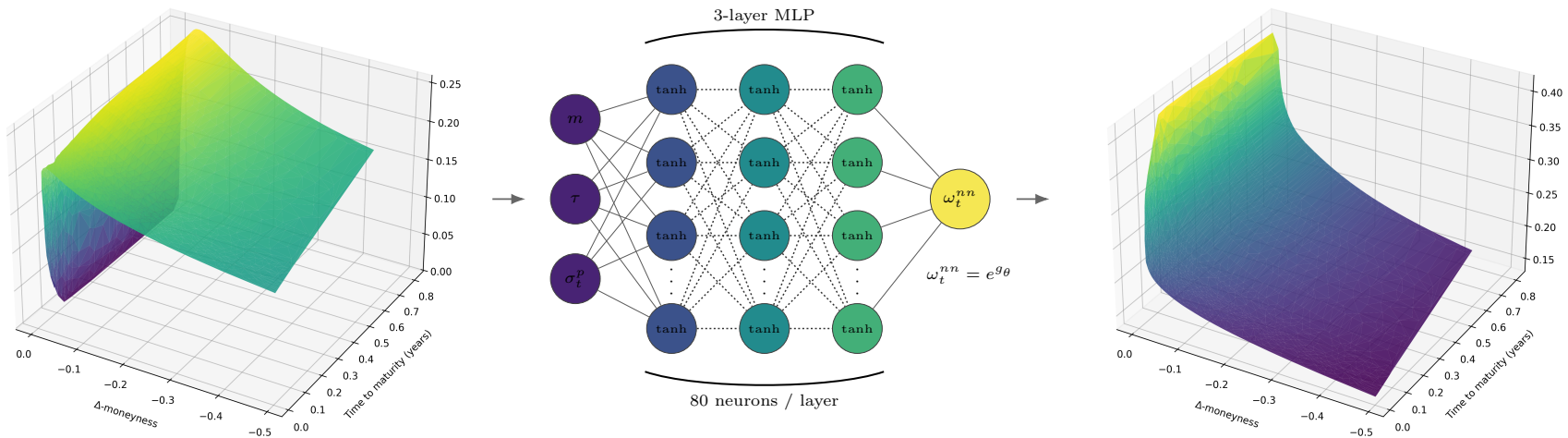


Figure 6: **Neural network correction of the spline-prior implied volatility surface.** The left plot shows the spline-prior put implied volatility surface prediction, while the right plot shows the corresponding neural network-corrected put implied volatility surface. The middle diagram illustrates the correction architecture: a three-layer MLP with 80 neurons per hidden layer and hyperbolic tangent activation functions maps moneyness (m), time-to-maturity (τ), and the prior implied volatility ($\bar{\sigma}_t^{prior}$) into the scalar correction factor. The exponential transformation imposes positivity so that the neural network applies a multiplicative correction in total-variance space while preserving the spline-prior structure.

5.2. Arbitrage consistency

The preceding results evaluate predictive accuracy. This subsection evaluates whether the neural smoother also produces economically admissible implied volatility surfaces. This is important because a model can forecast implied volatilities accurately while still generating surfaces that violate basic no-arbitrage restrictions. The analysis therefore studies the trade-off between statistical accuracy and arbitrage consistency by increasing the soft no-arbitrage penalties in the smoother objective.

5.2.1. Economic admissibility

Table 7 shows that the no-arbitrage penalties substantially improve the economic admissibility of the smoother. Without penalization, the model produces 44,789 calendar arbitrage violations and 122,759 butterfly arbitrage violations, corresponding to 1.57% and 4.31% of evaluated implied volatilities, respectively. Although the unpenalized specification is therefore not severely inconsistent, especially given the no-arbitrage cleaning procedure applied before training, a clear scope for further improvements remains. Introducing the penalties leads to an immediate and economically meaningful reduction in violations. At $\lambda = 10$, calendar and butterfly arbitrage fall to 2,556 and 28,134 violations, corresponding to rates of 0.09% and 0.99%, respectively. Further increases in the penalty continue to generally improve admissibility, with the largest and most consistent reductions observed for butterfly arbitrage, the dominant source of inconsistency in the unpenalized model. The lowest total violations are obtained at $\lambda = 100$, where calendar and butterfly arbitrage decline to 1,302 and 6,984 violations, or just 0.05% and 0.25% of evaluated implied volatilities. Relative to the unpenalized smoother, this corresponds to reductions of approximately 97% and 94%.

Model	λ^{14}	Calendar	Butterfly
Smoother	0	44,789 (1.57%)	122,759 (4.31%)
	10	2,556 (0.09%)	28,134 (0.99%)
	20	2,392 (0.08%)	26,156 (0.92%)
	50	2,120 (0.07%)	15,980 (0.56%)
	100	1,302 (0.05%)	6,984 (0.25%)

Table 7: Smoother arbitrage violations by penalty specification. The table reports the number of predicted implied volatilities classified as calendar and butterfly arbitrage violations for the smoother under different penalty specifications. Violation percentages are shown in parentheses.

5.2.2. Statistical adequacy

Model	λ	MAE (Agg.)	MAE (DOTM)
Smoother	0	0.01405	0.02463
	10	0.01424**	0.02468
	20	0.01437***	0.02493**
	50	0.01488***	0.02521***
	100	0.01569***	0.02666***

Table 8: Smoother forecast errors by penalty specification. The table reports aggregate and DOTM put mean absolute errors (MAE) for the smoother under different no-arbitrage penalty specifications. Stars indicate that the MAE is significantly different from the non-penalized smoother, corresponding to $\lambda = 0$, based on Diebold-Mariano tests. Statistical significance is denoted by ** and *** at the 5% and 1% level.

Table 8 evaluates how the no-arbitrage regularization affects the statistical accuracy of the smoother forecasts. The unpenalized specification attains the lowest aggregate MAE, which is expected because the network is fully optimized toward predictive accuracy without being constrained by economic shape restrictions. As the penalty increases, the aggregate forecast errors rise from 0.01405 at $\lambda = 0$ to 0.01569 at $\lambda = 100$. The significance markers indicate that the MAE increases are statistically distinguishable from the unpenalized specification across nearly all penalty levels. The deterioration is therefore visible and statistically significant but remains

¹⁴The same penalty value is used for both the calendar and butterfly no-arbitrage penalties, so that $\lambda = \lambda_{cal} = \lambda_{but}$.

moderate relative to the large reduction in arbitrage violations documented in Table 7. The same pattern is observed for the deep out-of-the-money put errors. The unpenalized smoother delivers the lowest MAE, while stronger penalization generally increases forecast errors in this region. The deep out-of-the-money put MAE rises from 0.02463 without penalization to 0.02521 at $\lambda = 50$ and 0.02666 at $\lambda = 100$, corresponding to increases of approximately 2.4% and 8.2%, respectively. These costs are small compared with the reduction in total arbitrage violations, which fall by approximately 89% at $\lambda = 50$ and 95% at $\lambda = 100$. This suggests that $\lambda = 50$ provides the most balanced specification, since it achieves most of the improvement in economic admissibility while preserving most of the regional forecasting performance. Nevertheless, even in the worst-performing penalized specification ($\lambda = 100$), the deep out-of-the-money MAE remains less than half that of the next-best benchmark specification and is lower than the aggregate MAE of that benchmark. Together with Table 7, these results show that the no-arbitrage penalty moves the smoother toward a more favorable accuracy-admissibility trade-off. Although higher penalty values increase forecast errors, the loss in accuracy remains modest compared with the improvement in economic admissibility. In fact, the smoother reduces total violations to levels below the restrictive benchmarks while retaining markedly better accuracy.¹⁵ The penalized smoother is therefore useful not only for reducing forecast errors but also for recovering implied volatility surfaces that are more economically coherent. These findings are broadly consistent with Zheng et al. (2021) and Akerer et al. (2020), who show that embedding no-arbitrage restrictions in neural network-based option pricing models can improve shape consistency while preserving predictive performance.

5.3. Statistical performance: risk-neutral density forecasts

This subsection evaluates whether improvements in implied-volatility forecasting translate into better risk-neutral density forecasts. The analysis first compares the models using out-of-sample density scoring rules, namely the log-density score and continuous ranked probability score. It then provides a visual interpretation of the recovered densities across maturities, illustrating the distributional features implied by the different volatility forecasts.

5.3.1. Statistical evidence

Model	LDS	DM statistic	CRPS	DM statistic
VAR	6.8490 ***	7.3766	148.1643 ***	16.2917
OU	6.8412 ***	6.5299	142.7723 ***	25.9604
Spline	6.8159 ***	3.0647	139.8738 ***	14.1889
Smoother	6.8052	—	138.6772	—

Table 9: **Out-of-sample risk-neutral density forecast evaluation.** The table reports the average log-density score (LDS) and continuous ranked probability score (CRPS) for each model, together with the corresponding Diebold-Mariano test statistics relative to the smoother model. Lower values indicate better density forecast performance. Significance stars are based on Diebold-Mariano tests conducted relative to the smoother model, with *, **, and *** denoting significance at the 10%, 5%, and 1% levels, respectively.

Interestingly, most studies on risk-neutral density forecasts evaluate the recovered densities directly, rather than asking whether improvements in the underlying implied volatility surface forecast translate into better density forecasts. The existing evidence on the forecasting performance of risk-neutral densities is mixed.

¹⁵Appendix G reports the number of arbitrage violations across the competing model specifications.

For example, Alonso et al. (2005) cannot reject accurate forecasting performance for IBEX 35 option-implied densities. The results in Table 9¹⁶¹⁷ add to this perspective by showing that forecast accuracy also depends on the quality of the surface forecast used to recover the density: the smoother obtains the lowest log-density score (~ 6.81) and continuous ranked probability score (~ 138.68), suggesting that its advantage extends to the full option-implied predictive distribution. The Diebold-Mariano tests further support this ranking. Relative to the smoother, the competing models perform significantly worse under both scoring criteria, suggesting that the neural smoothing correction not only improves point forecast accuracy but also the quality of the recovered density forecasts.

At the same time, Ornelas (2014) finds that option-implied densities perform poorly in the tails, consistent with the evidence in Jackwerth (2004), likely due to limited strike coverage. This is relevant in the present setting because risk-neutral density recovery is inherently sensitive to the shape of the option-price function. Since the density is obtained from the curvature of option prices, small irregularities in the forecasted implied-volatility surface can be magnified in the recovery step. The results therefore complement the literature on option-implied densities by showing that surface regularity is not only important for fitting implied volatilities but also for the quality of the recovered density forecasts. This is further reflected in Table 9 and the pairwise Diebold-Mariano tests of Appendix E9, where the ranking of the models follows their degree of flexibility: the VAR performs worst, the OU specification improves upon it, and the spline remains the second-best.

5.3.2. Visual interpretation

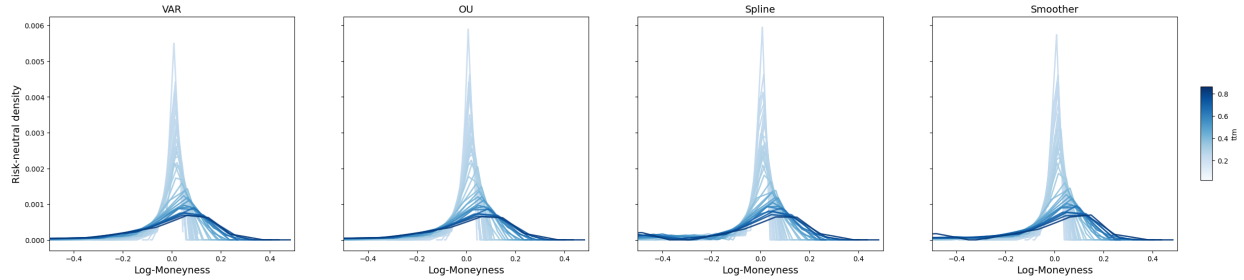


Figure 7: **Risk-neutral density forecasts across maturities for the VAR, OU, spline, and smoother specifications.** Densities are recovered from the one-step-ahead forecasted implied volatility surfaces using the Breeden-Litzenberger relation, after mapping implied volatilities into Black-Scholes call prices. Each curve corresponds to a different time-to-maturity, with darker shades indicating longer maturities.

Figure 7 complements the numerical density forecasts by showing the full shape of the recovered risk-neutral densities across maturities. Across all specifications, the densities are asymmetric, with visibly fatter left tails. This is consistent with evidence that option-implied risk-neutral distributions for equity indexes are negatively skewed and assign substantial probability mass to adverse market outcomes (e.g. Bakshi et al., 2003; Jackwerth, 2004). As time to maturity increases, the distributions become flatter and more dispersed, indicating greater uncertainty over longer horizons. This maturity pattern is in line with Bahra (1997), who notes that implied risk-neutral distributions typically become less spread out as expiration approaches, since shorter horizons leave less room for extreme outcomes. These patterns are economically meaningful

¹⁶Since the point forecast implied volatility results identify the smoother as the best-performing model, the Diebold-Mariano tests in Table 9 are conducted relative to the smoother.

¹⁷To ensure comparability between models and avoid additional estimation error, the risk-free rate, dividend yield, and related inputs are held fixed during the risk-neutral density forecasting exercise.

because the density recovery step depends on the curvature of the option-price curve, so local irregularities in forecasted implied volatilities can be amplified into distortions in the recovered densities.

5.4. Economic performance: Value-at-Risk backtest

This subsection evaluates the economic relevance of density forecasts through a Value-at-Risk backtest. First, empirical violation rates are reported and assessed using unconditional and conditional coverage tests, which examine whether VaR exceedances occur with the correct frequency and whether violations are independent over time. Second, average tick losses are computed to evaluate the accuracy of the predicted VaR quantiles themselves.

5.4.1. Value-at-Risk coverage and violation dynamics

Model ID	$\alpha = 0.01$			$\alpha = 0.05$			$\alpha = 0.10$		
	Rate	p_{UC}	p_{CC}	Rate	p_{UC}	p_{CC}	Rate	p_{UC}	p_{CC}
M1 (VAR)	0.01412	0.189	0.000	0.04678	0.615	0.000	0.07944	0.017	0.000
M2 (OU)	0.00000	0.000	0.000	0.00530	0.000	0.000	0.01147	0.000	0.000
M3 (Spline)	0.00088	0.000	0.000	0.00177	0.000	0.000	0.00353	0.000	0.000
M4 (Smoother)	0.00088	0.000	0.000	0.00265	0.000	0.000	0.01236	0.000	0.000

Table 10: **Coverage tests for density-based VaR forecasts.** The table reports empirical violation rates and p-values from unconditional and conditional coverage tests across nominal probability levels α . The unconditional coverage test evaluates whether the empirical violation frequency equals the nominal level, while the conditional coverage test additionally accounts for violation dependence. Larger p -values indicate weaker evidence against correct coverage. p -values are rounded to three decimals.

Table 10 reports the coverage results for the density-based Value-at-Risk forecasts. Overall, the empirical violation rates are mostly below the nominal levels, indicating conservative tail forecasts. At the same time, the low conditional coverage p -values indicate incorrect calibration, violation dependence, or both. This poor calibration is not entirely surprising in light of Zou et al. (2025). Their results show that even when implied volatility point forecasts are accurate, Value-at-Risk performance can remain poor if the predictive distribution is misspecified. In particular, they emphasize that Value-at-Risk depends on the full predictive distribution, including the covariance structure of forecast errors across the surface, rather than only on marginal forecast accuracy. The results in Table 10 point in a similar direction: the state-space covariance structure is useful for capturing common movements in implied volatility, but the resulting Gaussian predictive density remains too restrictive to deliver correctly calibrated tail forecasts.

The weak coverage, however, should not be interpreted as contradicting the strong forecasting performance reported earlier. The models are designed primarily to forecast implied volatility dynamics, not to serve as fully specified risk-management models. This distinction is consistent with Bams et al. (2017), who emphasize that forecasting volatility and forecasting tail quantiles are different tasks. This distinction is attributed to the volatility risk premium embedded in implied volatilities, while nonlinear and regime-changing dynamics make implied volatility less reliable for tail-quantile forecasting than for average volatility forecasting. The linear Gaussian structure of the models remains too restrictive for VaR evaluation, as Gaussian models typically fail to capture fat tails, jumps, volatility clustering, and regime shifts. Since VaR focuses on extreme quantiles, even small distributional misspecification can lead to rejection of correct coverage.

5.4.2. Quantile-accuracy: tick-loss evaluation

Table 11 presents the average tick loss for the Value-at-Risk forecasts. While the coverage tests evaluate whether violations occur with the correct frequency, the tick-loss criterion measures how close the VaR threshold is to the realized outcome. The smoother model obtains the lowest tick loss across all probability

Model ID	$\alpha = 0.01$	$\alpha = 0.05$	$\alpha = 0.10$
M1 (VAR)	0.00224 ***	0.00889 ***	0.01530 ***
M2 (OU)	0.00191 ***	0.00790 ***	0.01408 ***
M3 (Spline)	0.00291 ***	0.01251 ***	0.02288 ***
M4 (Smoother)	0.00153	0.00572	0.00938

Table 11: Average tick loss for Value-at-Risk forecasts. The table reports average tick loss across nominal probability levels α . Lower values indicate better quantile forecast performance. Significance stars indicate pairwise Diebold-Mariano test significance relative to the smoother model.

levels, with values of 0.00153, 0.00572, and 0.00938 for $\alpha = 0.01$, $\alpha = 0.05$, and $\alpha = 0.1$, respectively. The Diebold-Mariano tests indicate that these improvements are statistically significant relative to the competing specifications. Relative to the second-best model, the smoother reduces tick loss by approximately 20%, 28%, and 33% across the three probability levels. These results refine the coverage evidence: although none of the models achieves perfect Value-at-Risk calibration, the smoother produces Value-at-Risk thresh-

olds that are closest to the realized lower-tail outcomes. Hence, improved implied volatility forecasts do translate into better tail-quantile accuracy, even if the Gaussian density specification remains too restrictive to fully resolve the coverage problem.

These findings are broadly consistent with Zou et al. (2025), who show that improvements in density forecast performance for implied volatility surface models can also translate into better Value-at-Risk outcomes. Their results emphasize that VaR performance depends on the quality of the full predictive density, including the dependence structure of forecast errors, rather than on point forecast accuracy alone. The present results point in a similar direction from a different modeling perspective. Here, the improvement does not come from adjusting the covariance structure of a score-driven model but from using a neural smoothing correction to improve the recovered implied volatility surface and its associated Gaussian predictive distribution. The lower tick losses therefore suggest that the smoother provides economically useful lower-tail information, even though the coverage tests indicate that the Gaussian predictive distribution is not yet sufficiently calibrated to achieve correct VaR coverage.

6. Discussion

The empirical results suggest that the framework is valuable because it combines global dynamic structure with local nonlinear flexibility. The tensor-product spline state-space prior captures broad movements in the surface, but the full-contract results show that this structure is insufficient at the individual contract level. Remaining errors are concentrated in highly nonlinear regions, especially the deep out-of-the-money put region, where option prices are closely linked to downside tail-risk compensation and where buying pressure for index puts affects the shape of implied volatility functions (Bollen & Whaley, 2004; Bollerslev & Todorov, 2011). Rather than improving the forecasts uniformly, the neural smoother corrects local failures of the structured prior in the parts of the surface where dynamic factor models have been shown to struggle (van der Wel et al., 2016; van Vliet, 2017). The main contribution is therefore not simply lower forecast errors, but showing that neural residual learning improves economically important regions while retaining an interpretable dynamic prior. This also shows why aggregate forecast metrics are incomplete: a model

may perform well on average while still failing in the contracts most closely linked to downside-risk pricing. Furthermore, the accuracy-admissibility trade-off is especially important because no-arbitrage is a theoretical pricing condition, not a mechanism that prevents violations from appearing in observed option data. In frictionless markets, arbitrage should disappear immediately, but in real markets, apparent violations may arise because the surface is often constructed from bid-ask prices or derived from poor settlement data (Fengler, 2009). Strong no-arbitrage discipline is therefore not automatically the preferred empirical choice: it may remove implausible model outputs but also smooth away market imperfections that are genuinely present in the data. Conversely, giving the smoother too much flexibility may preserve or improve statistical fit while producing surfaces that are harder to interpret as valid option prices (Akerer et al., 2020; Zheng et al., 2021). A clear modeling choice is therefore required. Pure implied volatility forecasting may tolerate some residual violations, whereas density extraction and economic interpretation require greater admissibility because the surface must represent a coherent arbitrage-free pricing system (Breedon & Litzenberger, 1978; Gatheral & Jacquier, 2014; Roper, 2010). The penalty strength should therefore reflect the intended use of the model rather than being treated as a purely technical tuning parameter.

A first limitation is that the proposed framework encourages, rather than guarantees, arbitrage consistency. Because no-arbitrage is imposed through soft penalties in the neural smoothing objective, the resulting surfaces may still contain residual violations, unlike SVI/SSVI-type parameterizations that impose arbitrage-free structure directly (Gatheral & Jacquier, 2014). A related limitation is that the penalty structure remains selective and practically difficult to extend within the present architecture, since the construction of these penalties becomes less direct once they are applied to a more complex prior specification. While soft constraints provide a practical way to combine flexibility with economic discipline, as in neural arbitrage-aware approaches (Akerer et al., 2020; Zheng et al., 2021), the present paper focuses on a limited set of no-arbitrage restrictions, leaving open whether stronger economic discipline can be imposed without reducing the local benefits of the smoother. Further research could therefore use parameterizations that are arbitrage-free by construction or consider simpler prior models that make it easier to incorporate a broader set of no-arbitrage penalties.

A further limitation concerns the extent to which the forecasting gains can be translated into economic conclusions. Although the smoother improves quantile accuracy, as reflected in lower Value-at-Risk tick losses, the coverage tests show that the tail-risk forecasts remain imperfectly calibrated. This suggests that the predictive density may still be too restrictive for extreme outcomes, where volatility clustering, jumps, and regime shifts are important features. Further research could extend the underlying Gaussianity assumption with more flexible tail dynamics, for example, through heavy-tailed innovations (Roth et al., 2013) or regime-switching specifications (Guo et al., 2023). In addition, the present paper does not assess whether improved implied volatility forecasts translate into profitable trading returns. Because the data do not contain unique option identifiers, rebalancing rules and transaction-cost adjustments cannot be constructed directly. This matters because option-based trading profits are highly sensitive to market frictions and may disappear once transaction costs are imposed (e.g. Bernales & Guidolin, 2014; Gonçalves & Guidolin, 2006; van Vliet, 2017). Developing a profitable option-trading strategy therefore remains a separate question for further research.

These considerations suggest that future work should not treat statistical accuracy, arbitrage consistency, and economic usefulness as separate evaluation criteria. Instead, implied volatility surface models should be assessed by whether they jointly deliver local forecast accuracy, preserve economically admissible pricing relations, and provide reliable information about tail risk.

7. Conclusion

This paper develops an arbitrage-aware framework for forecasting the dynamics of the S&P 500 implied volatility surface. The proposed approach combines a tensor-product spline representation of the cross-section with mean-reverting Ornstein-Uhlenbeck factor dynamics in state-space form and augments this structured prior with a neural smoothing correction for the remaining residual component. In doing so, the paper addresses two central challenges in implied volatility surface modeling: capturing localized nonlinearities that arise in contract-level data, especially in the tails and corners of the surface, while improving economic consistency with no-arbitrage restrictions. The framework is evaluated against a sequence of benchmark specifications using bucketed and full-contract forecasts, arbitrage-consistency diagnostics, risk-neutral density forecasts, and Value-at-Risk backtesting.

The findings of this paper are threefold. First, the proposed framework substantially improves implied volatility surface forecasting by correcting the localized contract-level errors that remain after the spline-based state-space prior has captured the dominant surface dynamics. In the full-contract evaluation, the neural smoother reduces mean absolute error by approximately 66% for aggregate forecasts and by 71% for deep out-of-the-money put forecasts relative to the most restrictive benchmark. The latter result is particularly important because this region is both among the most difficult parts of the surface to forecast and one of the most economically relevant, as it is closely linked to downside-risk pricing and tail-risk premia. A similar improvement pattern is also observed under alternative sample splits, suggesting that the forecasting gains are not specific to a single estimation-evaluation period. These results show that the neural correction does not merely improve average forecast performance but adds predictive content precisely where global factor dynamics are least able to capture local nonlinearities in the implied volatility surface.

Second, the forecasting gains remain largely intact once no-arbitrage penalties are imposed, showing that the model’s improvement is not simply the result of unrestricted flexibility. Under the preferred penalty, the neural smoother reduces total arbitrage violations by approximately 89%, while sacrificing only a limited amount of forecast accuracy relative to the unpenalized model. Although stronger penalty weights lead to statistically significant increases in forecast errors, the magnitude of this deterioration remains modest compared with the substantial improvement in economic admissibility. The trade-off is therefore favorable: the penalty removes a large share of economically problematic surface behavior without substantially undermining the model’s predictive performance. This is important because a surface that fits option quotes well is not necessarily economically meaningful if it violates basic pricing restrictions. The results therefore show that the neural correction can retain most of its local predictive gains while producing surfaces that are substantially more consistent with no-arbitrage pricing theory.

Finally, the results show that the benefits of the proposed framework extend beyond implied volatility forecast accuracy to economically relevant distributional and risk-management applications. The neural smoother delivers the strongest risk-neutral density forecasts under both log-density and continuous ranked probability scores, indicating that improvements in the forecasted volatility surface translate into more accurate forecasts of the market-implied distribution. This is important because the risk-neutral density summarizes the market-implied assessment of future outcomes, including the relative weight assigned to downside states. At the same time, the Value-at-Risk backtesting results provide a more nuanced conclusion. While the coverage tests indicate that none of the models fully capture the realized tail dynamics, the smoother achieves the lowest tick loss across all probability levels. Hence, even when exact tail coverage remains difficult, the proposed framework produces the most accurate quantile forecasts among the competing specifications.

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Appendix

Appendix A: Data cleaning

This subsection presents the pseudo-code and reduction in data for the no-arbitrage filtering procedure used in the data cleaning stage. The algorithm sequentially enforces price bounds, monotonicity, and convexity conditions and returns an approximate maximal subset of option quotes that satisfies the imposed no-arbitrage conditions. This ensures that the subsequent models are estimated on option quotes that are consistent with the imposed no-arbitrage restrictions, rather than on observations potentially distorted by arbitrage violations.

A1. No-arbitrage algorithm

Algorithm 1 Iterative no-arbitrage filtering procedure

```
1: procedure NO-ARBITRAGE FILTER
2: Input: Preliminary filtered option quote dataset  $Q$ , including call quote  $C_t$ , put quote  $P_t$ , underlying
   price  $S_t$ , strike  $K$ , risk-free rate  $r$ , time-to-maturity  $T - t$ , quote date  $t$ , expiry date  $T$ , implied volatility
    $IV_t$ , and dividend yield  $q$ .
3: Output: Filtered dataset  $Q^{AF}$  satisfying all imposed no-arbitrage restrictions.
4:    $Q \leftarrow Q \setminus \text{quotes violating call or put price bounds}$ 
5:    $Q^{(0)} \leftarrow Q$ 
6:    $k \leftarrow 0$ 
7:   converged  $\leftarrow$  False
8:   while not converged do
9:      $Q^{old} \leftarrow Q^{(k)}$ 
10:    Sort  $Q^{(k)}$  by option type, quote date, maturity, and strike
11:     $Q^{(k)} \leftarrow Q^{(k)} \setminus \{\text{quotes violating strike monotonicity}\}$ 
12:     $Q^{(k)} \leftarrow Q^{(k)} \setminus \{\text{quotes violating strike convexity}\}$ 
13:     $Q^{(k+1)} \leftarrow Q^{(k)}$ 
14:    if  $|Q^{(k+1)}| = |Q^{old}|$  then
15:      converged  $\leftarrow$  True
16:    else
17:       $k \leftarrow k + 1$ 
18:    end if
19:  end while
20:   $Q^{AF} \leftarrow Q^{(k+1)}$ 
21:  return  $Q^{AF}$ 
22: end procedure
```

A2. Reduction of data

Step	Description	Obs. remaining	Obs. removed	% removed
0	Raw data	31,112,954	–	–
1	Removal of low prices	27,690,773	3,422,181	11.00
2	Time-to-maturity filter	23,331,733	4,359,040	15.74
3	Removal of ITM options	10,508,485	12,823,248	54.96
4	Removal of high IV quotes	10,502,621	5,864	0.06
5	Upper & lower bounds	10,367,444	135,177	1.29
6	Monotonicity in strike	10,155,818	211,626	2.04
7	Convexity in strike	7,033,371	3,122,447	30.75
8	Iteration until convergence	5,722,205	1,311,166	18.64
Final	Post-convergence clean-up ¹⁸	5,691,606	30,599	0.53

Table 12: **Reduction of data across key filtering and no-arbitrage steps.** The table reports the number of observations remaining and removed, and the percentage of observations removed after each sequential data cleaning and filtering procedure applied to the raw option dataset. The steps include the removal of low-priced options, restrictions on time-to-maturity, exclusion of moderately and deeply ITM contracts, exclusion of high IV quotes, and the enforcement of standard no-arbitrage conditions such as upper & lower bounds, monotonicity and convexity in strike.

Table 12 presents the sequential data cleaning process based on general constraints and no-arbitrage restrictions. Starting from roughly 31.1 million observations, the sample is progressively reduced as the algorithm moves through the constraints. The largest reductions come from the removal of ITM options ($\sim 55\%$), convexity in strikes ($\sim 31\%$), and the final iterations until convergence ($\sim 19\%$). Other steps, such as the removal of low prices, enforcing bounds, and monotonicity, contribute smaller reductions. Note that the statistics for steps 1-7 are based on an initial pass. Subsequently, steps 6 and 7 are iteratively reapplied until convergence, as the earlier steps are static. The final dataset, containing roughly 5.7 million observations, approximates the largest subset that satisfies all imposed no-arbitrage conditions. The magnitude of the reductions highlights that raw option data contains a substantial amount of internal inconsistencies, even after standard preliminary filters. Note that convergence is obtained after 39 iterations.

Appendix B: Test specifications and diagnostics

This subsection presents the formal statistical test specifications used throughout the empirical analysis. It provides the supporting econometric framework for the diagnostic checks and model-comparison procedures reported in the main text. By collecting these details separately, the appendix makes the implementation of the statistical analysis transparent while keeping the main discussion focused on the framework, results, and interpretation.

B1. Augmented Dickey-Fuller test

The Dickey and Fuller (1979) test checks whether a time series has a unit root, meaning whether the series is non-stationary. This is important because a non-stationary series can display persistent movements over time, so shocks to the series may have long-lasting or permanent effects rather than dying out. The null hypothesis is $H_0 : \gamma = 0$, which implies that the series has a unit root and is non-stationary. The alternative hypothesis is $H_A : \gamma < 0$, which implies that the series is stationary and mean-reverting. The regression estimated by the test is:

¹⁸Second removal of high implied volatility quotes and ITM options using the deltas and IVs computed in Subsection 3.3.

$$\Delta y_t = \alpha + \gamma y_{t-1} + \sum_{j=1}^p \delta_j \Delta y_{t-j} + u_t, \quad u_t \sim WN(0, \sigma^2). \quad (49)$$

Here, y_t denotes one series, Δy_t is its first difference, α is an intercept, y_{t-1} is the lagged level of the series, and the lagged differences are included to account for serial correlation in the errors. The ADF statistic reported by the test is the test statistic for γ , but it is compared to Dickey-Fuller critical values rather than the usual t -distribution. If the p -value is sufficiently small, the null hypothesis is rejected, providing evidence that the series is stationary over time. Conversely, if the null hypothesis is not rejected, there is insufficient evidence to conclude that the series is stationary.

B2. Diebold-Mariano test

The Diebold and Mariano (1995) test is used to compare the predictive accuracy of two competing forecasting models. It does this by looking at the loss differential $d_t = L(e_{1,t}) - L(e_{2,t})$, where $L(e_{1,t})$ and $L(e_{2,t})$ are the specified forecast losses of model 1 and model 2, respectively. The null hypothesis is $H_0 : \mathbb{E}[d_t] = 0$, meaning the two models have equal expected predictive accuracy. The alternative hypothesis is $H_A : \mathbb{E}[d_t] \neq 0$, meaning the models differ in expected predictive accuracy. If $\mathbb{E}[d_t] > 0$, model 2 performs better on average because model 1 has higher loss, and vice versa. The test statistic reads:

$$DM = \frac{\bar{d}}{\sqrt{\hat{S}_d/T}}. \quad (50)$$

Here, $\bar{d} = \frac{1}{T} \sum_{t=1}^T d_t$. Because forecast errors, and therefore loss differentials, can be serially correlated, the variance of $\bar{d}$ is estimated using a heteroskedasticity and autocorrelation consistent estimator, $\hat{S}_d = \hat{\gamma}_0 + 2 \sum_{j=1}^5 \hat{\gamma}_j$. Here, a fixed HAC truncation lag of five trading days is used, so autocovariances of the loss differential up to lag five are included. This HAC estimator corrects the standard error for both changing variance over time and autocorrelation in d_t . Under the null hypothesis and suitable regularity conditions, the Diebold-Mariano statistic is asymptotically standard normally distributed, permitting the use of standard normal critical values. For a usual 5% two-sided test, the null of predictive accuracy is rejected when $|DM| > 1.96$.

B3. Model Confidence Set

Hansen et al. (2011) propose an iterative procedure that compares a collection of forecasting models and retains those for which equal predictive ability cannot be rejected at a specified confidence level. Starting from a pool $\mathcal{M}$ of candidate models with losses $L_{m,T+s}$, pairwise loss differentials $d_{m,m',T+s}$ are computed for models m and m' at time $T+s$. The objective is to identify the subset $\mathcal{M}_{1-\alpha}^*$ of models that cannot be statistically distinguished in terms of predictive performance. The null hypothesis is $H_0 : \mathbb{E}[d_{m,m',T+s}] = 0, \forall m, m' \in \mathcal{M}$. If H_0 is rejected, the candidate model with the highest average loss among the remaining models is removed, and the procedure is continued until H_0 is not rejected. The remaining set $\mathcal{M}_{1-\alpha}^*$ is the Model Confidence Set. The test statistic reads:

$$T_{R,\mathcal{M}} = \max_{m,m' \in \mathcal{M}} |t_{m,m'}|. \quad (51)$$

Here, $t_{m,m'} = \bar{d}_{m,m'} / \sqrt{\widehat{\text{Var}}(\bar{d}_{m,m'})}$, where $\bar{d}_{m,m'}$ denotes the average loss differential between models m and m' . Since the statistics $t_{m,m'}$ are likely dependent across model comparisons, their covariance structure must

be accounted for. As such, p -values for $T_{R,\mathcal{M}}$ are obtained by bootstrap. First, $T_{R,\mathcal{M}}^{emp}$ is computed using the empirical data. Then, a block bootstrap is applied to account for autocorrelation in the forecast errors, and therefore also in $d_{m,m',T+s}$, yielding the bootstrapped loss differentials $d_{m,m',T+s}^*$. Next, bootstrapped Diebold-Mariano test statistics are computed from these resampled loss differentials:

$$t_{m,m'}^* = \frac{\bar{d}_{m,m'}^* - \bar{d}_{m,m'}}{\sqrt{\widehat{Var}^*(\bar{d}_{m,m'})}}. \quad (52)$$

Re-centering ensures the bootstrap is performed under the null hypothesis of EPA. Finally, $T_{R,\mathcal{M}}^*$ is computed for every bootstrap replicate using the bootstrapped statistics $t_{m,m'}^*$. The corresponding p -value is then given by $\hat{p}_{R,\mathcal{M}} = \frac{1}{B} \sum_{b=1}^B \mathbb{I}\{T_{R,\mathcal{M}}^{emp} < T_{R,\mathcal{M}}^{*,b}\}$. Thus, the MCS algorithm starts with a model set $\mathcal{M}$ containing all candidate models and tests the null hypothesis of EPA using the block bootstrap at significance level α . If the EPA test is rejected, the worst-performing model is removed from the set, and the test is repeated on the reduced model set. This elimination procedure continues until the EPA null is no longer rejected. The remaining models then form the MCS $\mathcal{M}_{1-\alpha}^*$ which contains the models that are statistically indistinguishable in terms of predictive ability at confidence level $1 - \alpha$.

B4. Unconditional coverage test

The Kupiec (1995) unconditional coverage test is used to evaluate the accuracy of Value-at-Risk forecasts. It tests whether the observed frequency of VaR violations is consistent with the nominal violation probability implied by the VaR level. Let the exceedance indicator be given by $I_{T+s} = \mathbb{I}\{r_{T+s} \leq \widehat{\text{VaR}}_{\alpha,t+s}\}$, where $I_{T+s} = 1$ if the realized return violates the VaR threshold and $I_{T+s} = 0$ otherwise. The empirical violation rate is defined as $\hat{p} = \frac{1}{|\mathcal{T}_{test}|} \sum_{s \in \mathcal{T}_{test}} I_{T+s}$. The Kupiec likelihood-ratio test statistic is then given by:

$$LR = 2 \cdot |\mathcal{T}_{test}| \cdot \log\left(\left(\frac{1-\hat{p}}{1-\alpha}\right)^{1-\hat{p}} \left(\frac{\hat{p}}{\alpha}\right)^{\hat{p}}\right). \quad (53)$$

Under the null hypothesis of correct unconditional coverage, the VaR model produces violations with probability α . Hence, the observed violation ratio should not differ systematically from the nominal violation probability implied by the chosen VaR level. Under this null hypothesis, the Kupiec test statistic is asymptotically distributed as $LR_{UC} \sim \chi^2(1)$. The null hypothesis is rejected at significance level α_{test} if LR_{UC} exceeds the corresponding critical value from the $\chi^2(1)$ distribution. A rejection indicates that the VaR model is not correctly calibrated in terms of unconditional coverage, meaning that it generates either too many or too few VaR violations relative to the expected frequency.

B5. Conditional coverage test

The P. F. Christoffersen (1998) conditional coverage test extends the Kupiec test by also assessing whether VaR violations are independent over time. The sequence of violations is modeled as a two-state Markov chain, which allows model adequacy to be decomposed into an unconditional coverage component and an independence component. This makes it possible to detect clustering in VaR violations, which is often a sign of misspecified volatility dynamics. The transition matrices are given by:

$$\Pi_1 = \begin{pmatrix} 1 - p_{0,1} & p_{0,1} \\ 1 - p_{1,1} & p_{1,1} \end{pmatrix}, \quad \Pi_2 = \begin{pmatrix} 1 - p_{0,1} & p_{0,1} \\ 1 - p_{0,1} & p_{0,1} \end{pmatrix}. \quad (54)$$

Here, Π_1 allows the probability of a violation to depend on whether a violation occurred in the previous period, whereas Π_2 imposes conditional independence by restricting this probability to be the same across

states. The corresponding likelihoods are denoted by $\mathcal{L}(\Pi_1)$ and $\mathcal{L}(\Pi_2)$, respectively. The conditional coverage likelihood-ratio statistic is defined as:

$$LR_{cc} = 2 [\log \mathcal{L}(\Pi_2) - \log \mathcal{L}(\alpha)] + 2 [\log \mathcal{L}(\Pi_1) - \log \mathcal{L}(\Pi_2)] \sim \chi^2(2). \quad (55)$$

This statistic combines the unconditional coverage component with the independence component and is asymptotically $\chi^2(2)$ -distributed under the null hypothesis of correct conditional coverage. Its computation is based on the transition counts $n_{i,j} = \#\{I_{T+s-1} = i, I_{T+s} = j\}$, for $i, j \in \{0, 1\}$, which record how often the exceedance indicator moves from state i in one period to state j in the next.

Under the alternative of first-order dependence, the exceedances follow a two-state Markov chain with estimated transition probabilities $\hat{p}_{0,1} = n_{01}/(n_{00} + n_{01})$, $\hat{p}_{1,1} = n_{11}/(n_{10} + n_{11})$. The corresponding likelihood is given by $\mathcal{L}(\Pi_1) = (1 - \hat{p}_{0,1})^{n_{00}} (\hat{p}_{0,1})^{n_{01}} (1 - \hat{p}_{1,1})^{n_{10}} (\hat{p}_{1,1})^{n_{11}}$. Under the null hypothesis of conditional independence, exceedances are independently Bernoulli distributed with probability $p = \hat{p}$, yielding the likelihood $\mathcal{L}(\Pi_2) = \hat{p}^{S\hat{p}} (1 - \hat{p})^{S(1-\hat{p})}$, where S denotes the sample size. Finally, $\mathcal{L}(\alpha)$ denotes the likelihood evaluated under the nominal violation probability $p = \alpha$, as used in the unconditional coverage test.

Appendix C: Preliminary statistics

This subsection presents the preliminary statistics used to describe and better understand the data introduced in Section 3. It first examines the correlation structure of the implied volatility observations in order to assess the degree of co-movement across maturities and moneyness categories. The analysis then summarizes the main patterns in the data using Principal Component Analysis, which helps identify the dominant sources of variation. Finally, the estimated coefficients are reported and supported by corresponding time-series plots.

C1. Implied volatility cross-correlations

Maturity	Moneyness	7-45						180-360					
		DOTM Put	OTM Put	ATM Put	ATM Call	OTM Call	DOTM Call	DOTM Put	OTM Put	ATM Put	ATM Call	OTM Call	DOTM Call
7-45	DOTM Put	1.000	0.975	0.953	0.942	0.926	0.902	0.883	0.902	0.893	0.891	0.871	0.853
	OTM Put		1.000	0.992	0.983	0.973	0.946	0.880	0.915	0.920	0.919	0.904	0.880
	ATM Put			1.000	0.990	0.985	0.962	0.868	0.909	0.922	0.922	0.913	0.887
	ATM Call				1.000	0.994	0.974	0.867	0.907	0.921	0.927	0.919	0.895
	OTM Call					1.000	0.988	0.864	0.904	0.923	0.930	0.927	0.909
	DOTM Call						1.000	0.862	0.899	0.918	0.927	0.929	0.925
180-360	DOTM Put							1.000	0.983	0.964	0.958	0.939	0.919
	OTM Put								1.000	0.991	0.985	0.971	0.949
	ATM Put									1.000	0.992	0.986	0.962
	ATM Call										1.000	0.993	0.969
	OTM Call											1.000	0.979
	DOTM Call												1.000

Table 13: **Correlation matrix across moneyness and maturity groups.** The table reports pairwise correlations of log implied volatility series across moneyness categories (deep out-of-the-money, out-of-the-money, at-the-money) and option types (put and call) for two maturity groups (7-45 days and 180-360 days). Each block of the table corresponds to within- and across-maturity correlations, with the upper-left and lower-right blocks representing within-group correlations and the off-diagonal blocks capturing cross-maturity relationships. All entries are rounded to three decimal places where needed.

C2. Implied volatility autocorrelations

Moneyness	Maturity	ACF					PACF				
		1	2	3	5	10	1	2	3	5	10
DOTM Put	7-45	0.956	0.924	0.896	0.849	0.773	0.956	0.111	0.060	0.052	0.015
	180-360	0.988	0.980	0.972	0.956	0.922	0.989	0.144	0.030	0.041	-0.025
OTM Put	7-45	0.964	0.938	0.916	0.875	0.800	0.965	0.121	0.053	0.024	0.003
	180-360	0.987	0.978	0.969	0.952	0.915	0.988	0.137	0.009	0.042	-0.030
ATM Put	7-45	0.966	0.941	0.920	0.881	0.810	0.966	0.133	0.055	0.033	0.013
	180-360	0.987	0.978	0.969	0.951	0.913	0.987	0.146	0.024	0.019	-0.053
ATM Call	7-45	0.964	0.942	0.920	0.882	0.811	0.964	0.172	0.045	0.023	0.014
	180-360	0.987	0.979	0.969	0.953	0.916	0.987	0.185	-0.009	0.057	-0.021
OTM Call	7-45	0.969	0.950	0.930	0.897	0.834	0.969	0.184	0.023	0.034	0.009
	180-360	0.988	0.981	0.972	0.957	0.923	0.988	0.179	-0.003	0.042	-0.032
DOTM Call	7-45	0.973	0.955	0.939	0.911	0.855	0.973	0.162	0.054	0.030	-0.008
	180-360	0.989	0.983	0.975	0.961	0.930	0.989	0.202	-0.009	0.031	-0.040

Table 14: **Autocorrelation and partial autocorrelation of log implied volatility across moneyness and maturity groups.** The table reports the sample autocorrelation function (ACF) and the partial autocorrelation function (PACF) values computed for log implied volatility series grouped by moneyness (DOTM, OTM, ATM) and option type (put, call) separately for two maturity ranges (7-45 days and 180-360 days). Both the ACF and PACF report values for lags 1, 2, 3, 5, and 10. All entries are rounded to three decimal places where needed.

C3. Principal component analysis

	Log-scale		Regular scale		Excl. 2020	
	%	Cum. %	%	Cum. %	%	Cum. %
PC 1	95.49%	95.49%	94.88%	94.88%	94.87%	94.87%
PC 2	1.89%	97.38%	2.61%	97.49%	2.08%	96.94%
PC 3	1.45%	98.83%	1.54%	99.02%	1.73%	98.67%
PC 4	0.40%	99.23%	0.28%	99.30%	0.45%	99.12%
PC 5	0.21%	99.44%	0.23%	99.53%	0.26%	99.38%
PC 10	0.03%	99.83%	0.03%	99.87%	0.04%	99.81%

Table 15: **Explained variance of the first ten principal components on log-scale, regular scale, and excluding 2020.** The table reports the proportion of total variance explained by each principal component under the log-transformed sample (left panel), regular sample (middle panel), and sample excluding 2020 (right panel), along with the cumulative contribution across components. Note that the sample excluding 2020 is based on the log scale of implied volatilities.

C4. Principal component autocorrelations

	ACF						PACF					
	1	2	3	4	5	10	1	2	3	4	5	10
PC1	0.980	0.965	0.950	0.936	0.923	0.871	0.980	0.106	0.029	0.013	0.031	-0.013
PC2	0.931	0.883	0.842	0.805	0.773	0.660	0.931	0.127	0.042	0.037	0.028	0.038
PC3	0.940	0.904	0.870	0.838	0.813	0.737	0.940	0.182	0.033	0.015	0.057	0.033

Table 16: **Autocorrelation and partial autocorrelation of the first three principal components.** The left panel presents the ACF values, and the right panel presents the PACF values. For each principal component, the reported values correspond to lags 1, 2, 3, 4, 5, and 10, with rows indicating the respective components and columns indicating the lag structure for both ACF and PACF.

C5. Plot of estimated coefficients over time

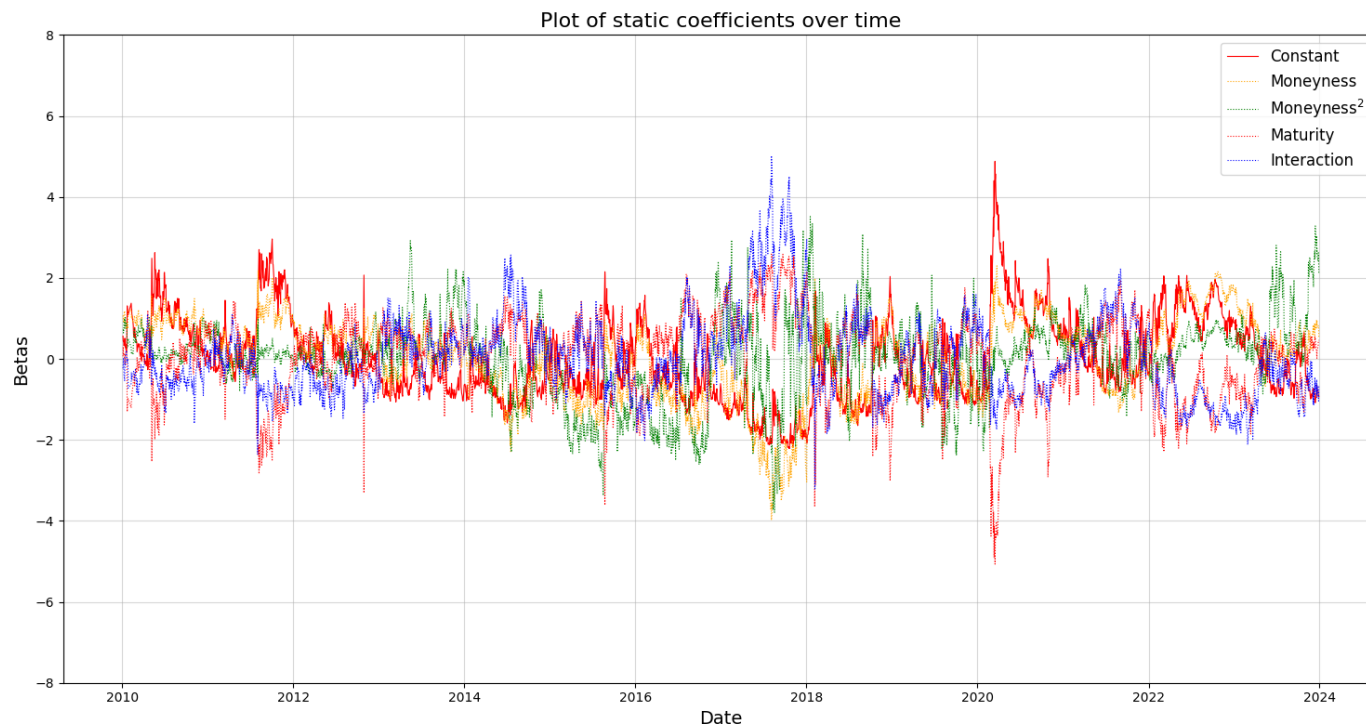


Figure 8: **Time series of standardized coefficient estimates from the five-factor specification obtained from cross-sectional OLS regressions.** Each line represents the estimated coefficient associated with one of the regressors in the specification: constant, moneyness, squared moneyness, time-to-maturity, and their interaction. The coefficients are plotted at daily frequency.

C6. Autocorrelations and ADF tests for daily estimated cross-sectional coefficients

	ACF				PACF				ADF	
	1	2	3	10	1	2	3	10	Stat.	p-val
$\hat{\beta}_{0,t}$ (c)	0.974	0.954	0.935	0.837	0.974	0.114	0.036	-0.006	-5.089	1.47e-05
$\hat{\beta}_{1,t}$ (m)	0.974	0.959	0.944	0.882	0.975	0.189	0.054	0.019	-4.171	7.36e-04
$\hat{\beta}_{2,t}$ (m^2)	0.937	0.886	0.841	0.640	0.938	0.062	0.036	0.027	-5.271	6.30e-06
$\hat{\beta}_{3,t}$ (τ)	0.943	0.903	0.868	0.697	0.943	0.125	0.050	0.012	-6.948	9.88e-10
$\hat{\beta}_{4,t}$ ($m\tau$)	0.954	0.926	0.903	0.787	0.954	0.187	0.078	0.025	-4.937	2.95e-05

Table 17: **Autocorrelation, partial autocorrelation, and Augmented Dickey-Fuller test results for the estimated coefficient series.** The table reports ACF and PACF values at lags 1, 2, 3, and 10 for each coefficient series, together with the ADF test statistic and corresponding p-value. The coefficients correspond to the level ($\hat{\beta}_{0,t}$), moneyiness ($\hat{\beta}_{1,t}$), squared moneyiness ($\hat{\beta}_{2,t}$), maturity ($\hat{\beta}_{3,t}$), and interaction ($\hat{\beta}_{4,t}$) factors.

C7. VAR(1) estimated coefficients

	Level		Moneyiness		Moneyiness ²		Maturity		Interaction	
	Coef.	Sign.	Coef.	Sign.	Coef.	Sign.	Coef.	Sign.	Coef.	Sign.
Intercept	-0.001		0.054	**	-0.026		-0.017		-0.088	***
$\hat{\beta}_{0,t-1}$ (c)	0.997	***	0.030		-0.057		-0.005		-0.021	
$\hat{\beta}_{1,t-1}$ (m)	0.013		1.011	***	0.068	*	-0.021	*	-0.043	**
$\hat{\beta}_{2,t-1}$ (m^2)	-0.001		-0.021	***	0.918	***	0.005		0.019	***
$\hat{\beta}_{3,t-1}$ (τ)	0.053	***	0.020		-0.072		0.920	***	-0.006	
$\hat{\beta}_{4,t-1}$ ($m\tau$)	0.008		0.050	***	0.061	*	-0.013		0.900	***
Stability	Stable (all eigenvalues inside unit circle)									

Table 18: **Estimated VAR(1) coefficients for the time series of estimated cross-sectional coefficients.** The table reports the coefficient estimates and corresponding significance levels for each equation in the VAR(1) system, where the dependent variables are the level, moneyiness, squared moneyiness, maturity, and interaction factors. Rows correspond to the intercept and the first lag of each factor. Statistical significance is denoted by ***, **, and * at the 1%, 5%, and 10% levels, respectively.

C8. Principal components and coefficient series correlations

	$\hat{\beta}_{0,t}$	$\hat{\beta}_{1,t}$	$\hat{\beta}_{2,t}$	$\hat{\beta}_{3,t}$	$\hat{\beta}_{4,t}$
PC1	0.992	0.809	0.045	-0.828	-0.714
PC2	0.119	0.027	-0.201	-0.528	-0.294
PC3	0.010	0.501	0.534	-0.016	-0.360

Table 19: **Correlation between principal components and estimated coefficient series.** The table reports contemporaneous correlations between the first three principal components of the implied volatility surface and the daily cross-sectional regression coefficients. The regression coefficients include the constant term ($\hat{\beta}_{0,t}$), the linear moneyiness ($\hat{\beta}_{1,t}$) and quadratic moneyiness ($\hat{\beta}_{2,t}$) coefficients, the maturity coefficient ($\hat{\beta}_{3,t}$), and the interaction coefficient ($\hat{\beta}_{4,t}$).

C9. Implied volatility surface residual analysis

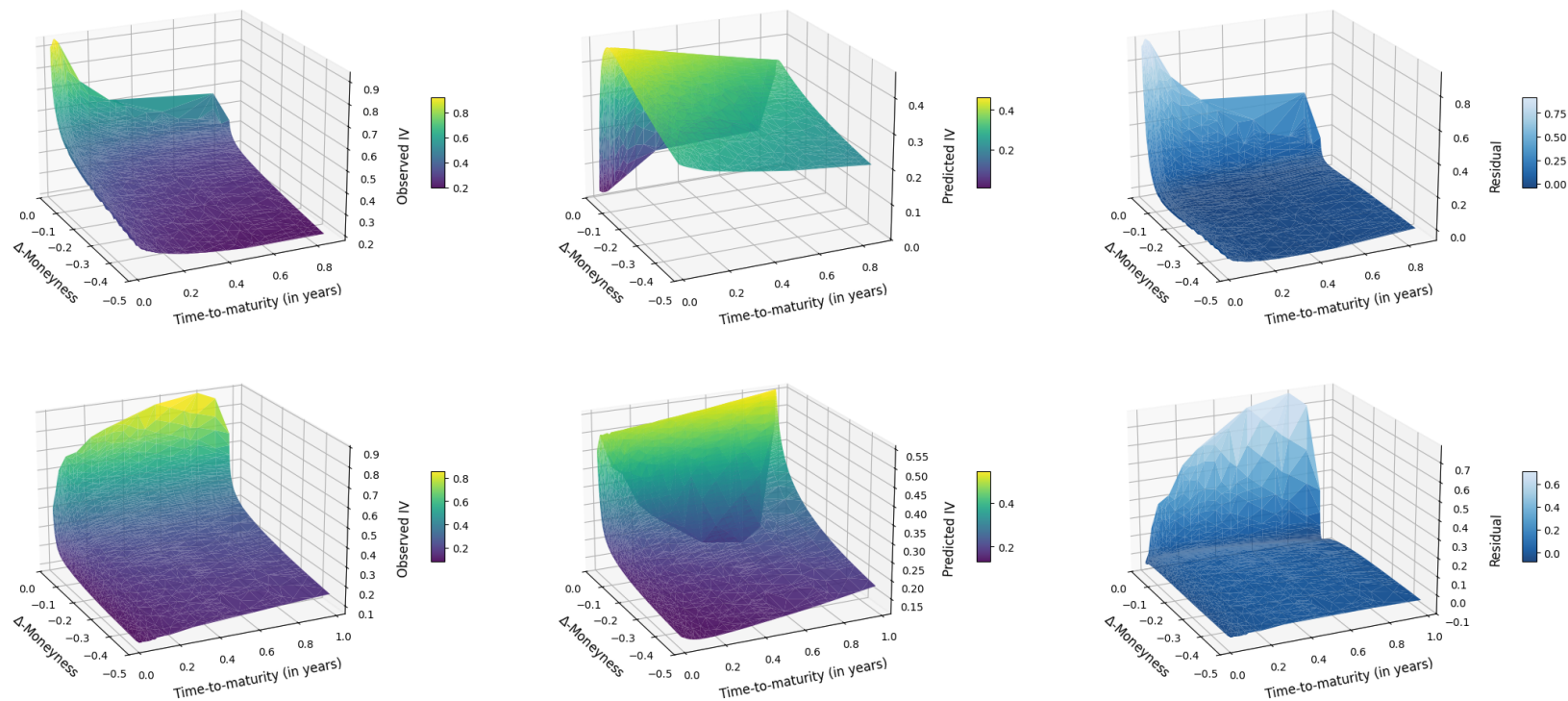


Figure 9: **Observed, predicted, and residual put option implied volatility surfaces on March 4, 2020, and June 25, 2021.** The top row presents the observed, predicted, and residual put option implied volatility surfaces for March 4, 2020, while the bottom row presents the corresponding surfaces for June 25, 2021. All surfaces are plotted as functions of Black-Scholes delta and time to maturity. The predicted implied volatility is computed using the VAR state-space model.

Appendix D: Residual heatmaps

This subsection presents the residual heatmaps encountered throughout the preliminary analysis and in the evaluation of the competing model specifications. The heatmaps provide a visual diagnostic of where each model systematically over- or under-predicts implied volatility across the moneyness and maturity dimensions. They therefore complement the numerical forecast-error measures reported in the main results by showing whether forecast errors are concentrated in specific regions of the implied volatility surface, such as deep out-of-the-money puts or short-maturity contracts.

D1. Basic five-factor VAR model

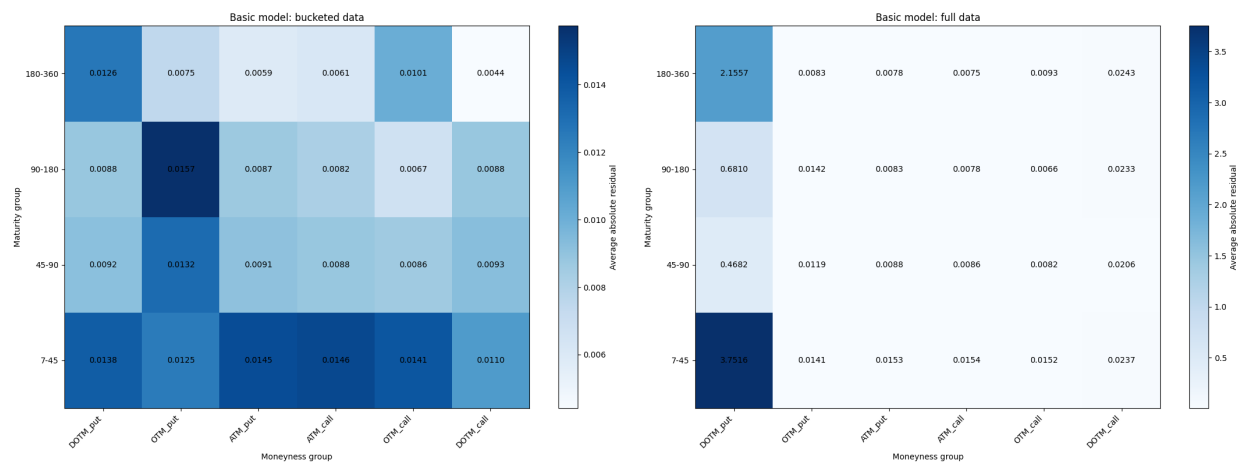


Figure 10: **Average absolute residual heatmaps for the basic VAR model.** The left panel reports residuals for the bucketed contract data, while the right panel reports residuals for the full contract data. Residuals are shown across moneyness and maturity groups. Darker shades of blue denote larger average absolute residuals.

D2. Five-factor VAR model (state-space)

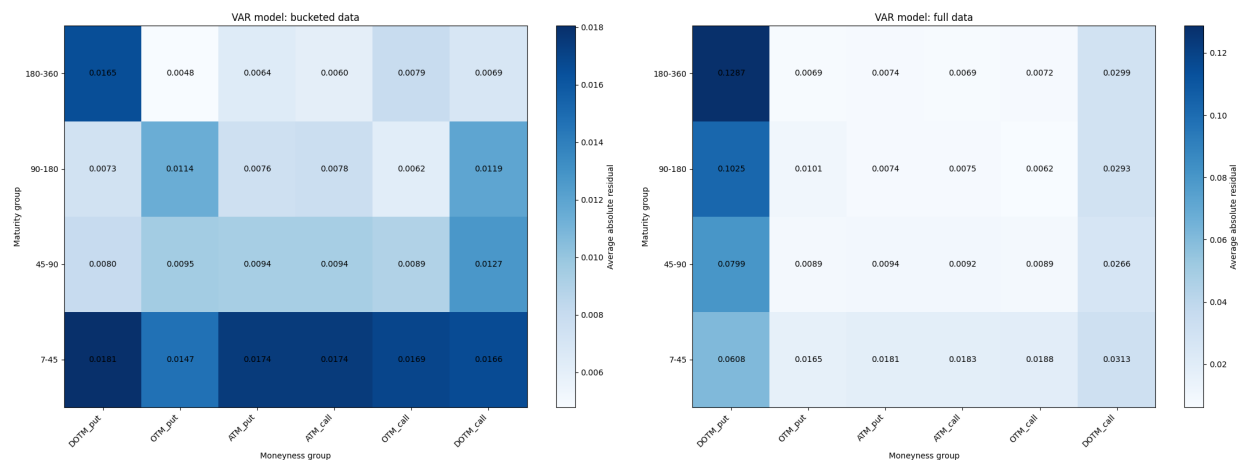


Figure 11: **Average absolute residual heatmaps for the five-factor VAR state-space model.** The left panel reports residuals for the bucketed contract data, while the right panel reports residuals for the full contract data. Residuals are shown across moneyness and maturity groups. Darker shades of blue denote larger average absolute residuals.

D3. Five-factor Ornstein-Uhlenbeck model

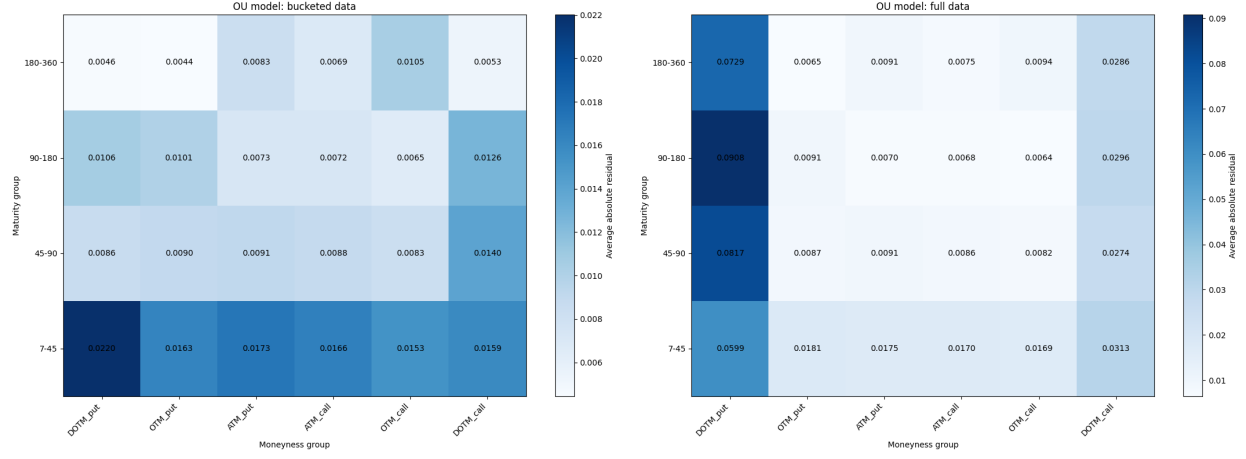


Figure 12: **Average absolute residual heatmaps for the five-factor Ornstein-Uhlenbeck model.** The left panel reports residuals for the bucketed contract data, while the right panel reports residuals for the full contract data. Residuals are shown across moneyness and maturity groups. Darker shades of blue denote larger average absolute residuals.

D4. Tensor spline Ornstein-Uhlenbeck model

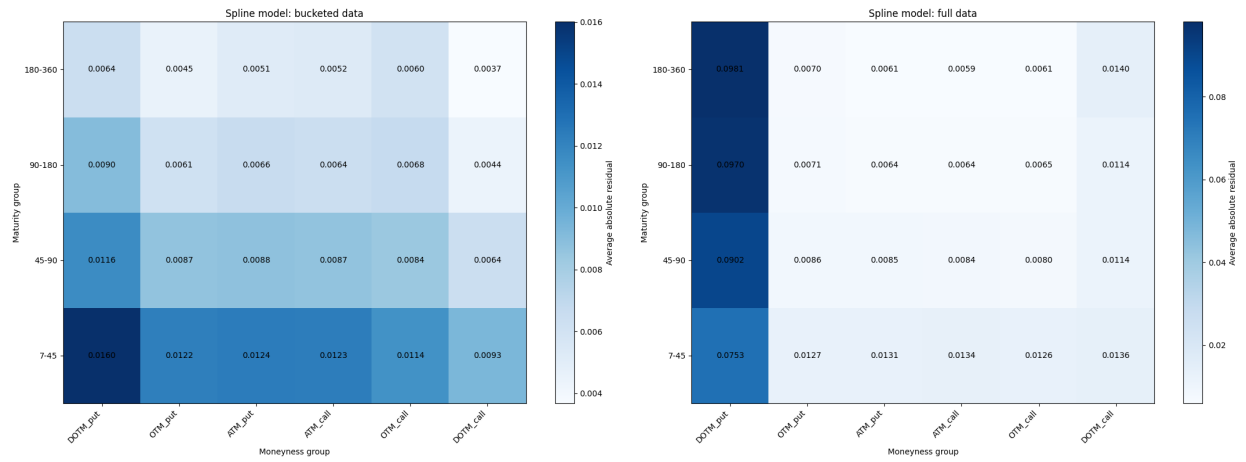


Figure 13: **Average absolute residual heatmaps for the tensor spline Ornstein-Uhlenbeck model.** The left panel reports residuals for the bucketed contract data, while the right panel reports residuals for the full contract data. Residuals are shown across moneyness and maturity groups. Darker shades of blue denote larger average absolute residuals.

D5. Neurally smoothed tensor spline Ornstein-Uhlenbeck model

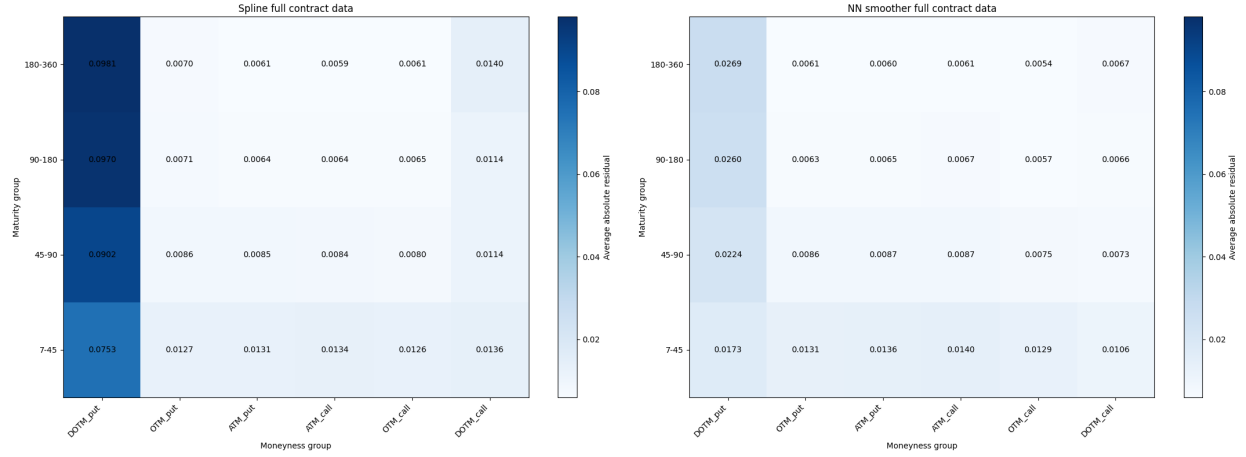


Figure 14: **Average absolute residual heatmaps for the tensor spline Ornstein-Uhlenbeck and the neural smoother models.** The left panel reports residuals for the spline prior full-contract data, while the right panel reports residuals for the neural smoother full contract data. Residuals are shown across moneyiness and maturity groups. Darker shades of blue denote larger average absolute residuals.

Appendix E: Pairwise Diebold-Mariano tests and Model Confidence Sets

This subsection reports the additional pairwise Diebold-Mariano tests and Model Confidence Sets used to assess forecast accuracy. The heatmaps summarize pairwise loss-differential comparisons, while the MCS tables show the sequential elimination procedure and the final set of models that remain under each loss function. The Diebold-Mariano tests are conducted on the full sample for both error metrics, in both the aggregate sample and the region-specific sample, while the MCS extends this to all evaluated subsamples.

E1. Pairwise Diebold-Mariano tests for bucketed aggregate data

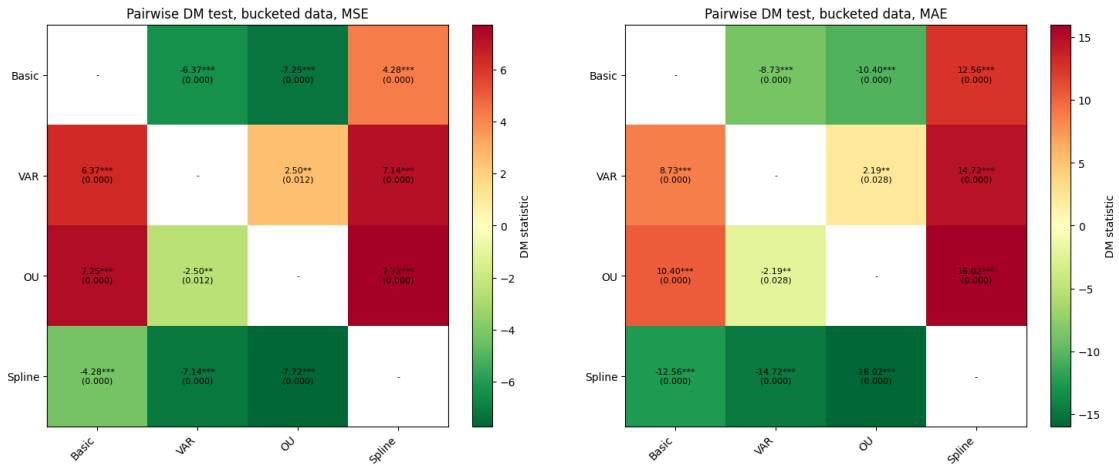


Figure 15: **Pairwise Diebold-Mariano test heatmaps for bucketed aggregate data.** The figure reports pairwise Diebold-Mariano test statistics for the bucketed-data forecasts using MSE loss on the left and MAE loss on the right. Rows denote the first model in the loss differential, and columns denote the comparison model. Values in parentheses report the corresponding p -values, and significance stars indicate conventional significance levels.

E2. Pairwise Diebold-Mariano tests for bucketed region-specific data

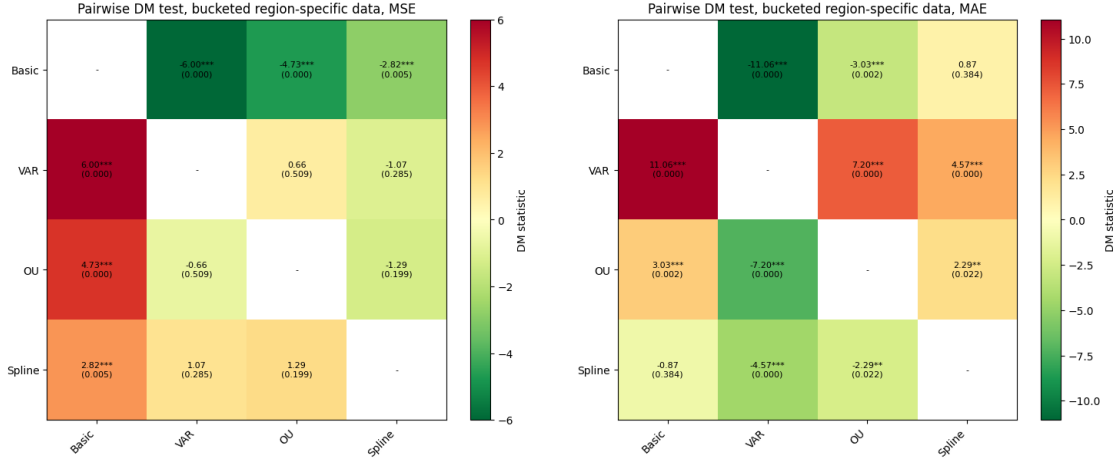


Figure 16: **Pairwise Diebold-Mariano test heatmaps for bucketed region-specific data.** The figure reports pairwise Diebold-Mariano test statistics for the bucketed data forecasts in the DOTM put region using MSE loss on the left and MAE loss on the right. Rows denote the first model in the loss differential, and columns denote the comparison model. Values in parentheses report the corresponding p -values, and significance stars indicate conventional significance levels.

E3. Pairwise Diebold-Mariano tests for full-contract aggregate data

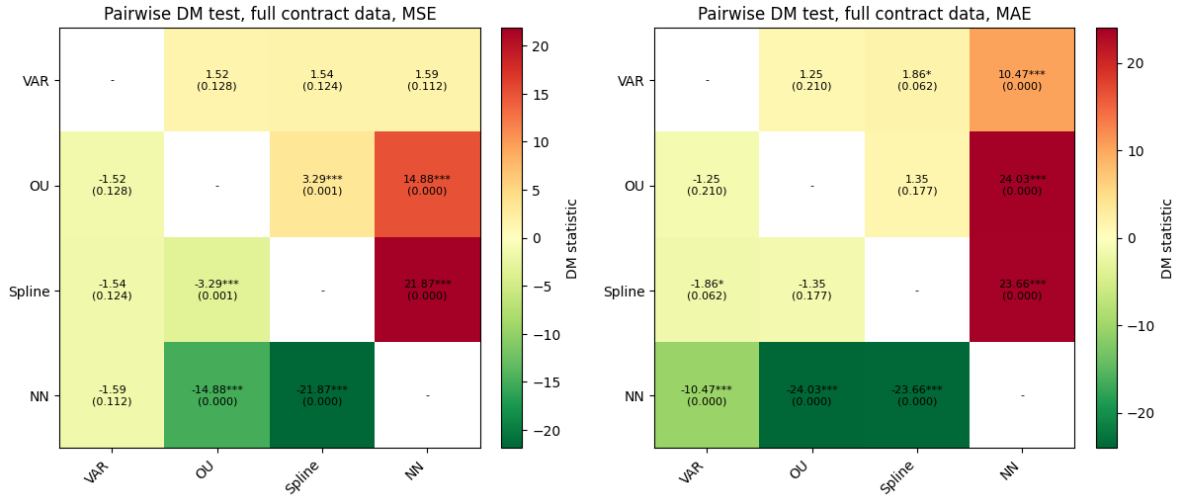


Figure 17: **Pairwise Diebold-Mariano test heatmaps for full-contract aggregate data.** The figure reports pairwise Diebold-Mariano test statistics for the full-contract-data forecasts using MSE loss on the left and MAE loss on the right. Rows denote the first model in the loss differential, and columns denote the comparison model. Values in parentheses report the corresponding p -values, and significance stars indicate conventional significance levels.

E4. Pairwise Diebold-Mariano tests for full-contract region-specific data

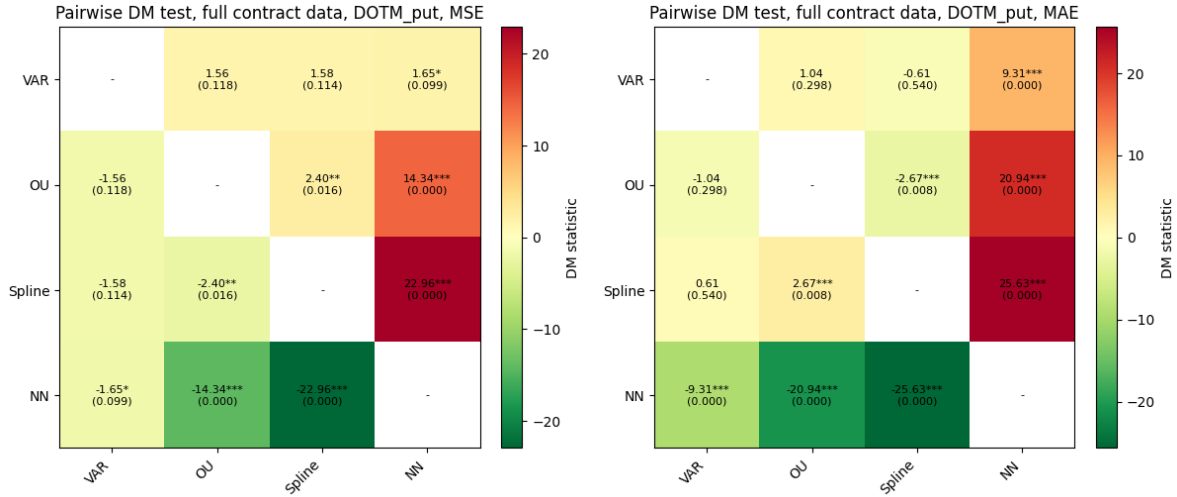


Figure 18: **Pairwise Diebold-Mariano test heatmaps for full-contract region-specific data.** The figure reports pairwise Diebold-Mariano test statistics for the full-contract-data forecasts in the DOTM put region using MSE loss on the left and MAE loss on the right. Rows denote the first model in the loss differential and columns denote the comparison model. Values in parentheses report the corresponding p -values, and significance stars indicate conventional significance levels.

E5. Model Confidence Sets for bucketed aggregate data

MSE				MAE			
Removed	Average loss	Statistic	p-value	Removed	Average loss	Statistic	p-value
Full sample							
VAR	0.000246	7.722	0.000	VAR	0.010826	16.022	0.000
OU	0.000235	7.722	0.000	OU	0.010649	16.022	0.000
Basic	0.000205	4.276	0.000	Basic	0.010089	12.559	0.000
Final MCS: Spline				Final MCS: Spline			
Non-COVID period							
VAR	0.000109	7.976	0.000	VAR	0.007785	10.659	0.000
OU	0.000105	7.976	0.000	OU	0.007688	10.659	0.000
Basic	0.000093	6.410	0.000	Basic	0.007303	8.563	0.000
Final MCS: Spline				Final MCS: Spline			
COVID period							
VAR	0.000323	6.503	0.000	VAR	0.012568	13.144	0.000
OU	0.000309	6.503	0.000	OU	0.012345	13.144	0.000
Basic	0.000268	3.223	0.000	Basic	0.011683	10.056	0.000
Final MCS: Spline				Final MCS: Spline			

Table 20: **Model Confidence Set results for bucketed aggregate data.** The table reports the sequential MCS elimination results for the bucketed aggregate forecasting exercise across three evaluation samples: full sample, COVID period, and the sample excluding COVID. Results are shown separately for MSE and MAE loss functions. For each loss function, rows list the models removed from the confidence set, together with the corresponding average loss, test statistic, and bootstrap p -value. The final MCS row reports the model or models that remain in the superior set after the elimination procedure.

E6. Model Confidence Sets for bucketed region-specific data

MSE				MAE			
Removed	Average loss	Statistic	p-value	Removed	Average loss	Statistic	p-value
Full sample							
Spline	0.000342	6.002	0.000	VAR	0.012466	11.063	0.000
VAR	0.000302	6.002	0.000	OU	0.011461	3.027	0.002
OU	0.000298	4.733	0.000	Final MCS: Basic, Spline			
Final MCS: Basic							
Non-COVID period							
VAR	0.000176	9.871	0.000	VAR	0.010269	10.168	0.000
OU	0.000126	4.532	0.000	Basic	0.008663	4.439	0.000
Final MCS: Basic, Spline				OU	0.008174	4.084	0.000
				Final MCS: Spline			
COVID period							
Spline	0.000477	4.523	0.000	VAR	0.013724	7.266	0.000
OU	0.000396	4.523	0.001	OU	0.013343	5.286	0.000
VAR	0.000374	4.196	0.000	Final MCS: Basic, Spline			
Final MCS: Basic							

Table 21: **Model Confidence Set results for bucketed region-specific data.** The table reports the sequential MCS elimination results for the bucketed DOTM put forecasting exercise across three evaluation samples: full sample, COVID period, and the sample excluding COVID. Results are shown separately for MSE and MAE loss functions. For each loss function, rows list the models removed from the confidence set, together with the corresponding average loss, test statistic, and bootstrap p -value. The final MCS row reports the model or models that remain in the superior set after the elimination procedure.

E7. Model Confidence Sets for full-contract aggregate data

MSE				MAE			
Removed	Average loss	Statistic	p-value	Removed	Average loss	Statistic	p-value
Full sample							
VAR	0.213779	21.873	0.000	VAR	0.041675	24.029	0.000
OU	0.009985	21.873	0.000	OU	0.038484	24.029	0.000
Spline	0.007516	21.873	0.000	Spline	0.036426	23.657	0.000
Final MCS: Smoother				Final MCS: Smoother			
COVID period							
OU	0.013166	14.952	0.000	OU	0.045865	20.930	0.000
VAR	0.011885	14.271	0.000	VAR	0.041172	17.182	0.000
Spline	0.007011	14.271	0.000	Spline	0.035360	15.876	0.000
Final MCS: Smoother				Final MCS: Smoother			
Excluding 2023							
OU	0.011157	18.050	0.000	OU	0.041656	21.686	0.000
VAR	0.009970	18.050	0.000	Spline	0.037696	18.995	0.000
Spline	0.007702	18.050	0.000	VAR	0.037356	18.190	0.000
Final MCS: Smoother				Final MCS: Smoother			

Table 22: **Model Confidence Set results for full-contract aggregate data.** The table reports the sequential MCS elimination results for the full-contract aggregate forecasting exercise across three evaluation samples: full sample, COVID period, and the sample excluding 2023. Results are shown separately for MSE and MAE loss functions. For each loss function, rows list the models removed from the confidence set, together with the corresponding average loss, test statistic, and bootstrap p -value. The final MCS row reports the model or models that remain in the superior set after the elimination procedure.

E8. Model Confidence Sets for full-contract region-specific data

MSE				MAE			
Removed	Average loss	Statistic	p-value	Removed	Average loss	Statistic	p-value
Full sample							
VAR	0.492299	22.958	0.000	Spline	0.087626	25.627	0.000
OU	0.026862	22.958	0.000	VAR	0.083392	20.940	0.000
Spline	0.021868	22.958	0.000	OU	0.077022	20.940	0.000
Final MCS: Smoother				Final MCS: Smoother			
COVID period							
OU	0.035760	15.054	0.000	OU	0.093487	17.984	0.000
VAR	0.031905	15.054	0.000	Spline	0.085577	16.675	0.000
Spline	0.021326	15.054	0.000	VAR	0.080110	13.279	0.000
Final MCS: Smoother				Final MCS: Smoother			
Excluding 2023							
OU	0.030144	18.982	0.000	Spline	0.090002	20.545	0.000
VAR	0.026644	18.982	0.000	OU	0.083538	18.618	0.000
Spline	0.022455	18.982	0.000	VAR	0.071515	14.216	0.000
Final MCS: Smoother				Final MCS: Smoother			

Table 23: **Model Confidence Set results for full-contract region-specific data.** The table reports the sequential MCS elimination results for the full-contract DOTM put forecasting exercise across three evaluation samples: full sample, COVID period, and the sample excluding 2023. Results are shown separately for MSE and MAE loss functions. For each loss function, rows list the models removed from the confidence set, together with the corresponding average loss, test statistic, and bootstrap p-value. The final MCS row reports the model or models that remain in the superior set after the elimination procedure.

E9. Pairwise Diebold-Mariano tests for density forecasts

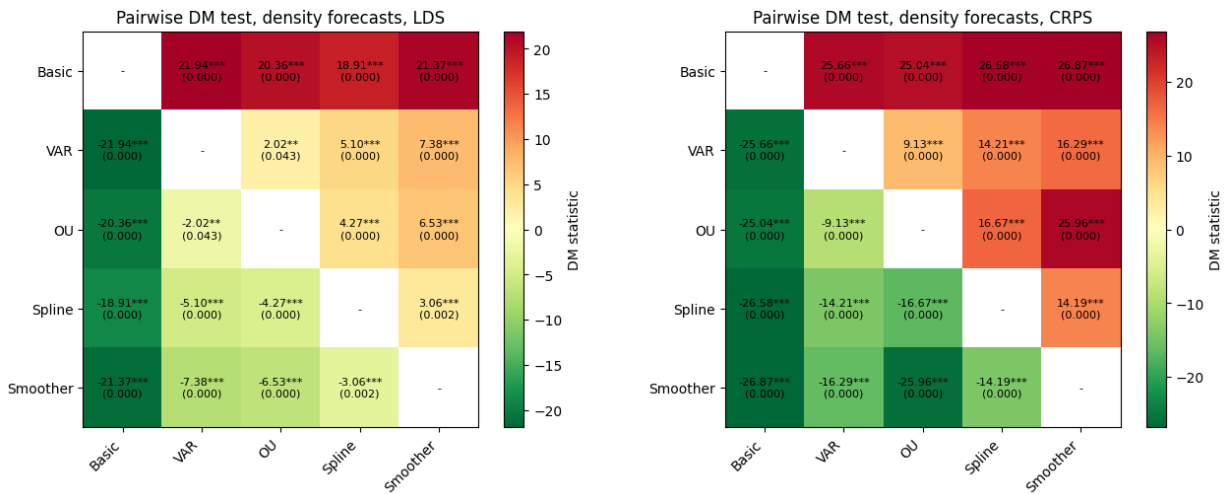


Figure 19: **Pairwise Diebold-Mariano test heatmaps for risk-neutral density forecasting scores.** The figure reports pairwise Diebold-Mariano test statistics for the full-contract risk-neutral density forecasts using log-density score (LDS) on the left and continuous ranked probability score (CRPS) on the right. Rows denote the first model in the loss differential and columns denote the comparison model. Values in parentheses report the corresponding p-values, and significance stars indicate conventional significance levels.

E10. Pairwise Diebold-Mariano tests for tick losses

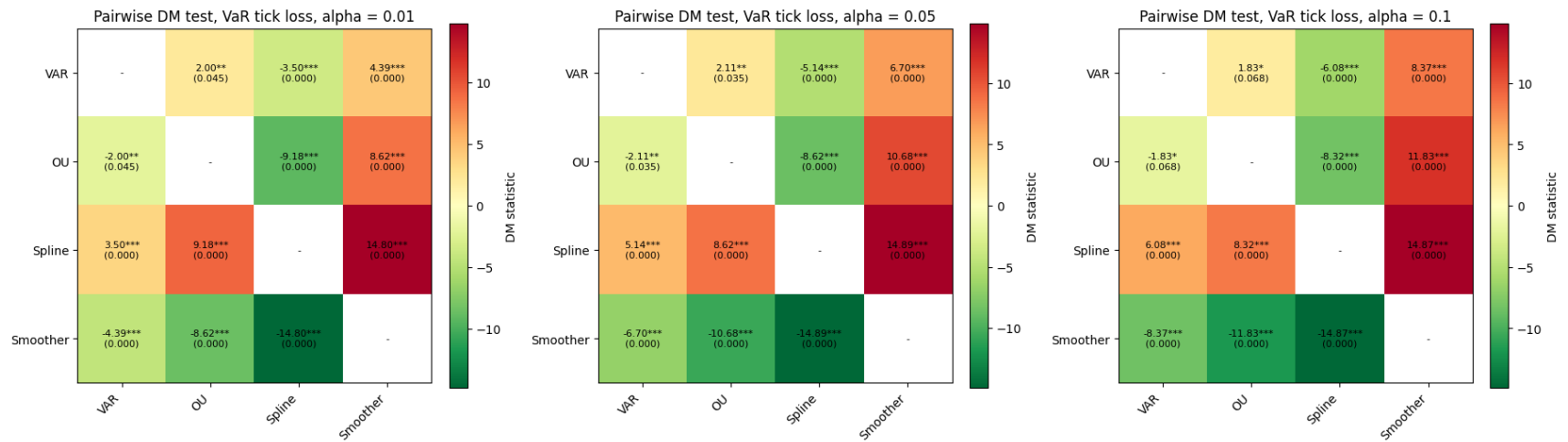


Figure 20: **Pairwise Diebold-Mariano test heatmaps based on tick losses.** The figure reports pairwise Diebold-Mariano test statistics for the full-contract Value-at-Risk tick losses at significance levels of 1%, 5%, and 10%. Rows denote the first model in the loss differential, and columns denote the comparison model. Values in parentheses report the corresponding p -values, and significance stars indicate conventional significance levels.

Appendix F: Robustness checks

This subsection reports the robustness-check results obtained from the rolling-window estimation design described in Subsubsection 5.1.6. Following a rolling estimation window of 500 observations and an out-of-sample evaluation window of 250 observations, model parameters are repeatedly re-estimated through time to account for potential structural changes in market conditions. Forecast performance is evaluated using full-contract implied volatility data and compared across alternative sample periods and the deep out-of-the-money put region.

F1. Table of error metrics

Model ID	All regions				Region-specific (DOTM put)			
	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
Full test sample (2012-2023)								
M1 (VAR)	207.86372	0.06746 ***	9.32×10^5	7.42	465.58903	0.14752 ***	1.03×10^6	11.06
M2 (OU)	0.28105	0.03569 ***	1.26×10^3	3.93	0.64902	0.07181 ***	1.44×10^3	5.38
M3 (Spline)	0.00487 ***	0.02719 ***	2.18×10^1	2.99	0.01374 ***	0.06318 ***	3.04×10^1	4.74
M4 (Smoother)	0.00022	0.00909	1.00	1.00	0.00045	0.01334	1.00	1.00
COVID period (2020-2022)								
M1 (VAR)	0.01533 ***	0.04980 ***	3.66×10^1	3.82	0.04034 ***	0.09853 ***	4.25×10^1	4.60
M2 (OU)	0.02945 ***	0.05569 ***	7.03×10^1	4.27	0.07653 ***	0.11507 ***	8.06×10^1	5.37
M3 (Spline)	0.00848 ***	0.03413 ***	2.02×10^1	2.61	0.02530 ***	0.08401 ***	2.66×10^1	3.92
M4 (Smoother)	0.00042	0.01305	1.00	1.00	0.00095	0.02141	1.00	1.00
Excluding 2023								
M1 (VAR)	0.00673 ***	0.03060 ***	2.98×10^1	3.39	0.01736 ***	0.05816 ***	3.80×10^1	4.40
M2 (OU)	0.01085 ***	0.03138 ***	4.80×10^1	3.48	0.02796 ***	0.06094 ***	6.12×10^1	4.61
M3 (Spline)	0.00502 ***	0.02776 ***	2.22×10^1	3.07	0.01410 ***	0.06407 ***	3.09×10^1	4.85
M4 (Smoother)	0.00023	0.00903	1.00	1.00	0.00046	0.01321	1.00	1.00

Table 24: **Model comparison using full-contract implied volatility surface data with alternative estimation-evaluation split.** The table reports aggregate forecast errors and DOTM put region-specific forecast errors for each model specification over the full test sample, the COVID period, and excluding 2023. Forecast accuracy is measured using the mean squared error (MSE) and mean absolute error (MAE). For each sample period, relative MSE and MAE are computed against the smoother, which is normalized to one. Stars indicate significance from Diebold-Mariano tests comparing each model against the smoother model. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. When a relative error metric is statistically significant, values below one indicate statistically superior forecast performance relative to the smoother, while values above one indicate significantly inferior performance.

Table 24 presents the error metrics for the full-contract data one-step-ahead forecasts under the rolling-window estimation design. Relative to the regular estimation procedure, the ranking of the models remains largely unchanged. The smoother consistently delivers the lowest MSE and MAE across all selected sample splits and regions. The Diebold-Mariano tests relative to the smoother model confirm this pattern, showing that the improvements are statistically significant against the competing models. Overall, these results indicate that the main conclusions are robust to updating the estimation window through time rather than relying on a single fixed training sample. The heatmaps in Appendix F2 further show that the rolling estimation design is particularly effective in reducing errors in the deep out-of-the-money regions.

F2. Neurally smoothed tensor spline Ornstein-Uhlenbeck model heatmap

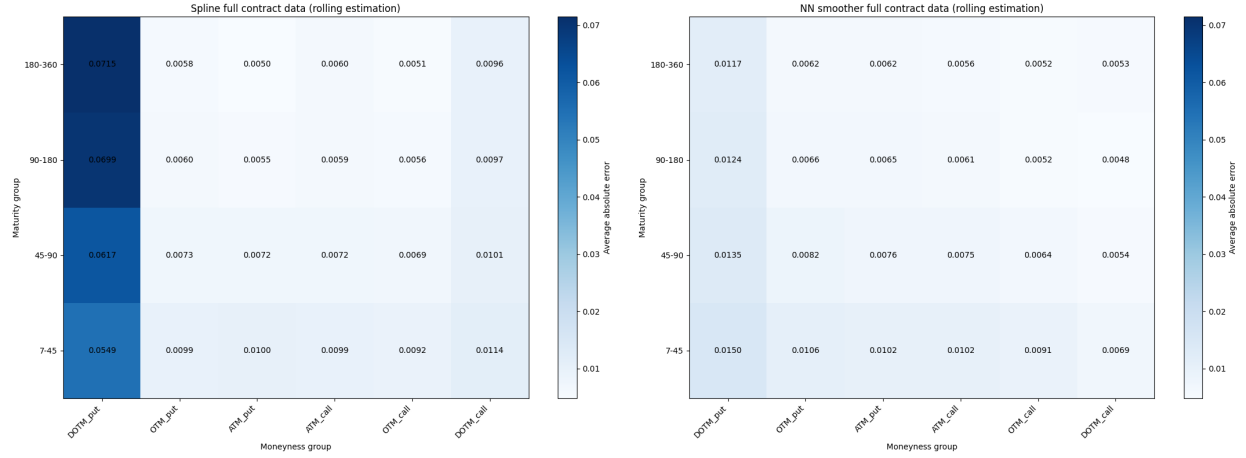


Figure 21: **Average absolute residual heatmaps for the spline and neural smoother models under rolling-window estimation.** The left panel reports average absolute errors for the prior spline model, while the right panel reports average absolute errors for the neural smoother model. Full-contract errors are shown across moneyness and maturity groups using full-contract data. Darker shades of blue denote larger average absolute errors.

F3. Pairwise Diebold-Mariano tests for full-contract aggregate data

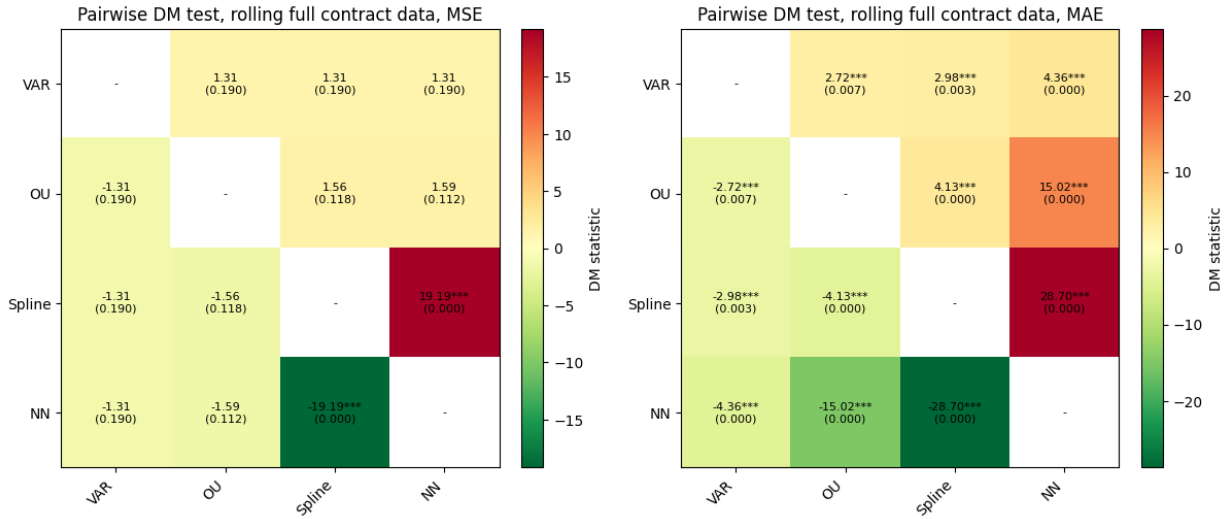


Figure 22: **Pairwise Diebold-Mariano test heatmaps for rolling-window full-contract aggregate data.** The figure reports pairwise Diebold-Mariano test statistics for the rolling full-contract-data forecasts using MSE loss on the left and MAE loss on the right. Rows denote the first model in the loss differential, and columns denote the comparison model. Values in parentheses report the corresponding p-values, and significance stars indicate conventional significance levels.

F4. Pairwise Diebold-Mariano tests for full-contract region-specific data

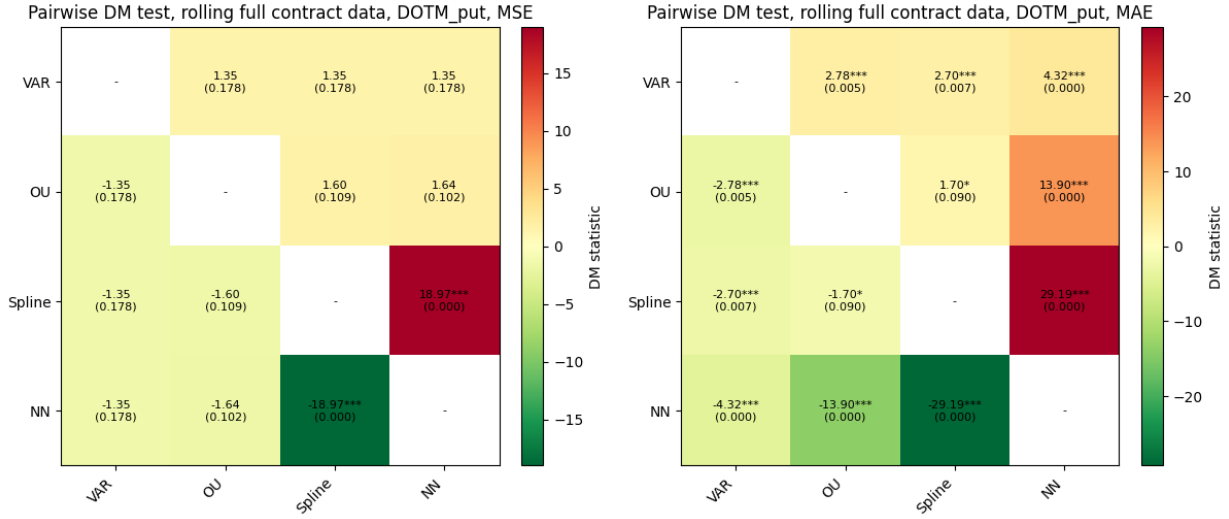


Figure 23: **Pairwise Diebold-Mariano test heatmaps for rolling-window full-contract region-specific data.** The figure reports pairwise Diebold-Mariano test statistics for the rolling full-contract-data forecasts in the DOTM put region using MSE loss on the left and MAE loss on the right. Rows denote the first model in the loss differential, and columns denote the comparison model. Values in parentheses report the corresponding p -values, and significance stars indicate conventional significance levels.

Appendix G: Arbitrage consistency

Model	Calendar arbitrage	Butterfly arbitrage	MAE
VAR	39,870 (1.23%)	31,431 (0.97%)	0.04168
OU	15,985 (0.49%)	13,249 (0.41%)	0.03848
Spline	11,309 (0.40%)	76,640 (2.69%)	0.03643

Table 25: Arbitrage violations by benchmark models. The table reports, for each benchmark model, the number of predicted option prices classified as calendar arbitrage violations and butterfly arbitrage violations. The corresponding violation rates are shown in parentheses. The table also reports the mean absolute error (MAE) for each specification.